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National Aeronautics and Space Administration

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**FLAMMABILITY, OFFGASSING, AND COMPATIBILITY
REQUIREMENTS AND TEST PROCEDURES**

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		2	2016-04-21	Administrative/Editorial Changes—Moved Appendix A, Requirements Compliance Matrix, to Appendix E and returned Appendices B-E to Appendices A-D to prevent revision of existing internal documents.
Revalidated		3	2025-6-9	Administrative change: This NASA Technical Standard is revalidated “as-is”.

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FOREWORD

This NASA Technical Standard is published by the National Aeronautics and Space Administration (NASA) to provide uniform engineering and technical requirements for processes, procedures, practices, and methods that have been endorsed as standard for NASA programs and projects, including requirements for selection, application, and design criteria of an item.

This NASA Technical Standard is approved for use by NASA Headquarters and NASA Centers and Facilities and may be cited in contract, program, and other Agency documents as a technical requirement. It may also apply to the Jet Propulsion Laboratory and other contractors only to the extent specified or referenced in applicable contracts.

This NASA Technical Standard establishes requirements for evaluation, testing, and selection of materials that are intended for use in space vehicles and associated Ground Support Equipment (GSE). These requirements may also apply to facilities used during assembly, test, and flight operations when imposed contractually to ensure safety of personnel and flight hardware. Included are requirements, criteria, and test methods for evaluating the flammability, offgassing, and compatibility of materials. Provisions of this NASA Technical Standard will be included, as applicable, in all future contracts and programs involving space vehicles, payloads, and associated support equipment. This NASA Technical Standard was previously published as NASA Handbook (NHB) 8060.1, Flammability, Odor, Offgassing, and Compatibility Requirements and Test Procedures for Materials in Environments that Support Combustion.

Requests for information should be submitted via “Feedback” at <https://standards.nasa.gov>. Requests for changes to this NASA Technical Standard should be submitted via MSFC Form 4657, Change Request for a NASA Engineering Standard.

Original Signed By:

08/26/2011

Michael Ryschkewitsch
NASA Chief Engineer

Approval Date

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FLAMMABILITY, OFFGASSING, AND COMPATIBILITY REQUIREMENTS AND TEST PROCEDURES

1. SCOPE

1.1 Purpose

The purpose of this NASA Technical Standard is to establish National Aeronautics and Space Administration (NASA) program requirements for evaluation, testing, and selection of materials to preclude unsafe conditions related to flammability, offgassing, and fluid compatibility. Additional materials performance requirements may be specified in other program or NASA Center-specific documentation.

1.2 Applicability

1.2.1 [FOCR 1] Materials intended for use in space vehicles, specified test facilities, and contractually specified Ground Support Equipment (GSE) shall meet the requirements of this NASA Technical Standard.

1.2.2 [FOCR 2] Responsible NASA Centers' materials and processes (M&P) organizations shall include applicable requirements of this NASA Technical Standard in their materials control programs.

This NASA Technical Standard is approved for use by NASA Headquarters and NASA Centers and Facilities and may be cited in contract, program, and other Agency documents as a technical requirement. It may also apply to the Jet Propulsion Laboratory and other contractors only to the extent specified or referenced in applicable contracts.

Verifiable requirement statements are numbered and indicated by the word "shall"; this NASA Technical Standard contains 425 requirements. Explanatory or guidance text (other than definitions) is indicated in italics beginning in section 4. To facilitate requirements selection and verification by NASA programs and projects, a Requirements Compliance Matrix is provided in Appendix EA.

1.3 Tailoring

[FOCR 3] Tailoring of this NASA Technical Standard for application to a specific program or project shall be formally documented as part of program or project requirements and approved by the responsible Technical Authority in accordance with NPR 7120.5, NASA Space Flight Program and Project Management Requirements.

2. APPLICABLE DOCUMENTS

2.1 General

The documents listed in this section contain provisions that constitute requirements of this NASA Technical Standard as cited in the text.

2.1.1 [FOCR 4] The latest issuances of cited documents shall apply unless specific versions are designated.

2.1.2 [FOCR 5] Non-use of specifically designated versions shall be approved by the responsible Technical Authority.

The applicable documents are accessible at <https://standards.nasa.gov>, may be obtained directly from the Standards Developing Body or other document distributors, or information for obtaining the document is provided.

2.2 Government Documents

Department of Defense

MIL-PRF-25604	Propellant, uns-Dimethylhydrazine
MIL-PRF-26536	Propellant, Hydrazine
MIL-PRF-26539	Propellants, Dinitrogen Tetroxide
MIL-PRF-27402	Propellant, Hydrazine-uns-Dimethylhydrazine (50% N ₂ H ₄ – 50% UDMH)
MIL-PRF-27404	Propellant, Monomethylhydrazine
MIL-STD-2223	Test Methods for Insulated Electric Wire (Method 3007, Dry Arc-Propagation Resistance only)

NASA

NASA-STD-5005	Standard for the Design and Fabrication of Ground Support Equipment
NASA-STD-6016	Standard Materials and Processes Requirements for Spacecraft
NASA/TM-2007-213740	Guide for Oxygen Compatibility Assessments on Oxygen Components and Systems
NPD 7100.8	Protection of Human Research Subjects

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NPR 7100.1	Protection of Human Research Subjects
NPR 7120.5	NASA Space Flight Program and Project Management Requirements

2.3 Non-Government Documents

ASTM International (ASTM) (originally known as the American Society for Testing and Materials)

ASTM D240	Standard Test Method for Heat of Combustion of Liquid Hydrocarbon Fuels by Bomb Calorimeter
ASTM D2863	Standard Test Method for Measuring the Minimum Oxygen Concentration to Support Candle-Like Combustion of Plastics (Oxygen Index)
ASTM D3294	Standard Specification for Polytetrafluoroethylene (PTFE) Resin Molded Sheet and Molded Basic Shapes
ASTM D4809	Standard Test Method for Heat of Combustion of Liquid Hydrocarbon Fuels by Bomb Calorimeter (Precision Method)
ASTM E502	Standard Test Method for Selection and Use of ASTM Standards for the Determination of Flash Point of Chemicals by Closed Cup Methods
ASTM E1354	Standard Test Method for Heat and Visible Smoke Release Rates for Materials and Products Using an Oxygen Consumption Calorimeter
ASTM G72	Standard Test Method for Autogenous Ignition Temperature of Liquids and Solids in a High-Pressure Oxygen-Enriched Environment
ASTM G74	Standard Test Method for Ignition Sensitivity of Nonmetallic Materials and Components by Gaseous Fluid Impact
ASTM G86	Standard Test Method for Determining Ignition Sensitivity of Materials to Mechanical Impact in Ambient Liquid Oxygen and Pressurized Liquid and Gaseous Oxygen Environments

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ASTM G124-10	Standard Test Method for Determining the Combustion Behavior of Metallic Materials in Oxygen-Enriched Atmospheres
ASTM G125	Standard Test Method for Measuring Liquid and Solid Material Fire Limits in Gaseous Oxidants

International Organization for Standardization (ISO)

ISO 14624-1:2003	Space systems — Safety and compatibility of materials — Part 1: Determination of upward flammability of materials
ISO 14624-2:2003	Space systems — Safety and compatibility of materials — Part 2: Determination of flammability of electrical-wire insulation and accessory materials
ISO 14624-3:2005	Space systems — Safety and compatibility of materials — Part 3: Determination of offgassed products from materials and assembled articles

Society of Automotive Engineers International (SAE)

SAE AS4373	Test Methods for Insulated Electric Wire (Method 508, Dry Arc Propagation Resistance only)
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2.4 Order of Precedence

2.4.1 The requirements and standard practices established in this NASA Technical Standard do not supersede or waive existing requirements and standard practices found in other Agency documentation.

2.4.2 [FOCR 6] Conflicts between this NASA Technical Standard and other requirements documents shall be resolved by the responsible Technical Authority.

3. ACRONYMS AND DEFINITIONS

3.1 Acronyms and Abbreviations

~	approximately
°C	degree Celsius
°F	degree Fahrenheit
≥	equal to or greater than
≤	equal to or less than
>	greater than
<	less than

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μ	micro
μg	microgram(s)
μl	microliter(s)
-	minus
Ω	ohm(s)
%	percent
+	plus
±	plus or minus
ASTM	ASTM International (originally known as the American Society for Testing and Materials)
cal	calorie(s)
cc	cubic centimeter(s)
cm	centimeter(s)
CO	carbon monoxide
CO ₂	carbon dioxide
DSC	differential scanning calorimetry
e.g.	for example
Eq.	equation
ft	foot, feet
ft-lb	foot-pound(s)
FOCR	Flammability, Offgassing, and Compatibility Requirements
FTIR	Fourier transform infrared spectroscopy
g	gram(s)
GLP	good laboratory practice(s)
GN ₂	gaseous nitrogen
GOX	gaseous oxygen
GSE	ground support equipment
H ₂	hydrogen
hr	hour(s)
IEC	International Electrotechnical Commission
in	inch(es)
ISO	International Organization for Standardization
ISS	International Space Station
J	joule(s)
JSC	Johnson Space Center
k	kilo
kg	kilogram(s)
kJ	kilojoule(s)
kPa	kiloPascal(s)
l	liter(s)
lb	pound(s)
lbf	pound(s)-force

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LOX	liquid oxygen
M&P	materials and processes
M	mega
m	meter(s)
MAPTIS	Materials and Processes Technical Information System
mg	milligram(s)
MIL	military
min	minute(s)
MJ	megajoule(s)
ml	milliliter(s)
MLW	maximum limit weight
MPa	megaPascal(s)
mm	millimeter(s)
MMH	monomethylhydrazine
MOP	maximum operating pressure
MSDS	material safety data sheet
msec	millisecond(s)
MSFC	Marshall Space Flight Center
MUA	materials usage agreement
N ₂	nitrogen
N ₂ H ₄	hydrazine
N	newton(s)
NASA	National Aeronautics and Space Administration
NHB	NASA Handbook
NPD	NASA Policy Directive
NPR	NASA Procedural Requirements
NVR	nonvolatile residue
O ₂	oxygen
OCA	oxygen compatibility assessment
OSHA	Occupational Safety and Health Administration
oz	ounce(s)
P	minimum contact pressure
ppb	parts per billion
ppm	parts per million
PRF	performance
psi	pounds per square inch
psia	pounds per square inch absolute
psig	pounds per square inch gauge
RMS	root mean square
rpm	revolutions per minute
SAE	Society of Automotive Engineers International
sec	second(s)

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SI	Système Internationale or metric system of measurement
SMAC	spacecraft maximum allowable concentration
STD	standard
T	toxic hazard index
TM	technical memorandum
UDMH	unsymmetrical dimethylhydrazine
V	volt(s)
v	linear surface velocity
VCS	voluntary consensus standard
W	watt(s)

3.2 Definitions

Test-specific definitions are also presented in descriptions of individual tests in this NASA Technical Standard (section 7 and Appendix A).

Ambient (room) temperature: Room temperature is 23 (± 5) °C (73 (± 9) °F).

Arc: A continuous, luminous discharge of electric current crossing a gap between two conducting surfaces.

Arc tracking (arc propagation): That phenomena whereby an arc between two or more wires, once initiated, will sustain itself through a conductive path provided by degradation of the insulation for a measurable length.

Assembled article: An assembled article could be any component or assembly of components that is not a single material.

Autogenous ignition temperature (autoignition temperature): The lowest temperature at which a material will spontaneously ignite under specific test conditions.

Degradation: An adverse physical or chemical change in a substance.

Enriched oxygen environment: Any environment that contains >21 percent oxygen by volume.

Flash point: The lowest temperature, corrected to a barometric pressure of 101.3 Pa (14.7 psia) at which application of an ignition source causes the vapors of a specimen to ignite under specified test conditions.

Gaseous fluid impact: Heat delivered to a specimen by rapid fluid compression; may also be referred to as pneumatic impact, rapid pressurization, or adiabatic compression.

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Gaseous oxygen (GOX): Gaseous environment that contains ≥ 99.5 percent oxygen.

Good Laboratory Practices (GLP): A system of management controls for laboratories and research organizations to ensure the consistency and reliability of results as outlined in the Organisation for Economic Co-operation and Development Principles of GLP and national regulations.

Hermetically sealed: Contained such that the leak rate does not exceed 1×10^{-4} cc/sec.

Liquid oxygen (LOX): Cryogenic liquid environment that contains ≥ 99.5 percent oxygen.

Mechanical impact: Energy delivered by a plummet that has been dropped from a pre-established height onto a striker pin in contact with a specimen.

Offgassed product: An organic or inorganic compound evolved as a gas from a material or assembled article.

Offgassing: The evolution of gaseous products from a liquid or solid material into an atmosphere.

Oxygen Compatibility Assessment (OCA): A systematic approach for identifying and addressing the fire hazards in an oxygen system; this process emphasizes the evaluation of ignition mechanisms and the application of materials test data.

Reaction: A chemical change in which a substance decomposes, combines with other substances, or interchanges constituents with other substances.

Reactive fluids: Fluids that are capable of reacting with materials in such a way as to create a hazard. Incompatibility between fluids and materials can result in hazards such as materials degradation and exothermic reactions that can lead to fire or explosion. Testing required by this document for reactive fluids other than oxygen is limited to propellants (hydrazine, monomethylhydrazine, unsymmetrical dimethylhydrazine, Aerozine 50, nitrogen tetroxide) and ammonia.

Single-barrier failure: A leak through a barrier within a component that permits the fluid to contact the materials directly behind the barrier. Single barriers include mechanical joints (for example, B-nuts; O-rings, gaskets, and bladders; and metallic and nonmetallic diaphragms). Structural parts (for example, pressure lines and tanks, welded or brazed joints, and redundant seals in series that have been pressure-tested individually before use) are not considered to be single barriers.

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Spacecraft Maximum Allowable Concentration (SMAC): The maximum concentration of an offgassed product that is allowed in the habitable area of the spacecraft for a specified duration.

Technical Authority: The Technical Authority provides technical checks and balances by assuring that safety and mission success, relevant technical standards, engineering work, and safety and reliability analysis products are being conducted properly in accordance with established, high-reliability processes independent of nontechnical program/project constraints.

Voluntary Consensus Standard (VCS): Technical standard that is developed or adopted by voluntary consensus standard bodies, both domestic and foreign.

Worst-case anticipated use environment: Most hazardous pressure, temperature, material thickness, and/or fluid exposure conditions. These conditions can occur during flight or during other operational periods (for example, ground testing, checkout, preflight).

4. EVALUATION AND TESTING OF MATERIALS

4.1 Material Evaluation Requirements

- a. [FOCR 7] Materials located in habitable areas of spacecraft, including the materials of the spacecraft, stowed equipment, and experiments, shall be evaluated for flammability and offgassing.
- b. [FOCR 8] All materials located in other areas shall be evaluated for flammability.
- c. [FOCR 9] Materials that are exposed to LOX, GOX, and other reactive fluids shall be evaluated for compatibility with the fluid in their use application.
- d. [FOCR 10] Materials exposed to pressurized breathing gases also shall be evaluated for offgassing.
- e. [FOCR 11] Electrical insulation materials also shall be evaluated for arc tracking.
- f. [FOCR 12] The worst-case anticipated use environment (most hazardous pressure, temperature, material thickness, and fluid exposure conditions) shall be used in the evaluation of material suitability.

Materials that have been shown to meet the acceptance criteria of the required tests are acceptable for further consideration in design.

Whenever possible, materials should be selected that have already been shown to meet the test criteria in the use environment.

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Existing test data are compiled in the NASA Marshall Space Flight Center (MSFC) Materials and Processes Technical Information System (MAPTIS). MAPTIS can be accessed at <http://maptis.nasa.gov/>.

4.2 Materials Usage Agreement (MUA)

a. [FOCR 13] Systems containing materials that have not been tested or do not meet the criteria of the required tests shall be verified to be acceptable in the use configuration by analysis or configurational testing through the use of an MUA.

b. [FOCR 14] An MUA shall be prepared in accordance with the MUA system described in NASA-STD-6016, Standard Materials and Processes Requirements for Spacecraft, for flight hardware, or the latest version of NASA-STD-5005, Standard for the Design and Fabrication of Ground Support Equipment, for GSE.

c. [FOCR 15] The MUA shall document the acceptance rationale.

d. [FOCR 16] The MUA shall be submitted for approval as described in NASA-STD-6016 in accordance with the Program/Project Materials and Processes Selection, Control, and Implementation Plan.

5. APPLICABLE MATERIAL TESTS

The applicable material test depends on the environment to which the material is exposed. Applicable environments are habitable environments, LOX and GOX systems, breathing gases, and reactive fluids.

a. [FOCR 17] Required tests shall be conducted on materials in their worst-case exposure conditions and representative use thicknesses and product forms.

b. [FOCR 18] To be considered acceptable for use without restriction, the materials shall pass the required tests.

All nonmetals tested in accordance with NASA-STD-6001 should be retested every 10 years or as required by the responsible program/project.

Table 1, Required Tests for Each Material Use, lists the required tests for each material use. Descriptions of these test methods are provided in section 7 in this NASA Technical Standard.

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Table 1—Required Tests for Each Material Use

ENVIRONMENT			TEST NUMBER	TITLE
Habitable Flight Compartments			1	Upward Flame Propagation
			4	Electrical Wire Insulation Flammability
			7	Determination of Offgassed Products ¹
			18	Arc Tracking
Other Areas ²			1	Upward Flame Propagation
			4	Electrical Wire Insulation Flammability
			18	Arc Tracking
LOX and GOX Environments ^{3,4}			1	Upward Flame Propagation ⁵
			17	Upward Flammability of Materials in GOX ⁶
Breathing Gases ⁴			1	Upward Flame Propagation
			7	Determination of Offgassed Products
Reactive Fluids			15	Reactivity of Materials in Hydrazine, Monomethylhydrazine, Unsymmetrical Dimethylhydrazine, Aerozine 50, Nitrogen Tetroxide, and Ammonia
		¹ Not required for materials inside hermetically sealed containers. ² Includes all areas outside the habitable flight compartment. ³ [FOCR 19] Materials used in LOX or GOX environments shall be evaluated as described in section 6.3 in this NASA Technical Standard. ⁴ [FOCR 20] The need for an OCA, as described in section 6.3 in this NASA Technical Standard, shall be addressed for compressed air systems and pressurized systems containing enriched oxygen. ⁵ Required for nonmetals for pressures ≤ 345 kPa (≤ 50 psia). ⁶ Required for nonmetals for pressures > 345 kPa (> 50 psia) and for all metals used in GOX and LOX.		

c. [FOCR 21] Materials that do not meet the criteria of the required tests and that are to remain candidates for use shall be verified to be acceptable in the use configuration by analysis or test and approved through the MUA process as described in section 4.2 in this NASA Technical Standard.

Supplemental tests may be conducted to support the system hazard evaluation for materials that do not meet the criteria of the required tests; data based on other appropriate and applicable test methods also may be used to support the evaluation. In addition, supplemental tests may be required as an aid to the OCA process or as a result of the findings of the OCA process. Table 2, Supplemental Tests for Each Material Use, lists the supplemental tests for each material use.

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Table 2—Supplemental Tests for Each Material Use

ENVIRONMENT		TEST (APPENDIX A SECTION)	TITLE
Habitable Flight Compartments		2 (Appendix A.2.1)	Heat and Visible Smoke Release Rates
		3 (Appendix A.2.2)	Flash Point of Liquids
		6 (Appendix A.2.3)	Odor Assessment
		10 (Appendix A.2.4)	Simulated Panel or Major Assembly Flammability
		(Appendix A.2.11)	Electrical Arc
Other Areas		2 (Appendix A.2.1)	Heat and Visible Smoke Release Rates
		3 (Appendix A.2.2)	Flash Point of Liquids
LOX and GOX Environments		13A (Appendix A.2.5)	Mechanical Impact For Materials in Ambient Pressure LOX
		13B (Appendix A.2.5)	Mechanical Impact for Materials in Variable Pressure LOX and GOX
		ASTM G74 (Appendix A.2.6)	Gaseous Fluid Impact for Components
		ASTM G72 (Appendix A.2.8)	Autogenous Ignition Temperature
		ASTM D240; ASTM D4809 (Appendix A.2.9)	Heat of Combustion
		ASTM G125 (Appendix A.2.10)	Oxygen Index
		(Appendix A.2.11)	Electrical Arc
		(Appendix A.2.12)	Frictional Heating
		(Appendix A.2.13)	Particle Impact
Breathing Gases		6 (Appendix A.2.3)	Odor Assessment
Reactive Fluids		(Appendix A.2.7)	Reactivity and Penetration of Materials due to Incidental Exposure to Hydrazine, Monomethylhydrazine, Unsymmetrical Dimethylhydrazine, Aerozine 50, Nitrogen Tetroxide, and Ammonia
		3 (Appendix A.2.2)	Flash Point of Liquids

d. [FOCR 22] The responsible NASA program/project M&P organization shall determine when batch/lot testing is required.

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6. SYSTEM EVALUATIONS

a. [FOCR 23] System evaluations shall demonstrate the acceptability of configurations resulting from the use of flammable, incompatible, or offgassing materials.

b. [FOCR 24] System evaluations shall address single-barrier failures that can expose materials to reactive fluids.

6.1 System Flammability in Low-Pressure Air or Oxygen-Enriched Environments, Including Habitable Environments

6.1.1 [FOCR 25] A system flammability evaluation shall be conducted if materials that fail to meet the criteria of Test 1 are proposed for use in spacecraft, contractually specified test facilities, or contractually specified GSE.

This evaluation can be conducted by analysis or by testing. JSC 29353, Flammability Configuration Analysis for Spacecraft Applications, provides guidelines for hardware flammability assessment and mitigation.

6.1.2 [FOCR 26] If the analytical approach is used, the following fire control guidelines shall be addressed and the results documented and submitted to the responsible NASA program/project M&P organization for evaluation:

a. The amount and arrangement of flammable materials are controlled to prevent the spread of fire in the vehicle; any potential fire is isolated in defined areas by firebreaks or other techniques (described in JSC 29353) without propagation paths to other areas.

b. The design ensures that ignition sources are eliminated or controlled.

c. The effects of an isolated fire on personnel in spacecraft crew-habitable modules are considered.

d. The rate and magnitude of pressure and temperature increase in any potential fire will not cause structural damage to or compromise the pressure integrity of the vehicle.

6.1.3 [FOCR 27] If the test approach is used, materials that do not meet the criteria of Test 1 shall be evaluated by a configuration test, such as Test 10, which evaluates the worst possible failure modes in the worst-case operating conditions.

6.1.4 [FOCR 28] If the results of the evaluation are inconclusive and a major assembly flammability test, such as a crew module test, is required, the hardware developer shall submit a test plan to the responsible NASA program/project M&P organization for approval before test.

6.2 Spacecraft Offgassing

System evaluation requirements for bulk materials/assembled articles are specified in section 7.7.3 in this NASA Technical Standard.

Total spacecraft offgassing evaluation is the responsibility of the NASA Johnson Space Center (JSC) Toxicology Office.

6.3 Material and Component Compatibility in LOX and GOX Environments

6.3.1 [FOCR 29] Materials, components, and systems used in LOX and GOX environments shall be evaluated using the approach shown in figure 1, Approach for Selecting Materials, Components, and Systems used in LOX and GOX Environments.

This approach begins with pre-selection of materials based on flammability and combustion test data, followed by a flammability assessment. If the materials are determined to be nonflammable in their use configuration and environment and no additional material control is required, the materials may be used.

6.3.2 [FOCR 30] If the materials are determined to be flammable, an OCA shall be performed in accordance with NASA/TM-2007-213740, Guide for Oxygen Compatibility Assessments on Oxygen Components and Systems.

The OCA is used to determine whether the material may be used or whether there is a need to perform supplemental material, configuration, or component testing.

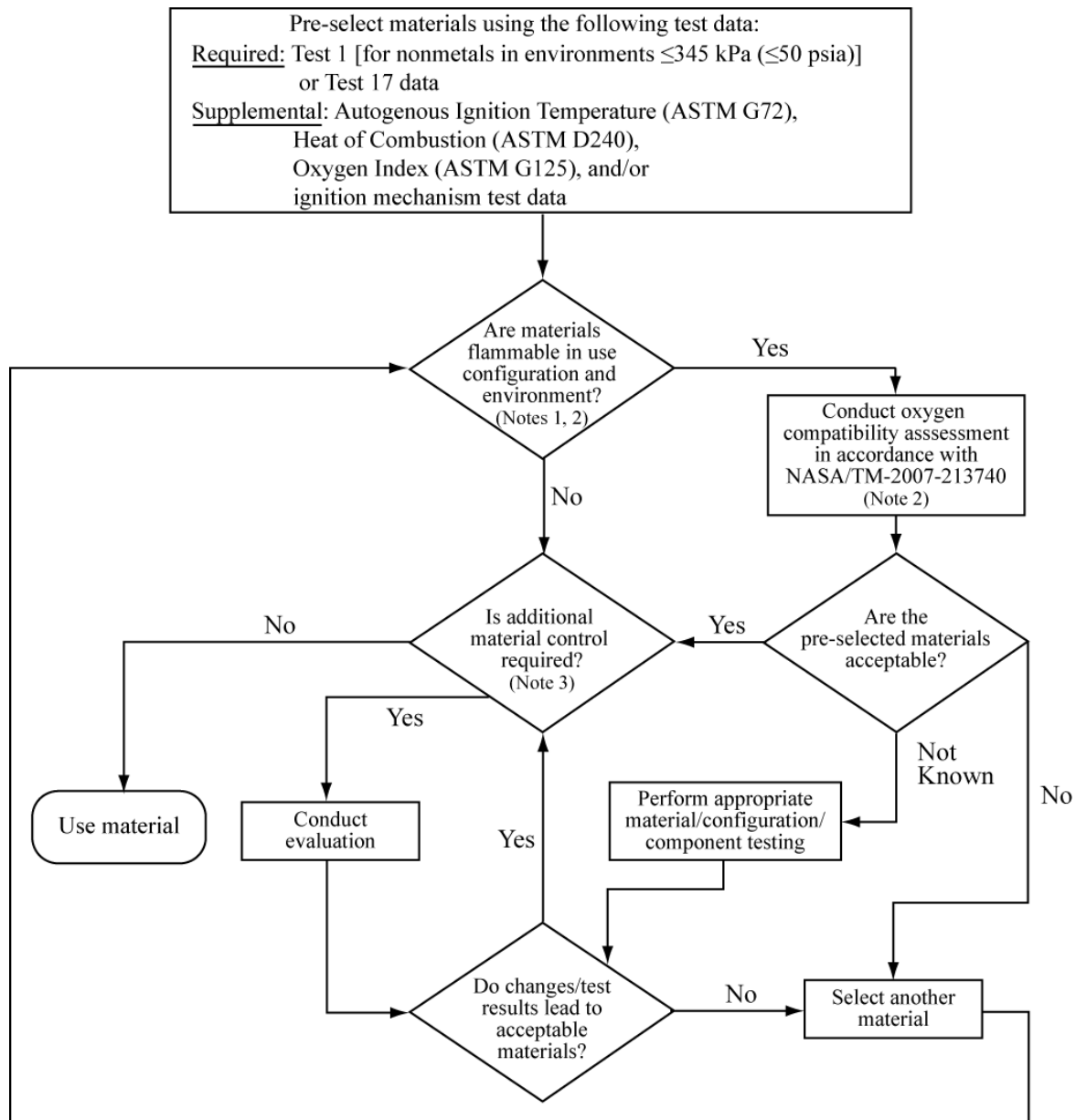
6.3.3 Compressed Air and Enriched Oxygen Systems

a. [FOCR 31] The need for an OCA shall be addressed for compressed air systems and pressurized systems containing enriched oxygen (greater than 21 percent oxygen by volume).

Such systems are inherently less hazardous than systems containing pure oxygen; the hazard increases with oxygen concentration and/or increasing pressure.

b. [FOCR 32] If it is determined that an OCA is not required, concurrence shall be obtained from the responsible program/project M&P organization.

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Notes:

1. Data used to assess material flammability has to be applicable to the specific configuration of the parts; therefore, data other than standard test data may be required.
2. Consult the responsible M&P organization to ensure that the testing data used to make flammability and ignitability assessments are applicable to the specific design configuration.
3. Additional material control refers to requirements from sources other than NASA-STD-6001, such as corrosion requirements.

Figure 1—Approach for Selecting Materials, Components, and Systems used in LOX and GOX Environments

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6.3.4 *Further details for several steps in the material and component evaluation process are:*

a. [FOCR 33] Pre-Selection of Materials. As a minimum, previously untested polymeric materials that fail Test 1 or Test 17 shall be subjected to autogenous ignition temperature (Appendix A.2.8) and heat of combustion testing (Appendix A.2.9).

The objective of the pre-selection process is to choose materials that exhibit desirable flammability and combustion characteristics. For instance, materials that do not burn in the specific environment and configuration are preferred for oxygen service. In addition, nonmetals that have high autogenous ignition temperatures, low heats of combustion, and high oxygen indices are preferred for oxygen service. Other data that may be useful in the pre-selection of materials for LOX and GOX environments include data for specific ignition mechanisms, such as particle impact, gaseous fluid impact, and mechanical impact.

b. [FOCR 34] Flammability Assessment. Data used to assess material flammability shall be applicable to the specific environment and part/component configuration.

(1) [FOCR 35] If the flammability or configuration of the materials is unknown, the materials shall be considered flammable.

(2) [FOCR 36] The responsible NASA program/project M&P organization shall approve that the data used to make flammability assessments are applicable to the specific design configuration.

In general, as pressure increases, materials such as metals, polymers, lubricants, and contaminants become flammable in 100 percent oxygen. In addition, almost all polymers are flammable in 100 percent oxygen at atmospheric pressure. Furthermore, the flammability of a material is dependent upon its configuration. For example, metals, including those that normally exhibit high resistance to ignition, are more flammable in oxygen when they have thin cross-sections (for example, thin-walled tubing) or when they are finely divided (for example, wire mesh and sintered filters). When assessing flammability, it is important to reference a cross-sectional view of each component that shows the configuration of the materials of construction.

c. OCA

(1) [FOCR 37] The results of the OCA shall be documented and submitted to the responsible NASA program/project M&P organization for concurrence.

(2) [FOCR 38] An MUA, if applicable, shall be used to submit the OCA results.

(3) [FOCR 39] The hardware developer shall verify that the materials assessed in the OCA reflect the as-built design.

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The OCA process described in NASA/TM-2007-213740 is a systematic approach that focuses on fire hazards. This process emphasizes the evaluation of ignition mechanisms and the application of materials test data.

Supplemental tests may be performed as an aid to the OCA process or as a result of findings of the OCA process. These tests may be material, configuration, or component tests. Guidance on supplemental tests can be obtained from the responsible NASA program/project M&P organization.

6.4 Materials Compatibility in Systems Containing Reactive Fluids

Test 15 is intended to be a short-term exposure test.

6.4.1 [FOCR 40] A system evaluation shall be conducted for all systems containing reactive fluids.

6.4.2 [FOCR 41] Materials used in system construction shall be evaluated for compatibility with the fluid to which they are to be exposed, as well as compatibility with the fluid decomposition products.

The effects of single-barrier failures should be considered as part of this system evaluation.

6.4.3 [FOCR 42] Evaluations shall be supported by analysis, by Test 15 data, and when necessary, by configuration-specific testing.

6.4.4 [FOCR 43] The analytical approach shall examine possible sources of uncontrolled exothermic reactions, develop control methods, and document the results.

Possible initiation sources for uncontrolled exothermic reactions include mechanical impact, rapid pressurization, sources of electrical energy, and other sources of heat.

6.4.5 [FOCR 44] The rationale for use of materials that fail to meet the criteria of Test 15 shall be documented and submitted to the responsible NASA program/project M&P organization for approval.

Guidance on configuration tests to evaluate uncontrolled reactions in reactive fluid systems can be obtained from the responsible NASA program/project M&P organization.

7. TEST METHODS

Methods for required flammability, offgassing, and compatibility tests are presented in this section.

a. [FOCR 45] Tests shall be performed by test facilities certified in accordance with Appendix B in this NASA Technical Standard.

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- (1) [FOCR 46] The test facility shall demonstrate good laboratory practices to produce accurate and repeatable test data.
- (2) [FOCR 47] Good laboratory practices shall include calibration and maintenance procedures.
- (3) [FOCR 48] At least every 2 years, test facilities shall demonstrate testing proficiency to maintain certification in accordance with Appendix B in this NASA Technical Standard.

b. [FOCR 49] Properly identified material for testing shall be provided by the responsible NASA Center or hardware developer and accompanied by a test request form similar to that provided in Appendix D in this NASA Technical Standard.

Alternatively, test facilities can be authorized by the test requester to procure the appropriate materials.

c. Material and Specimen Receipt and Preparation

- (1) [FOCR 50] Materials shall be accompanied by Material Safety Data Sheets (MSDSs) to comply with materials-handling requirements defined by the Occupational Safety and Health Administration (OSHA).
- (2) [FOCR 51] Material specimens shall be inspected visually and any flaws or contamination noted in the test report.

Materials and configured system characteristics can be significantly compromised by sources of contamination (such as exposure to solvents, cleaning agents, environmental pollutants, particulates, and handling), abnormal temperatures, and/or variations in humidity. It is important that exposure of the material to degrading environments be controlled sufficiently to minimize variation in test results.

- (3) [FOCR 52] Specimens shall be prepared in the proper dimensions.
- (4) [FOCR 53] Specimens shall be weighed and identified individually.
- (5) [FOCR 54] Specimens shall be cleaned to end-use specifications.
- (6) [FOCR 55] Specimens shall be inspected after preparation to ensure suitability for the specific test requested.
- (7) [FOCR 56] As a minimum, all fluids used for testing shall meet or exceed user specifications.

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d. [FOCR 57] When there is a deviation from standard test parameters such as nonstandard specimen preparation or test conditions, the test shall be identified as nonstandard and the results annotated as nonstandard.

Table 3, Minimum Quantities of Materials Required for Testing for Each Atmosphere or Fluid, lists the minimum quantities of material required to perform each test properly.

Table 3—Minimum Quantities of Materials Required for Testing for Each Atmosphere or Fluid

MATERIAL TEST FORM ¹	TEST	MINIMUM QUANTITIES
Solids (sheet, rod, etc.)	1	6 specimens, 30 by 6.4 cm (12 by 2.5 in) by required thickness [thin films, 30 by 7.5 cm (12 by 3.0 in) by required thickness ²]
	2	18 specimens, 10 by 10 cm (4 by 4 in) by required thickness
	6	1500 cm ² (232.5 in ²) total area or 15 g (0.5 oz)
	7	4000 cm ² (620 in ²) total area or 40 g (1.4 oz)
	13	1000 cm ² (155 in ²) total area, 0.64 cm (0.25 in) maximum thickness or individual specimen diameter of 1.75 cm (0.69 in)
	15	50 cm ² (7.75 in ²) total area or 20 g (0.7 oz) by required thickness
	17	15 specimens, 10.2-cm (4-in) (minimum) by 0.32-cm (0.125-in) diameter rods
Coatings (paints, inks, adhesives, etc.)	1	Sufficient to cover [paints, 6 specimens, 30 by 6.4 cm (12 by 2.5 in) by required thickness]
	2	18 specimens, 10 by 10 cm (4 by 4 in) by required thickness
	6	1500 cm ² (232.5 in ²) total area or 500 ml (16.9 oz)
	7	4000 cm ² (620 in ²) total area or 500 ml (16.9 oz)
	13	500 ml (16.9 oz)
	15	500 ml (16.9 oz)
Foams	1	6 specimens, 30 by 6.4 cm (12 by 2.5 in) by required thickness
	2	18 specimens, 10 by 10 cm (4 by 4 in) by required thickness
	6	1500 cm ² (232.5 in ²) total area
	7	4000 cm ² (620 in ²) total area
	13	1000 cm ² (155 in ²) total area [0.64 cm (0.25 in) maximum thickness]
	15	50 cm ² (7.75 in ²) total area or 20 g (0.7 oz)
Insulated Wires	1	2 m (6.6 ft) in length
	4	10 m (33 ft) in length
	6	15 g (0.5 oz)
	7	40 g (1.4 oz)
	13	250 cm (98 in) in length
	15	1 m (3.3 ft) in length
	18	70 m (230 ft) in length
Special	3	As recommended by selected ASTM test method

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MATERIAL TEST FORM ¹	TEST	MINIMUM QUANTITIES
Requirement	Gaseous Fluid Impact	As recommended by the OCA or by the test requester
	Autogenous Ignition Temperature	10 g (0.35 oz)
	Heat of Combustion	10 g (0.35 oz)
	Oxygen Index	As recommended by ASTM G125; when using ASTM D2863, 25 specimens, 15.2 cm by 6.4 cm (6 by 2.5 in) or equivalent amount of material
	Frictional Heating	As recommended by the OCA or by the test requester
	Particle Impact	As recommended by the OCA or by the test requester
Notes: ¹ [FOCR 58] Actual test configurations and material quantities for material forms other than those listed, e.g., O-rings and seals, shall be established and approved by the responsible NASA program/project M&P organization. ² When influence of test frame on results is of concern, optional thin films of 30 by 20 cm (12 by 8 in) by required thickness may be used, subject to the approval of the responsible NASA program/project M&P organization.		

7.1 Upward Flame Propagation (Test 1)

VCS adopted and tailored.

7.1.1 Purpose

The purpose of this test is to determine if a material, when exposed to a standard ignition source, will self-extinguish and not transfer burning debris, which can ignite adjacent materials.

7.1.2 Test Summary

- a. [FOCR 59] Test 1 shall evaluate the candidate material in the worst-case atmosphere to which the material could be exposed.
- b. [FOCR 60] Five test specimens, each with dimensions of 30 cm by 6.5 cm (12 in by 2.5 in) and tested one at a time, shall be mounted vertically in a test stand.
- c. [FOCR 61] The specimens shall be ignited at the bottom by an ignition system and allowed to burn until each self-extinguishes (figure 2, Flammability Post-Test Specimen).

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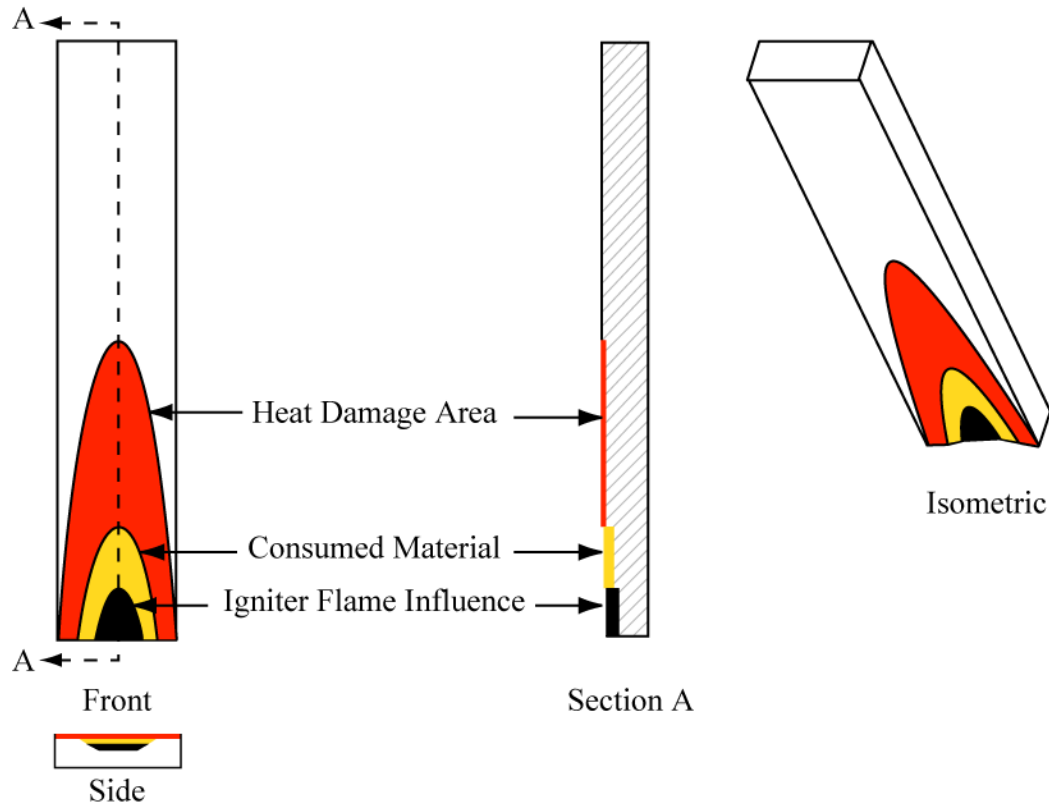


Figure 2—Flammability Post-Test Specimen

d. [FOCR 62] The burn length, i.e., the linear distance of consumed specimen, shall be measured and recorded in the test report.

e. [FOCR 63] Whether or not ignition of the paper occurred by a transfer of burning debris shall be recorded on the test report.

7.1.3 Test Criteria

a. [FOCR 64] The material shall meet the test acceptance criteria and be considered self-extinguishing if both of the following conditions are met:

- (1) No test specimen of the five standard-sized specimens burns >15 cm (6 in).
- (2) No test specimen propagates a flame by the transfer of burning debris.

7.1.4 Test Method

[FOCR 65] The test method defined in ISO 14624-1:2003, Space systems — Safety and compatibility of materials — Part 1: Determination of upward flammability of materials, shall be followed for this test, with the following exceptions, clarifications, and additions:

The relevant ISO 14624-1:2003 section is included in parentheses.

a. Exception (throughout): One test shall consist of five replicate, identical specimen ignitions.

The term “specimen” is used to indicate one standard-sized section of a material used for testing. The Upward Flame Propagation Test requires five specimens to produce one standard test.

b. Clarification (section 3.1): Burn length: the distance from the bottom of the specimen to the farthest evidence of damage to the test specimen caused by flame consumption as determined by visual observation, post-test examination, video of burn, and/or other means.

c. Addition (section 3.8): Maximum Oxygen Concentration: The maximum oxygen concentration at which at least five specimens pass the acceptance criteria of this test.

d. Exception, addition (section 4):

(1) In addition, the ignited specimens shall not ignite the paper (produce flaming combustion) below the test specimens, which would indicate that the transfer of burning debris would have sufficient energy to ignite adjacent materials.

(2) If, during a test, the paper ignites because of burning debris, subsequent burns during the same material test shall be conducted without the paper.

This is done because the burning paper can inhibit the burning of the test specimen. Individual specimen test results performed without the paper should be identified in the test report.

Failure of any one specimen constitutes failure of the material in that test environment.

e. Addition (section 4): Materials shall be tested in the worst-case configuration.

f. Additions (section 5.1):

(1) The test atmosphere shall consist of a mixture of GOX and GN₂, with the oxygen level being the highest that the material could witness in use conditions.

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If the worst-case environment is uncertain, determination of the maximum oxygen concentration is recommended.

- (2) The test gases shall be mixed thoroughly before testing a specimen.

The gases may be premixed before introduction into the test chamber or mixed inside the test chamber.

- g. Additions (section 5.1):

- (1) When gases are mixed in the chamber, they shall be circulated with a fan until a homogeneous mixture is attained, as determined by a gas analyzer.

- (2) The test gases shall be verified for conformity with the specification (including accuracy) for oxygen concentration to within +1.0 percent -0 percent of reading.

h. Addition (section 6.1): *Testing may be conducted in a fume hood in air if the above conditions can be met and the test results are verified against test chamber testing results.*

- i. Addition (section 6.1): Air shall not be allowed to flow during tests.

j. Clarification (section 6.3): *The term “ambient conditions” refers to an oxygen concentration of 20.9 percent, a pressure of 101.4 kPa (14.7 psia), and a temperature of 23 (±5) °C (73 (±9) °F).*

k. Exception (section 6.4): Bare nickel-chromium wire 0.8 to 1.0 mm in diameter, with a nominal resistivity of 2 to 2.5 $\Omega \cdot m$ and of sufficient length to wrap three equally spaced turns around the chemical igniter.

l. Exception (section 6.8): The paper surface density shall be between 150 g/m² and 250 g/m².

m. Exception (section 7.1, table 1): The minimum quantity of materials required for testing shall be as indicated in table 3 of this document.

- n. Exception (section 8.1b): All specimens shall be video recorded during testing.

o. Exception (section 8.2.1): Except when testing in air, the test specimen shall be subjected to vacuum for no less than 1 min but for no more than 3 min.

p. Exception (section 8.2.1): The igniter shall be retracted from the test specimen once the igniter extinguishes.

- q. Additions (section 8.2.2):

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- (1) Any flames emanating from the paper below the specimen shall be observed and recorded.
- (2) The paper shall be supported by a non-flammable, non-conducting screen.
- (3) Flame jets and sparks emanating from the specimen during combustion shall be observed and recorded.

r. Addition: The test report shall be submitted to MAPTIS in a MAPTIS-compatible format.

7.2 Heat and Visible Smoke Release Rates (Test 2)

Test 2 is now a supplemental test. A description of the test method is provided in Appendix A.2.1 in this NASA Technical Standard.

7.3 Flash Point of Liquids (Test 3)

Test 3 is now a supplemental test. A description of the test method is provided in Appendix A.2.2 in this NASA Technical Standard.

7.4 Electrical Wire Insulation Flammability (Test 4)

VCS adopted and tailored.

7.4.1 Purpose

The purpose of this test is to determine whether an electrical wire insulation material, when exposed to a standard ignition source, will self-extinguish and not transfer burning debris, which can ignite adjacent materials.

7.4.2 Test Summary

- a. [FOCR 66] The test shall evaluate the candidate wire insulation material in the worst-case atmosphere to which the wire exposure is anticipated.
- b. [FOCR 67] Five test specimens, each a wire equivalent to the candidate wire and having a length of 1.2 m (48 in), with the active area being 30 cm (12 in), shall be tested one at a time and mounted at a 15-degree angle from vertical in a test stand.
- c. [FOCR 68] The specimens shall be ignited at the bottom by an ignition system and allowed to burn until each self-extinguishes.
- d. [FOCR 69] The burn length, i.e., the linear distance of the consumed specimen, shall be measured and recorded.

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e. [FOCR 70] Whether ignition of the paper occurred by a transfer of burning debris shall be recorded on the test report.

7.4.3 Test Criteria

a. [FOCR 71] The material shall meet the test acceptance criteria and be considered self-extinguishing, if both of the following conditions are met:

(1) No test specimen of the five tested burns >15 cm (6 in).

(2) No test specimen propagates a flame by the transfer of burning debris.

The term “specimen” is used to indicate one standard-sized strand of a wire used for testing. Five specimens produce a standard Electrical Wire Insulation Flammability Test.

7.4.4 Test Method

[FOCR 72] The test method described in ISO 14624-2:2003, Space systems — Safety and compatibility of materials — Part 2: Determination of flammability of electrical-wire insulation and accessory materials, shall be followed for this test, with the following exceptions, clarifications, and additions:

The relevant ISO 14624-2:2003 sections are included in parentheses.

a. Exception (throughout): One test shall consist of five replicate, identical specimen ignitions.

b. Exception (throughout): The minimum quantity of material required for testing is listed in table 3 of this document.

c. Clarification (section 4.1): Burn length: the distance from the bottom of the specimen as it sits in the test stand to the farthest evidence of damage to the specimen’s test insulation material caused by flame consumption as determined by visual observation, post-test examination, video of burn, and/or other means.

d. Addition (section 4.3): Transfer of burning debris: movement of burning particles from a burning specimen to adjacent materials with sufficient energy to ignite the paper below the test specimen.

e. Addition (section 4): Maximum Oxygen Concentration: The maximum oxygen concentration at which at least five specimens pass the acceptance criteria of this test.

f. Exception (section 6.1): A standard test shall consist of five wire specimens tested.

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g. Addition (section 6.1): In addition, the ignited wire test specimens shall not ignite the paper (produce flaming combustion) below the test specimens, which would indicate that the transfer of burning debris would have sufficient energy to ignite adjacent materials.

h. Addition (section 6.1): If, during a test, the paper ignites because of burning debris, subsequent burns during testing of the same wire specimens shall be conducted without the paper.

This is done because the burning paper can inhibit the burning of the wire test specimen.

i. Additions (section 6.1):

(1) Failure of any one specimen shall constitute failure of the wire in that test environment.

(2) Wires shall be tested in the worst-case configuration.

j. Additions (section 6.2.1):

(1) The test atmosphere shall consist of a mixture of GOX and GN₂, with the oxygen level being the highest that the material could witness in use conditions.

If the worst-case environment is uncertain, determination of the maximum oxygen concentration is recommended.

(2) The test gases shall be mixed thoroughly before testing a specimen.

The gases may be premixed before introduction into the test chamber or may be mixed inside the chamber.

k. Additions (section 6.2.1):

(1) When gases are mixed in the chamber, they shall be circulated with a fan until a homogeneous mixture is attained, as determined by a gas analyzer.

(2) The test gases shall be verified for conformity with the specification (including accuracy) for oxygen concentration to within +1.0 percent -0 percent of reading.

l. Addition (section 6.3.1): *Testing may be conducted in a fume hood if the above conditions can be met and the test results are verified against test chamber testing results.*

m. Addition (section 6.3.1): Air shall not be allowed to flow during tests.

n. Addition (section 6.3.3): *The term “ambient conditions” refers to an oxygen concentration of 20.9 percent, a pressure of 101.4 kPa (14.7 psia), and a temperature of 23 (±5) °C (73 (±9) °F).*

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o. Exception (section 6.3.5): Bare nickel-chromium wire 0.8 to 1.0 mm in diameter, with a nominal resistivity of 2 to 2.5 $\Omega\cdot\text{m}$ and of sufficient length to wrap three equally spaced turns around the chemical igniter.

p. Exception (section 6.3.8): The paper shall have the following characteristics:

(1) Dimensions: $(200\text{ mm} \pm 50\text{ mm}) \times (300\text{ mm} \pm 50\text{ mm})$.

(2) Surface density: between 150 g/m^2 and 250 g/m^2 .

(3) Type: chemical wood processed.

(4) Color: uniformly white.

(5) Condition: clean, free from dirt spots, oil spots, and foreign matter (lint, fuzz, etc.); free from holes, tears, cuts, folds, and scuff marks; and containing no splices.

q. Exception (section 6.5.1.4): All specimens shall be video recorded during testing.

r. Addition (section 6.5.2.1): The test specimen shall be subjected to vacuum for no less than 1 min but for no more than 3 min.

s. Exception (section 6.5.2.1): The igniter shall be retracted from the test specimen once the igniter extinguishes.

t. Additions (section 6.5.2.2):

(1) Flames emanating from the paper below the wire specimen shall be observed and recorded.

(2) The paper shall be supported by a non-flammable, non-conducting screen.

(3) Flame jets and sparks emanating from the specimen during combustion shall be observed and recorded.

u. Exception (section 7): Alternate Wire Insulation Flammability Test in a Gas Flow Environment, Test B, shall not be performed as Test 4.

v. Addition: The test report shall be submitted to MAPTIS in a MAPTIS-compatible format.

7.5 Electrical Connector Potting and Conformal Coating Flammability (Test 5)

Deleted in previous version.

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7.6 Odor Assessment (Test 6)

Test 6 is now a supplemental test. A description of the test method is provided in Appendix A.3 in this NASA Technical Standard.

7.7 Determination of Offgassed Products (Test 7)

VCS adopted and tailored.

7.7.1 Purpose

The purpose of this test is to determine the identity and quantity of volatile offgassed products from materials and assembled articles.

7.7.2 Test Summary

- a. [FOCR 73] Test 7 shall determine the offgassing characteristics under standardized conditions for materials and assembled articles to be located within habitable environments.
- b. [FOCR 74] Specimens shall be placed into certified-clean sealed containers and thermally conditioned for 72 (± 1) hr at 50 (± 3) °C (122 (± 5) °F).
- c. [FOCR 75] After the thermal conditioning, the atmosphere inside the specimen container shall be analyzed to determine the offgassed compounds.
- d. [FOCR 76] Using the SMAC for each offgassed compound, the overall toxicity rating shall be determined.
- e. [FOCR 77] The overall rating of each material or assembled article shall determine the quantity of each material or the number of assembled articles that can be flown concurrently.
- f. [FOCR 78] For a material or assembled article, the total Toxic Hazard Index (T) values for all volatile offgassed products shall be less than 0.5.

7.7.3 Test Criteria

- a. [FOCR 79] For a material or assembled article, the total T values for all volatile offgassed products shall be less than 0.5.
- b. [FOCR 80] All materials used in habitable flight compartments shall meet the offgassing requirements, using one of the following methodologies:

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- (1) [FOCR 81] Assembled article: Summation of T values (total concentration in mg/m^3 /SMAC in mg/m^3) of all offgassed constituent products shall not exceed 0.5.
- (2) Hardware components evaluated on a materials basis:
 - A. [FOCR 82] Individual materials used to make up an assembled article shall be evaluated based on the actual or estimated mass of the material used in the hardware component.
 - B. [FOCR 83] The total T value for all materials used to make up the assembled article shall be less than 0.5.
- (3) [FOCR 84] More than one assembled article: If a single assembled article is tested or evaluated for toxicity but more than one is to be flown, the T value obtained for one unit times the number of flight units shall be less than 0.5.
- (4) [FOCR 85] Bulk materials and other materials not inside a sealed container: All materials shall be evaluated individually using the ratings in the MAPTIS database (<http://maptis.nasa.gov/>).

MAPTIS specifies the maximum quantity and associated rating for each material code.

7.7.4 Test Method

[FOCR 86] The test method described in the baseline ISO 14624-3:2005, Space systems — Safety and compatibility of materials — Part 3: Determination of offgassed products from materials and assembled articles, shall be followed for this test, with the following exceptions, clarifications, and additions:

The relevant ISO 14624-3:2005 sections are included in parentheses.

- a. Addition (section 3.4 note): *A current listing of SMAC values is maintained in MAPTIS at <http://maptis.nasa.gov/>.*
- b. Addition (section 3): Primary Gas Standard: gas mixtures that have gravimetric or analytical traceability and to which all measurements are ultimately compared.
- c. Addition (section 3): Trace: the result reported when the identified offgassed compound is present in less than reportable quantities. This category includes compounds that are labeled as unidentified components, because the concentration is too low for the spectral information to allow for identification. It does not include compounds that have adequate spectral information but are labeled as unidentified components because suitable standard spectra for identification are not available.

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d. Addition (section 3): Reportable quantities: This amount shall be determined by each analytical laboratory and based on analyzed concentrations of the specific compound.

Compounds that have been identified, but for which the specific compound is unavailable as a standard, may have reportable quantities based on analyzed concentrations of a representative compound.

e. Addition (section 3.5): T Value Calculations:

The calculation of the toxic hazard index uses the free volume (spacecraft volume) of the habitable area of the craft under consideration. For the Space Shuttle Orbiter, this volume is 65 m³ (2300 ft³); for the International Space Station (ISS), the total volume has increased as assembly has proceeded, but the volume used in T value calculation is set at 118 m³ (4170 ft³), which is the volume of the US Laboratory (Destiny) Module; and for the Multipurpose Crew Vehicle Orion vehicle, this volume is anticipated to be 15 m³ (530 ft³). For habitable volumes not defined herein, contact the responsible NASA program/project M&P organization.

SMAC is the maximum concentration of an offgassed product that is allowed in the habitable area of the spacecraft for a specified duration.

SMAC values for each offgassed constituent are reported in mg/m³. (Unless otherwise specified, all calculations use 7-day SMACs as identified by the NASA JSC Toxicology Office.)

(1) Calculations for 45.36 kg (100 lb) of material shall be:

Q = Amount for each offgassed constituent in micrograms

M = Mass of material tested (g)

TX1 = Reported amount for each offgassed constituent in micrograms of constituent per gram of material tested (µg/g)

TX2 = Amount for each offgassed constituent in milligrams per cubic meter (mg/m³) for 45.36 kg (100 lb) of material

TC = T value for each offgassed constituent for 45.36 kg (100 lb) of material (dimensionless)

T100 = Summation of the TC values of all offgassed constituents detected for 45.36 kg (100 lb) of material (dimensionless)

MLW_{kg} = Maximum Limit Weight in kilograms of material

MLW_{lb} = Maximum Limit Weight in pounds of material

Calculation of TX1:

$$TX1 \left(\frac{\mu g}{g} \right) = \frac{Q(\mu g)}{M(g)} \quad (\text{Eq. 1})$$

Calculation of TX2: Convert µg/g to mg/m³

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$$TX2\left(\frac{\text{mg}}{\text{m}^3}\right) = TX1\left(\frac{\mu\text{g}}{\text{g}}\right) \times \left(\frac{45.36 \text{ kg}}{\text{Spacecraft volume } (\text{m}^3)}\right) \quad (\text{Eq. 2})$$

Simplification yields:

$$TX2\left(\frac{\text{mg}}{\text{m}^3}\right) = TX1\left(\frac{\mu\text{g}}{\text{g}}\right) \times \left(\frac{45.36 \left(\text{kg} = \left(\frac{\text{g} \times \text{mg}}{\mu\text{g}}\right)\right)}{\text{Spacecraft volume } (\text{m}^3)}\right) \quad (\text{Eq. 3})$$

$$TX2\left(\frac{\text{mg}}{\text{m}^3}\right) = \left(\frac{(TX1 \times 45.36) (\text{mg})}{\text{Spacecraft volume } (\text{m}^3)}\right) \quad (\text{Eq. 4})$$

Calculation of TC:

$$TC = \frac{TX2\left(\frac{\text{mg}}{\text{m}^3}\right)}{SMAC\left(\frac{\text{mg}}{\text{m}^3}\right)} \quad (\text{Eq. 5})$$

Calculation of T100:

$$T100 = \sum_{n=1}^n TC_n, \text{ for } n \text{ TC values} \quad (\text{Eq. 6})$$

Calculation of the maximum limit weight (MLW):

$$MLW_{\text{kg}} = \frac{0.5}{T100} \times 45.36 \text{ kg} \quad (\text{Eq. 7})$$

$$MLW_{\text{lb}} = \frac{0.5}{T100} \times 100 \text{ lb} \quad (\text{Eq. 8})$$

(2) Calculation for assembled articles shall be:

All assembled articles should be tested in the intended flight configuration.

Q = Amount for each offgassed constituent in micrograms

TC = T value for each offgassed constituent (dimensionless)

T = Summation of the TC values of all offgassed constituents detected (dimensionless)

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Calculation of the TC:

$$TC = \left(\frac{\left(Q(\mu g) \left(\frac{1(\text{mg})}{1000(\mu g)} \right) \right)}{\text{Spacecraft volume (m}^3\text{)}} \right) \times \left(\frac{1}{\text{SMAC} \left(\frac{\text{mg}}{\text{m}^3} \right)} \right) \quad (\text{Eq. 9})$$

Calculation of the T:

$$T = \sum_{n=1}^n TC_n, \text{ for } n \text{ TC values} \quad (\text{Eq. 10})$$

Calculation of the maximum number of assembled articles (round down):

$$\text{Total Number of Assembled Articles} = \frac{0.5}{T} \quad (\text{Eq. 11})$$

- f. Exception (section 3.11): *Room temperature is equal to 23 (±5) °C (73 (±9) °F).*
- g. Clarification (section 6.1): *The test atmosphere should be at least a volume fraction of 20.9 (±2) percent for GOX with the balance GN₂ or argon, and the test pressure should be ±15 kPa (±2.18 psi) of the test facility ambient pressure.*
- h. Addition (section 6.1 note): *Batteries or assembled articles containing batteries should be tested in an inert atmosphere to reduce the risk of generating an explosive gas mixture. The batteries or assembled articles containing batteries tested in an inert atmosphere do not need to be tested again in an oxygen atmosphere for toxic offgassing.*
- i. Clarification, addition (section 7.3): *Analytical instrumentation is not specified; however, it should be capable of the identification and quantification of all offgassed products at, or below, their SMAC concentrations when tested at a test-material-mass-to-specimen-container-volume ratio of 5.0 (±0.25) g/l.*
 - (1) If the instrumentation cannot achieve this sensitivity, the minimum reportable concentration (reporting limit) for those offgassed products (except for formaldehyde) shall be reported.
 - (2) For formaldehyde, the analytical technique shall be capable of detecting a concentration of 0.1 ppm or current SMAC.

The recommended analytical instruments include a gas chromatograph, primarily using a flame ionization detector, a gas chromatograph/mass spectrometer, and an infrared spectrophotometer. Some analytical compounds may be more difficult to determine; therefore, special methods may be required to identify and quantify these compounds. For example, the

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determination of formaldehyde may be performed using the proposed method of trapping on 2,4-dinitrophenylhydrazine cartridges for derivation and subsequent analysis by High Performance Liquid Chromatography.

In some cases, the reporting limit for a compound is greater than the SMAC, but the ability to meet the SMAC requirement is known to be crucial. To determine if the compound offgasses at the SMAC, it is necessary to test more than the standard test-material-mass-to-specimen-container-volume ratio of 5.0 (± 0.25) g/l. The quantity of test material should be increased proportionally from the standard test-material-mass-to-specimen-container-volume ratio to a quantity that will allow the analysis to meet the SMAC requirement.

Example:

Benzene SMAC = 100 ppb

Reporting limit = 500 ppb

Standard material weight per chamber volume = 5 g/l

$$\frac{5 \text{ g/l}}{100 \text{ ppb}} = \frac{X \text{ g/l}}{500 \text{ ppb}}$$

X = Necessary material weight per chamber volume = 25 g/l

j. Addition (section 8.2.1): *Materials should meet a test-material-mass-to-specimen-container-volume ratio of 5.0 (± 0.25) g/l. If the specimen weight cannot be met, the maximum practical quantity of specimen, at least 750 (± 50) cm²/l of test chamber volume is tested, with the actual specimen weight and surface area reported.*

k. Additions (section 8.2):

(1) Unless the requester specifies a post-cure duration beyond the cure duration specified in the manufacturer's procedure, the material shall be aged by performing a post-cure under open-air, ambient conditions for a minimum of 14 days.

(2) The duration of the post-cure shall be reported.

l. Additions (section 8.2.3):

(1) Liquids and semi-solids shall be placed in non-reactive vessels.

(2) The approximate total specimen surface area shall be reported.

m. Addition (section 8.3): Pre- and post-test photographic images shall be acquired when needed or requested by the test requester or test lead.

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- n. Additions (section 9.4):
 - (1) The methods of quantitative analysis shall be traceable to primary gas standards.
 - (2) When available, standards used to quantify specific compounds shall be traceable to the national, international, or intrinsic standard.
- o. Exception (section 10.2): The specimen container shall be cooled to 23 (± 5) °C (73 (± 9) °F).
- p. Addition (section 10.2): If strong odors are detected during the Test 7 procedures, the test lead shall note this and recommend that Test 6 (Odor Assessment) be performed.
- q. Addition: The test report shall be submitted to MAPTIS in a MAPTIS-compatible format.
- r. Exception (section 13.1): Ethyl alcohol, methyl alcohol, tetrachloroethylene, tetrachloromethane, and acrylonitrile shall be measured to within 30 percent of the specified concentrations.
- s. Exception, addition (table 1): *The analysis of furfural in Mix B (as defined in ISO 14624-3:2005) is not required; however, this compound is useful as a diagnostic tool because it presents a meaningful challenge to the analytical system.*
- t. Exception (Annex A.1): Laboratories should be accredited to perform the determination of offgassed products from materials and assembled articles test method contained in ISO 14624-3:2005.
- u. Exception (Annex B): *Annex B is not applicable to Test 7.*
- v. Clarification (Annex C.3): Ratings shall be based on the MLW:

Materials:

K = MLW of 45 kg (100 lb) or greater
H = MLW of 22.5 to 45 kg (50 to 100 lb)
A = MLW of 4.5 to 22.5 kg (10 to 50 lb)
V = MLW of 2.3 to 4.5 kg (5 to 10 lb)
X = MLW of 0 to 2.3 kg (0 to 5 lb)

Assemblies:

A = Summation of $T \leq 0.5$
X = Summation of $T > 0.5$

7.8 Flammability Test for Materials in Vented or Sealed Containers (Test 8)

Deleted.

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7.9 Electrical Overload for Sealed Containers (Test 9)

Deleted in previous version.

7.10 Configurational Flammability (Test 10)

Test 10, formerly titled Simulated Panel or Major Assembly Flammability Test, is now a supplemental test. A description of the test method is provided in Appendix A.2.4 in this NASA Technical Standard.

7.11 Guidelines for Simulated Crew Bay Configuration Flammability Verification Test (Test 11)

Deleted in previous version.

7.12 Total Spacecraft Offgassing (Test 12)

Deleted.

7.13 Mechanical Impact for Materials in Ambient Pressure LOX (Test 13A) and Mechanical Impact for Materials in Variable Pressure LOX and GOX (Test 13B)

Tests 13A and 13B are now supplemental tests. Descriptions of the test methods are provided in Appendix A.2.5.

7.14 Pressurized Gaseous Oxygen Pneumatic Impact for Nonmetals (Test 14)

Deleted. A similar test method for components is described in Appendix A.2.6 in this NASA Technical Standard.

7.15 Reactivity of Materials in Hydrazine, Monomethylhydrazine, Unsymmetrical Dimethylhydrazine, Aerozine 50, Nitrogen Tetroxide, and Ammonia (Test 15)

7.15.1 Purpose

Test 15 is a short-term exposure test that identifies changes resulting from exposure to fluids that degrade either the material or the fluid or produce a reaction, which would cause the pressure in a closed system to rise. As described herein, Test 15 is applicable to the hydrazine family of fuels, nitrogen tetroxide and mixed oxides of nitrogen, and ammonia. Requirements for compatibility of materials with other aerospace fluids are contained in other documents, such as NASA-STD-6016.

This test is comprised of a screening test lasting 2 hr, followed by an immersion test lasting 48 hr.

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[FOCR 87] Appropriate long-term tests shall be conducted for materials with long-term exposure to fuels, oxidizers, and other hazardous fluids.

The responsible NASA program/project M&P organization should be consulted for guidance related to long-term exposure. Information on reactivity of materials with fluids may also be obtained using other tests such as accelerated rate calorimetry, isothermal microcalorimetry, and differential scanning calorimetry (DSC).

The test method described in Appendix A.2.7 is a supplemental test procedure for determining the effects of incidental exposure (minor amounts, such as a splash) of fluids on materials. Incidental exposure time is considered to be ≤ 240 min.

7.15.2 Test Criteria

This test is used to determine and document the reactivity of a material and fluid in comparison to a reference material; therefore, test criteria are dependent on the intended use environment and material configuration.

Section 7.15.12 addresses failure criteria.

7.15.3 Definitions

Definitions of relevant terms used in the description of this test method are:

Degradation: An adverse physical or chemical change in a substance.

Screening (beaker) test: A test performed with a small amount of material (≤ 0.25 g) and fluid added in drops to screen for gross reactivity.

Immersion test: A test in which the fluid covers the entire specimen for the duration of the test in a sealed container with pressure measurement.

Reaction: A chemical change in which a substance decomposes, combines with other substances, or interchanges constituents with other substances.

7.15.4 Test Conditions

a. Screening Test.

(1) [FOCR 88] The standard test conditions for the screening test shall be the test facility ambient temperature and pressure.

(2) [FOCR 89] If a material is used above the test temperature, additional testing at the maximum use temperature shall be performed.

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The test duration is 2 hr; but in the case of gross reactivity, the test may be terminated early at the discretion of the test lead and the time until test termination be recorded.

b. Immersion Test.

- (1) [FOCR 90] For hydrazine, monomethylhydrazine (MMH), unsymmetrical dimethylhydrazine (UDMH), aerazine 50, and nitrogen tetroxide and the mixed oxides of nitrogen, the standard test conditions for the immersion test shall be 71 (± 3) °C (160 (± 5) °F) and the vapor pressure of the fluid at that temperature.

The elevated temperature of the immersion test is used to allow comparison of results with historical data. Elevated temperatures accelerate a reaction, make it measurable in a short time, and/or show the effects of elevated temperature on the pressure and post-test material and fluid analysis.

- (2) [FOCR 91] For ammonia, the standard test conditions for the immersion test shall be 30 (± 3) °C [86 (± 5) °F], and the vapor pressure at that temperature.

The test duration is 48 hr from the onset of heating, but the test may be terminated early in the case of gross reactivity or of a pressure rise that may cause the immersion tube to relieve and the time until test termination be recorded.

7.15.5 Test Fluids

Unless otherwise specified,

- a. [FOCR 92] Hydrazine shall meet the requirements of MIL-PRF-26536, Propellant, Hydrazine, high purity grade.
- b. [FOCR 93] MMH shall meet the requirements of MIL-PRF-27404, Propellant, Monomethylhydrazine.
- c. [FOCR 94] UDMH shall meet the requirements of MIL-PRF-25604, Propellant, uns-Dimethylhydrazine.
- d. [FOCR 95] Aerazine 50 shall meet the requirements of MIL-PRF-27402, Propellant, Hydrazine-uns-Dimethylhydrazine (50% N₂H₄ – 50% UDMH).
- e. [FOCR 96] Nitrogen tetroxide and mixed oxides of nitrogen shall meet the requirements of MIL-PRF-26539, Propellants, Dinitrogen Tetroxide, type MON-3.
- f. [FOCR 97] Ammonia shall be procured with a minimum purity of 99.99 percent (liquid phase).

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7.15.6 Specimen Preparation

The goal of material testing is to simulate, as close as possible, the actual usage. To this end, the material preparation should employ the same cleaning and preparation processes (for example, surface finish, porosity, and heat treatment) that are expected to be used on the materials in the field.

a. [FOCR 98] Solid and foam materials shall be cut to obtain three specimens 4 by 1 by 0.15 cm (1.6 by 0.4 by 0.06 in) to obtain a total surface area measuring approximately 25 (± 5) cm² (3.9 (± 0.8) in²).

During specimen preparation, consideration should be given to edge effects of the specimen, e.g., cut ends of composite structures, insufficient application of coatings.

Specimens are prepared to a geometric surface area; however, the actual surface area may be greater.

b. [FOCR 99] Greases, semisolids, and liquids shall be applied onto both sides of a tared, wedge-shaped 304L stainless steel substrate.

c. [FOCR 100] The tare weight of the 304L stainless steel substrate shall be determined to an accuracy of 1.0×10^{-4} g.

d. [FOCR 101] Electrical insulation from wires and cables shall be stripped to prepare a specimen measuring 25 (± 5) cm² (3.9 (± 0.8) in²) specimen of the insulation.

e. Inks, adhesives, coatings, and primers.

(1) [FOCR 102] Test materials shall be applied in the end-use thickness onto abraded 4- by 1- by 0.15-cm (1.6- by 0.4- by 0.06- in) 304L stainless steel substrates.

(2) [FOCR 103] Requested cures shall be performed.

(3) [FOCR 104] A default thickness of 0.013 cm (0.005 in) shall be used if the adhesive end-use thickness is not specified.

(4) [FOCR 105] Sufficient substrates shall be coated to achieve a 25- (± 5 -) cm² (3.9- (± 0.8 -) in²) specimen.

f. Heat-shrinkable tubing.

(1) [FOCR 106] Tubing shall be pre-shrunk in accordance with manufacturer's instructions to simulate actual-use conditions.

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(2) [FOCR 107] Tubing shall be cut to achieve a surface area of $25 (\pm 5) \text{ cm}^2$ ($3.9 (\pm 0.8) \text{ in}^2$).

g. [FOCR 108] Pressure-sensitive tapes shall be applied to both sides of three 4- by 1- by 0.15-cm (1.6- by 0.4- by 0.06-in) 304L stainless steel substrates to achieve a surface area of $25 (\pm 5) \text{ cm}^2$ ($3.9 (\pm 0.8) \text{ in}^2$).

h. [FOCR 109] Segments shall be cut or an appropriate number of O-rings and irregularly shaped items be used to achieve a surface area of $25 (\pm 5) \text{ cm}^2$ ($3.9 (\pm 0.8) \text{ in}^2$).

i. [FOCR 110] Specimens shall be cleaned to end-use specifications.

j. [FOCR 111] The specimen's mass shall be determined and the approximate surface area reported.

k. [FOCR 112] Flaws or residual contamination shall be reported.

l. [FOCR 113] A reference specimen shall be prepared for the immersion test.

7.15.7 Test System

This test requires the handling of hazardous fluids and has the potential to produce energetic events, such as fire, thermal runaway, and explosion.

a. [FOCR 114] The test system for the immersion test (figure 3, Immersion Test System for Test 15) shall consist of one reference and one specimen immersion tube, each constructed of borosilicate glass and instrumented with a pressure transducer.

b. [FOCR 115] The total ullage volume above the fill line shall be sufficient for accurate pressure measurement and recorded.

c. [FOCR 116] The remainder of the system shall be constructed of 304L stainless steel.

d. [FOCR 117] The immersion tubes shall be maintained in a temperature-controlled water bath instrumented with at least one thermocouple positioned in the bath near the immersion tubes.

(1) [FOCR 118] Thermocouples shall not touch the immersion bath container sides or bottom.

(2) [FOCR 119] The water bath level shall always be above the specimen and reference tube fill levels.

e. [FOCR 120] Test data, consisting of time, temperature, and pressure, shall be recorded at appropriate intervals (depending on the reaction rate) throughout the test.

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f. [FOCR 121] The immersion system shall have provisions for inert atmosphere purging, filling operations, and cleaning.

Analytical techniques for post-test analysis of the fluid and material include nonvolatile residue (NVR), gas chromatography, ion chromatography, DSC, or Fourier transform infrared spectroscopy (FTIR).

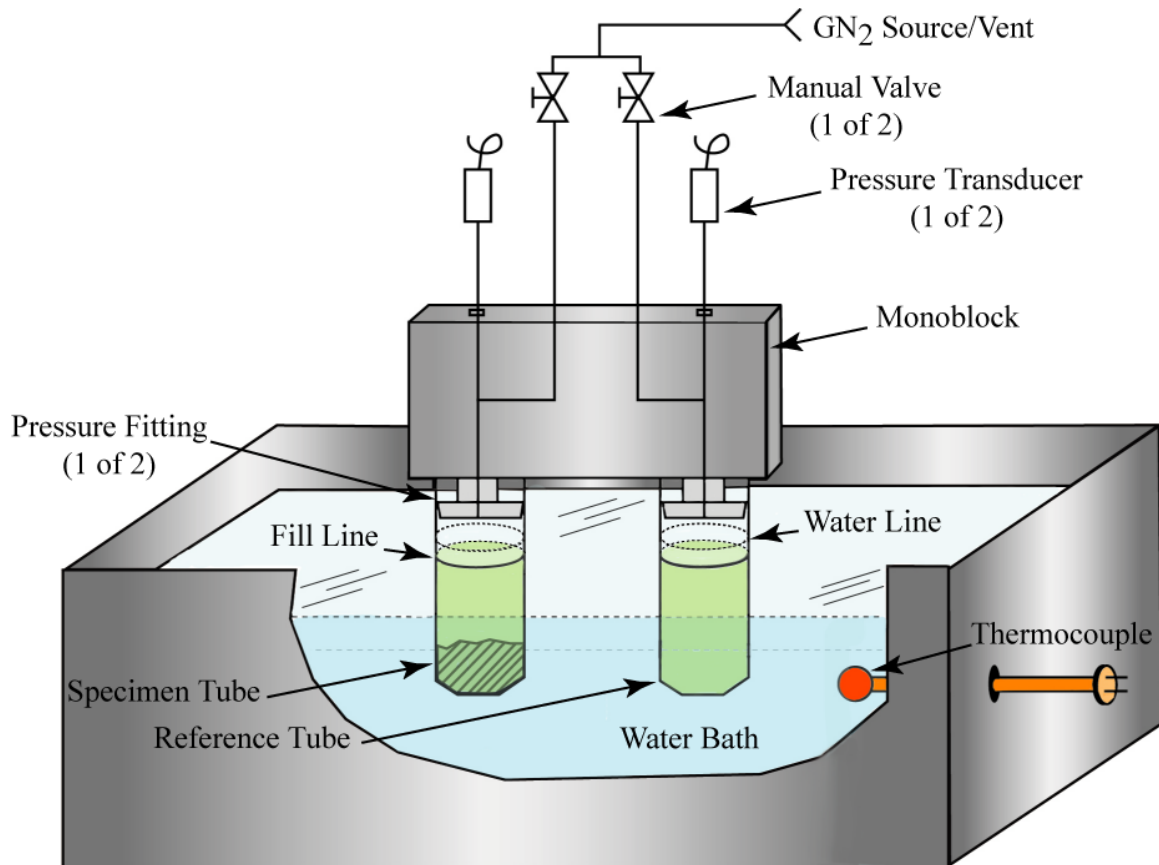


Figure 3—Immersion Test System for Test 15

7.15.8 Pre-Test Procedure

a. [FOCR 122] The fluid to be used in testing shall be analyzed and verified to meet the required specifications before use.

b. [FOCR 123] All pertinent information for the test (for example, specimen identification, composition, pre-test mass and dimensions, and fluid analysis results) shall be recorded.

c. [FOCR 124] Reference/pre-test photographic images of the specimens shall be acquired and retained.

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d. [FOCR 125] The test system shall be clean and all the measuring devices in current calibration.

7.15.9 Test Procedure

The test system for the screening test should consist of a nonreactive glass beaker in a fume hood approved for the handling of the test fluid.

a. Screening Test Procedure.

- (1) [FOCR 126] A test specimen weighing ≤ 0.25 g shall be placed in the glass container.
- (2) [FOCR 127] The test fluid shall be added drop by drop with observation for any evidence of gross reactivity.
- (3) [FOCR 128] If no gross reactivity is observed, the remainder of the test fluid (typically 10 ml total) shall be added.
- (4) [FOCR 129] The specimen shall be observed for visible signs of reaction at the beginning and appropriate intervals (depending on the reaction rate) throughout the test.

The screening test may be stopped at signs of gross reactivity.

- (5) At the end of the 2-hr exposure:
 - A. [FOCR 130] The specimen shall be removed from the fluid.
 - B. [FOCR 131] Observations shall be recorded.
 - C. [FOCR 132] Post-exposure mass and dimensions shall be recorded for the remaining material.
 - D. [FOCR 133] Post-test visual observations shall be made using defined standard terms (table 4, Standard Descriptive Terms and Definitions for Test 15 Post-Test Analysis).

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Table 4—Standard Descriptive Terms and Definitions for Test 15 Post-Test Analysis

DESCRIPTIVE TERM	DEFINITION
Adherent	Bonded or clinging to the surface of a material
Brittle	Easily fractured or broken, not malleable or ductile
Bubble	To form gaseous products
Char	To darken appreciably in color, as in oxidation
Decompose	To break down into component parts or disintegrate, either partially or completely
Degradation	An adverse physical or chemical change in a substance
Discolor	To alter the color
Dissolve	To pass into solution with little or no decomposition
Firm	Stiff and unyielding to pressure
Flexible	Capable of being bent; pliable
Friable	Property of a substance capable of being easily rubbed, crumbled, or reduced to powder
Frothing	A mass of bubbles adhering to the liquid surface; foam
Hard	Resistant to pressure, not readily penetrated, firm
Matte	Having a dull, non-reflective surface or finish
No Visible Reaction	No visual evidence of change
Opaque	Does not transmit light
Pitted	Marked by pits or small depressions, either regular or irregular
Powder	Ground, dispersed solid particles
Reaction	A chemical change in which a substance decomposes, combines with other substances, or interchanges constituents with other substances
Reactive Fluid	A fluid that readily responds to a stimulus through reaction
Rigid	Not bending; inflexible
Rough	A bumpy, uneven surface
Shape	The characteristic bulk configuration or form
Smooth	Having a surface free from irregularities, projections, or roughness
Soft	Yielding readily to pressure or weight
Solid	Of definite shape and volume; a single piece
Swell	To increase in size or volume as a result of liquid absorption
Tacky	Adhesive or gummy to the touch
Translucent	Transmits light but does not permit the perception of images
Transparent	Transmits light to the extent that images can be perceived
Woven	Constructed from an interlacing of webs or strands
Wrinkled	Ridges or creases on surface

If the material is determined to be reactive, it does not continue to the immersion test.

If there is slight or no reactivity observed, then the test lead has the option to continue to the immersion test.

- b. Immersion Test Procedure.

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- (1) [FOCR 134] Both the test specimen and the reference material specimen shall be exposed to the same fluid in their respective containers.
- (2) [FOCR 135] For polymeric specimens, polytetrafluoroethylene (ASTM D3294 Type 1, Grade 1, and a minimum of class B) shall be used as the reference material.
- (3) [FOCR 136] For metallic specimens, 304L stainless steel shall be the reference material.
- (4) [FOCR 137] The specimen and the reference materials shall be placed in the appropriate immersion tube and then purged with GN₂ to remove air.
- (5) [FOCR 138] The system shall be checked for leaks and sufficient test fluid (~25 ml) added up to the fill line to cover the specimen completely to produce the same ullage space in the specimen and the reference tubes.
- (6) [FOCR 139] The pressure and temperature recording system shall be activated.
- (7) [FOCR 140] The immersion tubes shall be placed in the water bath with the water bath level above the specimen and reference tube fill levels.
- (8) [FOCR 141] Heating of the bath shall be ~0.25 °C (~0.32 °F)/min.
- (9) [FOCR 142] When not in use, the system shall be maintained under dry GN₂ purge.

The duration of the test is 48 hr; however, the test may be terminated early if excessive pressure generation is noted and the time until test termination will be recorded.

- A. [FOCR 143] The closed system shall be held at 71 (±3) °C (160 (±5) °F).
- B. [FOCR 144] Pressure shall be recorded as a function of time.
- C. [FOCR 145] At the end of the test, the immersion tubes shall be vented, removed from the water bath, and allowed to cool.
- D. [FOCR 146] The test fluid shall be removed from the immersion tubes using inert atmosphere transfer and submitted for post-test analysis.
- E. [FOCR 147] The immersion tubes shall be removed from the system, inspected, rinsed with deionized water, and allowed to dry overnight.

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- F. *The types of post-test fluid analyses are determined by the fluid being tested.*
- i. [FOCR 148] Post-test analyses for hydrazine fuels shall include assay or purity, carbon dioxide (CO₂), and NVR.
 - ii. [FOCR 149] Nitrogen tetroxide and ammonia post-test analyses shall include NVR.

In each case, the NVR may be analyzed for the appropriate metals (metallic specimen) or anions (polymeric specimen).

- G. [FOCR 150] Post-test material analyses shall include photographic documentation, visual observations using the standard terms (table 4), and determination of the material mass and dimensions after a documented drying time.
- H. [FOCR 151] Post-test DSC and FTIR analyses shall be performed on specimens tested in nitrogen tetroxide as a screen for the formation of potentially energetic materials produced by nitration.

7.15.10 Reporting

- a. [FOCR 152] The test report shall include specimen identification, method of preparation, configuration, fluid identity and specification (including grade), test conditions, data, and observations from the test, including visual observations, photographic images, gas evolution rate or system pressure with time, and results of post-test analyses.
- b. [FOCR 153] The test report shall be submitted to MAPTIS in a MAPTIS-compatible format.
- c. [FOCR 154] If there is a deviation from standard test conditions (for example, test duration, temperature, and specimen surface area), the test shall be identified as nonstandard.

7.15.11 Precision

[FOCR 155] Measurements shall be made to the following precision:

- a. Absolute pressure: ± 3 kPa (± 0.5 psi).
- b. Temperature: ± 3 °C (± 5 °F).
- c. Specimen dimensions: ± 2 mm (± 0.08 in).
- d. Specimen Mass: ± 1 mg.
- e. Time: ± 10 sec.

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7.15.12 Failure Criteria

The criteria listed below are for materials failing the test. If the material does not fail, it still may not be appropriate for the proposed application without additional long-exposure application and configuration-specific testing.

[FOCR 156] The material shall be considered to have failed this test if any of the following occur in either the screening test or the immersion test:

- a. Burning, charring, or fire.
- b. Frothing.
- c. The material dissolves in the test fluid.
- d. The material crumbles, becomes friable, or generates particulate.
- e. The material changes shape or physical dimensions by more than 20 percent.
- f. The material suffers more than 20 percent degradation of the physical properties for which it was selected.

Physical property degradation should be determined by comparison of the physical properties with unexposed control specimens from the same lot of material, using test methods identified in the procurement specification. When the procurement specification does not identify test methods, the test methods in the manufacturer's product data sheet should be used.

- g. Visible changes in appearance of the test fluid (color and/or clarity).
- h. For nitrogen tetroxide, a pressure rise >68.9 kPa (>10 psi).
- i. For polymers exposed to nitrogen tetroxide, evidence of material nitration as indicated by DSC and/or FTIR.

Pressure rise in the system is indicative of a chemical reaction, which will vary depending on the fluid. Ammonia and nitrogen tetroxide do not degrade catalytically, and a pressure rise is an indication of a chemical reaction. For the hydrazine family of fuels, the reaction is usually catalytic; the rate is dependent on the material and the surface area. No failure criteria for the gas evolution rate for the hydrazine fuels has been established because surface area of the use parts may be small, as in the case of O-rings or gaskets; however, the gas evolution data should be considered in the material selection.

In most cases, the generation of ionic species in solution, especially halides, can increase the potential for corrosion in a system. Generation of NVR or particulate has the potential of reducing flow in a system or interfering with close-tolerance moving parts, such as valves.

7.15.13 Quality Control

Quality control is based on the system reference side pressure.

a. [FOCR 157] Based on historical data, the pressure rise on the system reference side shall be <0.4 kPa (<0.06 psia)/hr for nitrogen tetroxide and ammonia.

b. [FOCR 158] In the case of the hydrazine fuels, the gas evolution rate shall be $<7.0 \times 10^{-3}$ standard $\text{cm}^3/(\text{cm}^2 \text{ hr})$.

7.16 Determination of Offgassed Products from Assembled Article (Test 16)

Deleted in previous version.

7.17 Upward Flammability of Materials in GOX (Test 17)

VCS adopted and tailored.

7.17.1 Purpose

The purpose of this test is to determine the flammability of materials at the use pressure in GOX at ambient or elevated temperatures. This test can be used to provide supplemental information by testing at pressures other than the intended use pressure.

7.17.2 Test Summary

a. [FOCR 159] Test 17 shall determine the flammability of materials at the use pressure in GOX at ambient or elevated temperatures.

Test 17 may be used to provide supplemental information by testing at pressures other than the intended use pressure.

b. [FOCR 160] The test specimens shall be 0.32-cm (0.126-in) diameter rods with a minimum length of 10.2 cm (4 in).

c. [FOCR 161] The specimens shall be mounted vertically in a test chamber, ignited at the bottom by an ignition system, and allowed to burn until each self-extinguishes.

d. [FOCR 162] The burn length, e.g., the length of the specimen that has been consumed by combustion, shall be measured.

e. [FOCR 163] To determine a pressure at which the material is not flammable, at least 10 specimens shall be tested without consumption >3 cm (>1.2 in).

7.17.3 Test Criteria

A material is considered flammable at the maximum use pressure if at least one specimen burns >3 cm (>1.2 in).

a. [FOCR 164] To determine a pressure at which the material is not flammable, at least 10 specimens shall be tested without consumption greater than >3 cm (>1.2 in).

The results of this test are dependent upon the configuration of the test specimens. Specimens with high-surface-area-to-volume ratios, e.g., filter materials, burn at lower pressures than specimens in bulk configuration, e.g., 0.32-cm (0.126-in) rods.

b. [FOCR 165] The test shall be considered valid only if the promoter ignites, burns, and detaches from the specimen.

7.17.4 Test Method

[FOCR 166] The test method defined in ASTM G124, Standard Test Method for Determining the Combustion Behavior of Metallic Materials in Oxygen-Enriched Atmospheres, shall be followed for this test, with the following exceptions, clarifications, and additions.

The relevant ASTM G124 section is included in parentheses.

ISO 14624-4, Space systems — Safety and compatibility of materials — Part 4: Determination of upward flammability of materials in pressurized GOX or oxygen-enriched environments, describes a similar test procedure.

a. Exception (throughout): The specimen shall be tested at ambient temperature if its use temperature is ≤ 260 °C (≤ 500 °F).

b. Exception (section 8.2): For metallic materials, the promoter shall consist of a sufficient quantity of aluminum or magnesium to release at least 3.0 kJ (717 cal).

c. Exception (section 8.2): For polymeric materials, the promoter shall be sufficient for ignition of the material, as determined by the responsible NASA program/project M&P organization.

d. Exception (section 10.1): The test specimens shall be a minimum of 10.2 cm (4 in) (± 5 percent) in length.

A minimum length of 15 cm (6 in) (± 5 percent) is preferable.

e. Exception (section 10.1): If a material cannot be obtained or prepared in the cylindrical form, a nonstandard specimen shall be prepared.

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The nonstandard specimen should have 0.32-cm (0.126-in) (± 5 percent) sides and should be a minimum of 10.2 cm (4 in) (± 5 percent) in length.

f. Addition (section 16): The sample temperature measurement shall have precision of ± 5 percent of reading.

g. Addition (section 16): The test report shall be submitted to MAPTIS in a MAPTIS-compatible format.

h. Addition (section 16): If there is a deviation from the standard test parameters (for example, nonstandard specimen preparation, orientation, configuration, ignition source, and test pressure), the test shall be identified as nonstandard.

i. Addition (section 16.1.6): The pressure measurements shall have a precision of ± 1 percent of reading.

j. Addition (section 16.1.9): The measurement of the length of sample consumed shall have a precision of ± 0.25 cm (± 0.1 in).

7.18 Arc Tracking (Dry Arc Propagation Resistance) (Test 18)

7.18.1 Purpose

The purpose of this test is to provide an assessment of the ability of an insulation to prevent damage in an electrical arc environment. This test also evaluates the ability of the insulation to prevent further arc propagation when the electrical arc is re-energized.

7.18.2 Test Summary

This test method evaluates the general arc propagation resistance characteristics of wire insulations. The test consists of connecting five wires of a seven-wire bundle to a specified power supply and installing the bundle in a test fixture to hold the wire bundle perpendicular to an abrader blade. The abrader blade is used to initiate an arc through oscillation. The extent of damage caused by arc propagation and the capability of wires to maintain their dielectric strength are evaluated.

7.18.3 Test Criteria

a. [FOCR 167] To determine the arc propagation resistance of the wire configuration, at least 15 seven-wire bundles shall be tested.

b. [FOCR 168] A wire insulation material shall be considered to have met the acceptance criteria of this test if all the following requirements are met:

(1) [FOCR 169] At least 64 wires shall pass the dielectric test.

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(2) [FOCR 170] Three wires or less shall fail the dielectric test in any one bundle.

(3) [FOCR 171] Actual damage to the wire shall not be more than 7.6 cm (3 in) in any one bundle.

7.18.4 Definitions

Definitions of relevant terms used in the description of this test method are:

Arc: A continuous, luminous discharge of electric current crossing a gap between two conducting surfaces.

Arc tracking (arc propagation): That phenomena whereby an arc between two or more wires, once initiated, will sustain itself through a conductive path provided by degradation of the insulation for a measurable length.

7.18.5 Test Conditions

[FOCR 172] Test conditions shall be as defined in either MIL-STD-2223, Test Methods for Insulated Electric Wire, Method 3007, Dry Arc-Propagation Resistance, or SAE AS4373, Test Methods for Insulated Electrical Wire, Method 508, Dry Arc Propagation Resistance.

7.18.6 Specimen Preparation

a. [FOCR 173] A test specimen shall be a bundle of seven wires and of sufficient length (35.6 cm (14 in) minimum) to allow the bundle to be installed in the test fixture.

b. [FOCR 174] Fifteen bundles shall be required for a full test.

For this test, 20 American Wire Gauge wire is recommended.

c. [FOCR 175] The specimens shall be prepared according to MIL-STD-2223, Method 3007, or SAE AS4373, Method 508.

7.18.7 Test System

[FOCR 176] The test system shall be that described in MIL-STD-2223, Method 3007, or SAE AS4373, Method 508.

7.18.8 Pre-Test Procedure

[FOCR177] The pre-test procedures shall be those described in MIL-STD-2223, Method 3007, or SAE AS4373, Method 508.

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7.18.9 Test Procedure

[FOCR 178] The test procedures shall be those described in MIL-STD-2223, Method 3007, or SAE AS4373, Method 508.

7.18.10 Reporting

a. [FOCR 179] The test report shall include specimen identification, test conditions, total number of wires that pass the dielectric test, number of wires from each bundle that pass the dielectric test, and the length of physical damage to each individual wire in the bundle.

b. [FOCR 180] The test report shall be submitted to MAPTIS in a MAPTIS-compatible format.

For Test 18, test standards that meet or exceed the requirements of MIL-STD-2223, Method 3007, or SAE AS4373, Method 508, may be used with the approval of the responsible NASA program/project M&P organization.

Flight vehicle includes the launch vehicle itself, the payload (if the payload outer surfaces are exposed to the triboelectrification process), and fairings or shrouds protecting payloads.

APPENDIX A

SUPPLEMENTAL TESTS

A.1 Purpose and/or Scope

The purpose of this Appendix is to provide requirements for test methods that are selected to provide supplemental data.

A.2 Test Methods for Providing Supplemental Data

a. [FOCR 181] Tests shall be performed by test facilities certified in accordance with Appendix B in this NASA Technical Standard.

- (1) [FOCR 182] The test facility shall demonstrate GLPs to produce accurate and repeatable test data.
- (2) [FOCR 183] GLPs shall include calibration and maintenance procedures.
- (3) [FOCR 184] At least every 2 years, test facilities shall demonstrate testing proficiency to maintain certification in accordance with Appendix B in this NASA Technical Standard.

b. [FOCR 185] Properly identified material for testing shall be provided by the responsible NASA Center or contractor hardware supplier and accompanied by a test request form similar to that provided in Appendix D in this NASA Technical Standard.

Alternatively, certified test facilities can be authorized by the test requester to procure the appropriate materials.

c. Material and Specimen Receipt and Preparation

- (1) [FOCR 186] Materials shall be accompanied by MSDSs to comply with materials-handling requirements defined by OSHA.
- (2) [FOCR 187] Material specimens shall be visually inspected and any flaws or contamination noted in the test report.

Materials and configured system characteristics can be significantly compromised by sources of contamination (such as exposure to solvents, cleaning agents, environmental pollutants, particulates, and handling), abnormal temperatures, and/or variations in humidity. It is important that exposure of the material to degrading environments be controlled sufficiently to minimize variation in test results.

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- (3) [FOCR 188] Specimens shall be prepared in the proper dimensions.
- (4) [FOCR 189] Specimens shall be weighed and identified individually.
- (5) [FOCR 190] Specimens shall be cleaned to end-use specifications.
- (6) [FOCR 191] Specimens shall be inspected after preparation to ensure suitability for the specific test requested.
- (7) [FOCR 192] As a minimum, all fluids used for testing shall meet or exceed user specifications.

d. [FOCR 193] When there is a deviation from standard test parameters, e.g., nonstandard specimen preparation or test conditions, the test shall be identified as nonstandard.

Table 3 lists the minimum quantities of material required to perform each test properly.

A.2.1 Heat and Visible Smoke Release Rates (Test 2)

A.2.1.1 Purpose

The purpose of this test is to provide supplemental information on the flammability of materials.

The information obtained from this test includes effective heat of combustion, time to ignition, rate of heat released, smoke obscuration, and total heat released.

A.2.1.2 Test Summary

This test method consists of exposing specimens in flowing ambient air or oxygen-enriched environments, while subjecting them to a predetermined external heat flux. Burning may be either with or without spark ignition. The measurements include oxygen concentrations, exhaust gas flow rate, mass-loss rate of the specimen, length of time to sustained flaming, and smoke obscuration. The rate of heat release is determined by measurement of the oxygen consumption; the effective heat of combustion is determined from a simultaneous measurement of specimen mass-loss rate in combination with the heat release rate; and smoke development is measured by obscuration of light caused by the combustion product stream.

A.2.1.3 Test Criteria

[FOCR 194] The test shall be conducted on specimens that are representative of those in the intended use environment and at worst-case conditions.

There is no pass/fail criteria associated with this test. However, the test results may be used as elements of a fire risk assessment, which considers the factors that are pertinent to an

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assessment of the fire hazard of a particular end use (for example, scaling, surface morphology, and microgravity). The standard test is conducted under flow rates of GOX/ GN₂ mixtures, which may not be representative of real spacecraft applications.

A.2.1.4 Definitions

Definitions of relevant terms used in the description of this test method are:

Effective heat of combustion: The measured heat release divided by the mass loss for a specified time period.

Extinction coefficient: A measure of the reduction of light transmission through a medium.

Heat flux: The incident flux imposed externally from the radiant heater on the specimen.

Ignitability: The propensity of a material to ignite as measured by the time to ignition at a specified heating flux.

Rate of heat release: The heat evolved from the specimen per unit time.

Smoke obscuration: The reduction of light transmission by smoke, as measured by light attenuation, and reported as the extinction coefficient.

Total heat release: The quantity of heat released determined by integrating the rate of heat released as a function of time.

A.2.1.5 Test Conditions

[FOCR 195] Test conditions (incident heat flux and oxygen concentration) shall simulate the worst-case conditions in which ignition and combustion of the material are most likely to occur.

A.2.1.6 Specimen Preparation

a. [FOCR 196] Specimens shall be prepared in accordance with ASTM E1354, Standard Test Method for Heat and Visible Smoke Release Rates for Materials and Products Using an Oxygen Consumption Calorimeter.

b. [FOCR 197] Coatings or any viscous materials that cannot be mounted for testing without a substrate shall be applied to the end-use substrate material in the worst-case thickness.

Aluminum foil substrates, approximately 0.08 mm (0.003 in) thick, can be used if the end-use substrate is not known.

c. Cleaning of Specimens

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- (1) [FOCR 198] Contamination on the surfaces of solid, nonporous specimens shall be removed by washing with deionized water and any detergent, rinsing with deionized water, and drying with filtered GN₂.
- (2) [FOCR 199] Particulate on the surfaces of solid, porous specimens shall be removed with filtered GN₂.

A.2.1.7 Test System

[FOCR 200] The test system shall be as described in ASTM E1354 but with additional capabilities that will allow testing in GOX/ GN₂ mixtures other than air.

A typical test system is shown in figure 4, Test System for Test 2.

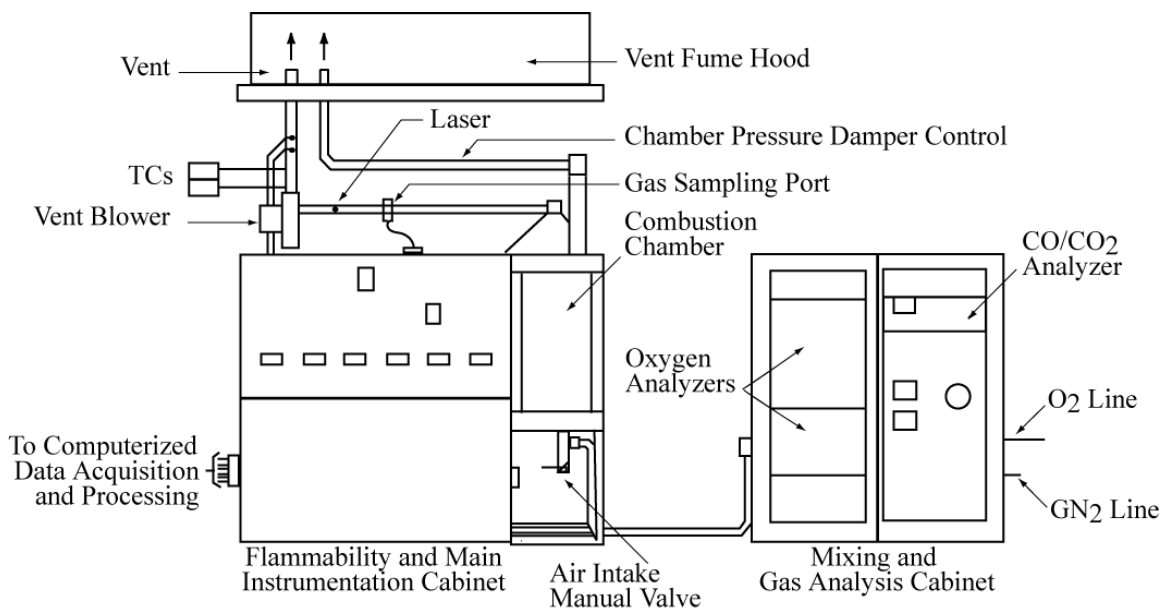


Figure 4—Test System for Test 2

A.2.1.8 Pre-Test Procedure

[FOCR 201] The pre-test procedures shall be as described in ASTM E1354.

A.2.1.9 Test Procedure

[FOCR 202] The test procedures shall be as described in ASTM E1354, with the following exceptions:

- a. The specimen shall be placed on the load cell in the test chamber and exposed to the test atmosphere by allowing the test gases to flow through the test chamber.
- b. The initial oxygen concentration shall be measured.

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A.2.1.10 Reporting

[FOCR 203] The test report content shall be as described in ASTM E1354.

Proper reporting of the test observations is critical.

A.2.2 Flash Point of Liquids (Test 3)

A.2.2.1 Purpose

The purpose of this test is to provide supplemental information on the flash point of liquids.

A.2.2.2 Test Summary

The test method for determining the flash point of liquids consists of placing a specimen in a specimen cup and heating it at a controlled rate, with or without stirring. A flame is directed into the vapor space of the cup at regular intervals to determine whether a flash occurs. The flame dimensions are specified by the test method; a typical size (described in the ASTM D1310, Standard Test Method for Flash Point and Fire Point of Liquids by Tag Open-Cup Apparatus,) is a 4-mm sphere.

A.2.2.3 Test Criteria

This test is used to determine the flash point of liquids under specified conditions of test; therefore, test criteria are dependent upon the intended use of the material.

The flash point temperature is a measure of the tendency of the test specimen to form a flammable mixture with air under controlled laboratory conditions. It is only one of a number of properties that are considered in assessing the overall flammability hazard of a material.

A.2.2.4 Definition

The definition of the relevant term used in the description of this test method is:

Flash point: The lowest temperature, corrected to a barometric pressure of 101.3 kPa (14.7 psia) at which application of an ignition source causes the vapors of a specimen to ignite under specified test conditions.

A.2.2.5 Selection of a Flash Point Test Method

[FOCR 204] The appropriate flash point test method for the application shall be selected by following the recommendations of ASTM E502, Standard Test Method for Selection and Use of ASTM Standards for the Determination of Flash Point of Chemicals by Closed Cup Methods.

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Depending on the application, an open-cup method, such as ASTM D92, Standard Test Method for Flash and Fire Points by Cleveland Open Cup Tester, or ASTM D1310 also may be used.

A.2.2.6 Test Systems and Procedures

[FOCR 205] The test systems and procedures shall be identical to those described in the selected ASTM test method(s).

A.2.2.7 Reporting

[FOCR 206] The test report shall include specimen identification, the method used, test conditions, the flash point of the liquid corrected to 101.3 kPa (14.7 psia), and observations from the test.

A.2.2.8 Precision

[FOCR 207] Measurements shall meet the precision requirements of the ASTM method(s) used.

A.2.3 Odor Assessment (Test 6)

A.2.3.1 Purpose

The purpose of this test is to determine if the odor from a material or assembled article is objectionable or revolting.

A.2.3.2 Test Criteria

The odor from a material or assembled article is objectionable or revolting if an average rating of 2.5 or higher (on an odor-characteristic scale of 0 to 4) is assigned from the test.

A.2.3.3 Definitions

Definitions of relevant terms used in the description of this test method are:

Assembled article: An assembled article could be any component or assembly of components that is not a single material.

Objectionable odor: An odor that is disagreeable (3 on an odor-characteristics scale of 4).

Offgassed product: An organic or inorganic compound evolved as a gas from a material or assembled article.

Offgassing: The evolution of gaseous products from a liquid or solid material into an atmosphere.

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Revolting odor: An odor that is extremely offensive and causes discomfort (4 on an odor-characteristics scale of 4).

Ambient (room) temperature: Room temperature is 23 (± 5) °C (73 (± 9) °F).

A.2.3.4 Test Conditions

- a. [FOCR 208] The test atmosphere shall be at least a volume fraction of 20.9 (± 2) percent for GOX, with the balance being GN₂ or argon.
- b. [FOCR 209] Test pressure shall be ± 3.5 kPa (± 0.5 psi) of the test facility ambient pressure.
- c. [FOCR 210] The maximum volume fraction limits (expressed as a volume fraction in $\mu\text{l/l}$) for impurities in the compressed gases shall be:
 - (1) Carbon monoxide: 1.0.
 - (2) Carbon dioxide: 3.0.
 - (3) Total hydrocarbons, as methane: 0.1.
 - (4) Halogenated compounds: 0.5.
 - (5) Water: 7.0.

Batteries or assembled articles containing batteries should be tested in an inert atmosphere to reduce the risk of generating an explosive gas mixture. The batteries or assembled articles containing batteries tested in an inert atmosphere do not need to be tested again in an oxygen atmosphere for toxic offgassing.

- d. [FOCR 211] The specimen shall be subject to a thermal exposure for 72 (± 1) hr at 50 (± 3) °C (122 (± 5) °F).

Specimens tested at one oxygen concentration do not have to be retested at a different oxygen concentration.

A.2.3.5 Specimen Preparation

- a. [FOCR 212] Test specimens shall be prepared from either materials or assembled articles.
- b. [FOCR 213] Handling of test specimens shall be in a manner that preserves the integrity of the specimen surface without adding contaminants.
- c. [FOCR 214] All materials shall meet the requirement of test-material-mass-to-specimen-container-volume ratio of 5.0 (± 0.25) g/l.
- d. Preparation of Specimens to Proper Dimensions

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(1) Specimen Preparation for Test Materials Based on Mass.

- A. [FOCR 215] Materials that are essentially two-dimensional and require application to a substrate (for example, coatings, primers, inks, paints, adhesives, tapes, and thin film lubricants) shall be applied at their thickness of use to clean aluminum substrates.
 - i. [FOCR 216] A sufficient number of substrates with applied specimen material shall be prepared to provide a net test-material-mass-to-specimen-container-volume ratio of 5.0 (± 0.25) g/l.
 - ii. [FOCR 217] The approximate total specimen surface area shall be recorded.

Specimens may be applied to both sides of the substrate.

- B. [FOCR 218] Materials that are essentially two-dimensional and are not applied to a substrate (for example, fabrics, photographic film plastic, plastic film, elastomers, and non-adhesive tape) shall be cut to convenient test dimensions.
 - i. [FOCR 219] Heat-shrinkable tubing shall be shrunk to simulate actual use configuration.
 - ii. [FOCR 220] A sufficient quantity of specimens shall be prepared to provide a test-material-mass-to-specimen-container-volume ratio of 5.0 (± 0.25) g/l.
- C. [FOCR 221] Non-curing liquids shall be placed in suitable non-reactive vessels and the approximate total specimen surface area reported.
- D. Specialized items and materials that do not meet the above requirements and that require special handling, e.g., non-homogeneous materials.
 - i. [FOCR 222] Such materials shall be tested in the manner designated by the responsible NASA program/project M&P organization.
 - ii. [FOCR 223] The manner of testing and specimen preparation shall be reported.

The desired test-material-mass-to-specimen-container-volume ratio for such materials is 5.0 (± 0.25) g/l.

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(2) Specimen Preparation for Flight Articles.

- A. [FOCR 224] Specimens that are assembled articles shall be inspected for parts that are not designated for flight, e.g., dust covers, tape, or test leads.
 - i. [FOCR 225] These items shall be removed before testing.
 - ii. [FOCR 226] The absence of items (for example, batteries or photographic film) that will be included during flight but that are not included with the specimen shall be recorded.
- B. [FOCR 227] The ratio of test-material-mass-to-specimen-container-volume shall be approximately 1:3.

When flight articles are required for other activities and have to be unloaded before the Test 7 results can be evaluated for toxicity, the offgas test atmosphere can be transferred to another specimen container and preserved for odor assessment.

- C. [FOCR 228] When the offgas test atmosphere is preserved for odor assessment, the odor assessment shall be initiated within 14 days from the start of the specimen gas preservation.
- e. [FOCR 229] Specimens shall be cleaned in accordance with section 8.4 of ISO 14624-3:2005.

A.2.3.6 Test System

- a. [FOCR 230] The test system shall consist of the specimen container, the test chamber with controlled temperature, and volunteer odor panel personnel.
- b. [FOCR 231] The specimen container, including any soft goods, shall have an odor rating of 0 (Undetectable), as defined in section A.2.3.8.g in this NASA Technical Standard, when an empty chamber is tested in accordance with the procedure in section A.2.3.8 in this NASA Technical Standard.
- c. [FOCR 232] The test chamber shall have the capability to maintain the test temperature to within ± 3 °C (± 5.4 °F) for the duration of the test.
- d. [FOCR 233] The test chamber instrumentation shall have the capability to record the temperature continuously.
- e. [FOCR 234] Glass syringes (30 cm³ (1.8 in³) capacity) shall be used for measuring and administering the odor specimen.

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f. [FOCR 235] When administering the specimen, a low-odor non-obtrusive mask that covers the nose and mouth shall be used.

g. [FOCR 236] The syringes and masks shall have an odor rating of 0 (Undetectable), as defined in section A2.3.8.g in this NASA Technical Standard, when an empty chamber or blank air specimen is tested in accordance with the procedure in section A.2.3.8 in this NASA Technical Standard.

A.2.3.7 Pre-Test Procedure

a. [FOCR 237] A NASA or NASA contractor volunteer shall perform the following:

(1) Read, understand, and sign a consent form as required in NPR 7100.1, Protection of Human Research Subjects, and NPD 7100.8, Protection of Human Research Subjects.

(2) Be examined and qualified by a registered medical professional.

(3) Be odor qualified by distinguishing the seven primary standards (table 5, Seven Primary Standards for Detection of Odor) from three odorless standards.

Table 5—Seven Primary Standards for Detection of Odor

ODOR	STANDARD COMPOUND	DILUTION IN WATER
Ether	Diethyl ether	90 µl in 333 ml
Camphor	1, 8-Cineole	5 µl in 500 ml
Musk	15-Hydroxypentadecanoic acid lactone	1 mg in 1000 ml
Floral	1-Methyl-1-ethyl-2-phenylpropanol	75 µl in 500 ml
Mint	dl-Menthone	2 µl in 333 ml
Pungent	Acetic acid	2 ml in 333 ml
Putrid	Methyl disulfide	1 µl in 10 l

b. [FOCR 238] An odor panel shall test each new compressed gas that is used for test atmospheres before use.

c. [FOCR 239] Gases used for test atmospheres shall be judged as being odorless.

d. [FOCR 240] The specimen containers shall be cleaned by washing with a diluted soap solution and rinsing with deionized water, allowed to dry, and then purged with filtered air or GN₂.

e. [FOCR 241] Before use, each container shall be certified to be free of odor by the odor panel.

f. [FOCR 242] Before odor testing, materials or assembled articles shall be tested for offgassed products (Test 7).

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g. [FOCR 243] If the gas from the specimen container has been determined to be unsafe to test, based on the offgassing analysis, the calculated toxic level and reasons for not performing the test shall be recorded.

h. [FOCR 244] Duplicate specimens of materials shall be offgassed—one specimen for offgassing analysis and the other for odor assessment.

i. [FOCR 245] If the specimen cannot be provided in duplicate, the same specimen shall be used in both offgassing analysis and odor assessment.

j. [FOCR 246] Before each test, members of the odor panel shall receive a nose and throat examination (for nasal irritation or other abnormal conditions) by a medical staff member.

k. [FOCR 247] Results of the examination shall be recorded.

l. [FOCR 248] Each odor panel member shall identify an odorless solution from two of the seven primary standards.

m. [FOCR 249] Members of the odor panel shall not be allowed to participate if their sense of smell has been affected in any manner, such as by recent smoking, ingestion of highly flavored foods, or exposure to pungent vapors.

A.2.3.8 Test Procedure

a. [FOCR 250] The specimen shall be weighed and placed in the specimen container.

b. [FOCR 251] The atmosphere in the specimen container shall be replaced with the test atmosphere, either by purging or by evacuation.

(1) [FOCR 252] The requesting organization shall indicate if the specimen can or cannot withstand a vacuum.

(2) [FOCR 253] Exposure of any specimen to vacuum shall be less than 3 min.

c. [FOCR 254] The specimen container, with the test atmosphere, shall be at the specified test pressure when the test temperature is achieved.

d. [FOCR 255] The specimen shall be placed in the test chamber and heated to the test temperature of 50 (± 3) °C (122 (± 5) °F), unless otherwise specified.

(1) [FOCR 256] The test temperature shall be maintained for 72 (± 1) hr.

(2) [FOCR 257] Then, the specimen container shall be cooled to ambient temperature and the pressure recorded.

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- e. [FOCR 258] The odor panel shall be convened.
- f. [FOCR 259] At least five qualified members shall be administered at least 30 cm³ of the gas from the specimen container.
- g. [FOCR 260] Using the following scale, each odor panel member shall assign an odor characteristic to the gas from the specimen container.

Undetectable:	0
Barely Detectable:	1
Easily Detectable:	2
Objectionable:	3
Revoltng:	4
- h. [FOCR 261] Between 2 and 4 hr after the test, the panel members shall be examined medically for nasal irritation or other abnormal conditions and the results of this examination recorded.
- i. [FOCR 262] Each odor panel member shall be exposed to only one odor specimen in 48 hr.

A.2.3.9 Reporting

- a. [FOCR 263] The test report shall include specimen identification, configuration, test conditions, and the average rating from the odor panel members.
- b. [FOCR 264] Any nasal irritation or other abnormal conditions from the medical examination shall be recorded.
- c. [FOCR 265] If the material is determined to be unsafe for testing (based on the offgassing analysis from Test 7), the reasons for the determination not to perform Test 6 on the specimen shall be included.
- d. [FOCR 266] The test report shall be submitted to MAPTIS in a MAPTIS-compatible format.

A.2.3.10 Precision

[FOCR 267] Measurements shall be made to the following precision:

- a. Absolute pressure: ± 1 percent of reading.
- b. Temperature: ± 3 °C (± 5.4 °F).
- c. Oxygen concentration: ± 0.5 percent of reading.
- d. Mass: ± 0.01 g.

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A.2.3.11 Odor Panel Member Requalification

[FOCR 268] Each odor panel member shall be requalified every 4 months or before conducting an odor test, using the procedure described in section A.2.3.7.a in this NASA Technical Standard.

A.2.4 Configurational Flammability (Test 10)

This test was formerly titled Simulated Panel or Major Assembly Flammability Test.

In situations in which materials do not meet the criteria of Test 1, Test 10 may be used to gather supplemental data.

A.2.4.1 Purpose

The purpose of this test is to determine if a hardware article, when exposed to a standard ignition source, will self-extinguish and not transfer burning debris, which can ignite adjacent materials or other hardware.

A.2.4.2 Test Criteria

[FOCR 269] Test 10 shall be performed in accordance with the test procedure described in Test 1.

A.2.4.3 Test Method

a. [FOCR 270] A test plan specific for the worst-case use environment (maximum temperature, pressure, and oxygen concentration), placement of ignition source, use configuration, and application shall be developed by the test organization in conjunction with the test requester.

b. [FOCR 271] The acceptance criteria shall be determined by the responsible NASA program/project M&P organization.

A.2.5 Mechanical Impact for Materials in Ambient Pressure LOX (Test 13A) and Mechanical Impact for Materials in Variable Pressure GOX and LOX (Test 13B)

A.2.5.1 Purpose

This test provides supplemental information on the reaction sensitivity of materials to ignition by mechanical impact in GOX or LOX. The test can be performed to meet any of the following objectives:

a. *To characterize the reaction sensitivity of a material at a specific impact energy.*

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- b. *To determine the impact energy threshold of a material at a given pressure.*
- c. *To determine the pressure threshold of a material at a given impact energy.*

A.2.5.2 Test Summary

- a. [FOCR 272] The test specimens shall be 17.5 mm (0.69 in) in diameter.
- b. [FOCR 273] Materials shall be tested in their use thickness for thicknesses up to 6.35 mm (0.25 in).
- c. [FOCR 274] If the use thickness is greater than 6.35 mm (0.25 in), the test specimen thickness shall be to 6.35 (± 0.13) mm (0.250 (± 0.006) in).

If the use thickness is not known or established, the test specimen thickness should be 1.52 ± 0.13 mm (0.060 (± 0.005) in).

- d. [FOCR 275] The specimens shall be subjected to mechanical impact at the desired energy and inspected for evidence of a reaction.

A.2.5.3 Test Criteria

This test evaluates the reaction sensitivity of materials when mechanically impacted in oxygen; therefore, the test criteria are dependent on the application. Typical test criteria are as follows:

- a. [FOCR 276] When impacted at 98 J (72 ft-lb), 20 specimens shall not react.
- b. [FOCR 277] If 1 specimen out of 20 reacts, 40 additional specimens shall be tested without any reactions.

When no test criteria are specified by the requesting organization, these typical criteria should normally be used. If 1 specimen out of 20 reacts, the 40 additional tests should be conducted only if there is sufficient test material and a desire to determine if the material will pass.

A.2.5.4 Definitions

Definitions of relevant terms used in the description of this test method are:

Mechanical impact: Energy delivered by a plummet that has been dropped from a pre-established height onto a striker pin in contact with a specimen.

Reaction: A chemical change in which a substance decomposes, combines with other substances, or interchanges constituents with other substances.

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In the mechanical impact test, reaction normally takes the form of ignition or oxidation. A reaction from mechanical impact can be determined by an audible report, an electronically or visually detected flash, or detectable charring of the specimen, specimen cup, or striker pin.

A.2.5.5 Test Conditions

a. [FOCR 278] For ambient pressure LOX tests, the test conditions (pressure and temperature) shall be the test facility ambient pressure and the LOX boiling point at that pressure.

b. [FOCR 279] For variable pressure LOX and GOX tests, the test conditions (pressure and temperature) shall be determined for each test as recommended in the OCA.

A.2.5.6 Specimen Preparation

a. [FOCR 280] Materials shall be prepared as stated in ASTM G86, Standard Test Method for Determining Ignition Sensitivity of Materials to Mechanical Impact in Ambient Liquid Oxygen and Pressurized Liquid and Gaseous Oxygen Environments.

b. [FOCR 281] The test material dimensions specified in ASTM G86 shall be used, unless otherwise specified by the responsible NASA program/project M&P organization.

As a minimum, unless otherwise specified or approved by the NASA program/project M&P organization, contamination on the surfaces of solid, nonporous specimens should be removed by washing with deionized water and any detergent, rinsing with deionized water, and drying with filtered GN₂.

Particulates on the surfaces of solid, porous specimens should be removed with filtered GN₂.

c. [FOCR 282] The specimen shall be inspected and flaws noted.

d. [FOCR 283] If the flaws result from specimen preparation at the test facility, new specimens shall be prepared.

If the flaws do not result from specimen preparation at the test facility, the customer is contacted to verify that the test should proceed.

e. [FOCR 284] When testing materials to determine relative ranking, specimen preparation (including cleaning) and condition (for example, thickness, surface finish, product form) shall be identical.

A.2.5.7 Test System

[FOCR 285] The test system shall be identical to that described in ASTM G86.

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A.2.5.8 Pre-Test Procedure

[FOCR 286] The pre-test procedure, including calibration of the test system, shall be identical to that described in ASTM G86.

A.2.5.9 Test Procedure

[FOCR 287] The test procedure shall be identical to that described in ASTM G86, with the exception that testing is performed at the requester-specified pressure, temperature, and impact energy.

A.2.5.10 Reporting

a. [FOCR 288] The test report shall include specimen identification, configuration, test conditions, number of reactions, and observations from the test.

Proper reporting of the test observations, especially of unusual behavior, is critical.

b. [FOCR 289] The test report shall be submitted to MAPTIS in a MAPTIS-compatible format.

A.2.5.11 Precision

[FOCR 290] Precision of measurements (for example, specimen dimensions, drop height, and time) shall be identical to ASTM G86.

A.2.6 Gaseous Fluid Impact for Components

A.2.6.1 Purpose

This test provides supplemental information on the reaction sensitivity of components to rapid pressurization heating when pneumatically impacted in a pressurized GOX system.

The test can be performed to meet any of the following objectives:

- a. To characterize the reaction sensitivity of a component at a specified pressure.*
- b. To determine the pressure threshold for ignition of a component over a range of pressures.*
- c. To determine the temperature reaction threshold of a component at a specified pressure.*

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A.2.6.2 Test Summary

- a. [FOCR 291] The test specimens shall be components in their normal use condition, including the end-use cleanliness level.
- b. [FOCR 292] The test specimens shall be subjected to gaseous fluid impact at 1.25 times Maximum Operating Pressure (MOP), followed by functional tests and inspection for evidence of ignition.

A.2.6.3 Test Criteria

This test evaluates the reaction sensitivity of components when pneumatically impacted in a pressurized GOX system. Therefore, the test criteria are dependent on the application.

- a. [FOCR 293] At least one component shall be subjected to a minimum of 60 pneumatic impacts.
- b. [FOCR 294] Pressurization shall be through the ports specified by an OCA.

Testing of up to three components is preferable.

A.2.6.4 Definitions

The definition of the relevant term used in the description of this test method is:

Gaseous fluid impact: Heat delivered to a specimen by rapid fluid compression; may also be referred to as pneumatic impact, rapid pressurization, or adiabatic compression.

A.2.6.5 Test Conditions

- a. [FOCR 295] Test pressure shall be 1.25 times MOP.
- b. [FOCR 296] Test gas temperature shall be 60 (± 3) °C (140 (± 5.4) °F).
- c. [FOCR 297] Pressurization rate to within 95 percent of the test pressure shall be 20 (+0/-5) msec to enhance reaction probability in the component.

A.2.6.6 Specimen Preparation

[FOCR 298] Components shall be tested in their normal use condition, including the end-use cleanliness level.

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A.2.6.7 Test System

[FOCR 299] The test system shall be as described in ASTM G74, Standard Test Method for Ignition Sensitivity of Materials to Gaseous Fluid Impact, with the exception that the test chamber subassembly is replaced by the component to be tested.

A.2.6.8 Test Procedure

[FOCR 300] The test procedure for testing components shall be as described in ASTM G74 with the following exceptions:

- a. The test chamber subassembly shall be replaced with the component to be tested.
- b. The configuration of the component to be tested shall be as recommended in the OCA.
- c. The component shall be subjected to impacts at intervals of approximately 30 sec in each test configuration recommended in the OCA.
- d. After testing, each component shall be subjected to functional tests and inspected for signs of ignition.

A.2.6.9 Reporting

[FOCR 301] The test report shall include specimen identification, configuration, test conditions, number of reactions, and observations from the test.

Proper reporting of the test observations, especially of unusual behavior, is critical.

A.2.6.10 Precision

[FOCR 302] Measurements shall be made to the following precision:

- a. Absolute pressure: ± 1 percent of reading.
- b. Temperature: ± 3 °C (± 5.4 °F).

A.2.7 Reactivity and Penetration of Materials due to Incidental Exposure to Hydrazine, Monomethylhydrazine, Unsymmetrical Dimethylhydrazine, Aerozine 50, Nitrogen Tetroxide, and Ammonia

A.2.7.1 Purpose

The purpose of this test is to identify changes resulting from incidental exposure (minor amounts, such as a splash) to fluids that degrade or penetrate the material or produce a reaction.

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A.2.7.2 Test Criteria

This test is used to determine and document the effects of the hydrazine family of fuels, nitrogen tetroxide and mixed oxides of nitrogen, and ammonia on materials; therefore, any test criteria are dependent on the intended use of the material.

The acceptability of the material is determined by the test requester and the responsible NASA program/project M&P organization.

A.2.7.3 Definitions

Definitions of relevant terms used in the description of this test method are:

Degradation: An adverse physical or chemical change in a substance.

Reaction: A chemical change in which a substance decomposes, combines with other substances, or interchanges constituents with other substances.

A.2.7.4 Test Conditions

a. [FOCR 303] Unless otherwise specified, the test conditions shall be the test facility ambient temperature and pressure.

Incidental exposure time is considered to be ≤ 240 min.

Test exposure time is specified by the test requester, based on the intended use for the material and expected duration of field exposure.

In the case of gross reactivity, the test may be terminated early.

b. Test Fluids

(1) [FOCR 304] Hydrazine shall meet the requirements of MIL-PRF-26536.

(2) [FOCR 305] UDMH shall meet the requirements of MIL-PRF-25604.

(3) [FOCR 306] MMH shall meet the requirements of MIL-PRF-27404.

(4) [FOCR 307] Aerozine 50 shall meet the requirements of MIL-PRF-27402.

(5) [FOCR 308] Nitrogen tetroxide and mixed oxides of nitrogen shall meet the requirements of MIL-PRF-26539.

(6) [FOCR 309] Ammonia shall be procured with a stated purity of 99.99 percent (liquid phase).

A.2.7.5 Specimen Preparation

A.2.7.5.1 Films, Fabrics, Sheets, Metals, and Composites

Preparation of these specimens for testing involves the following tasks:

- a. [FOCR 310] Preparation of a specimen of the proper mass or dimensions: specimens shall be prepared to a geometric surface area of 10.2 by 10.2 cm (4 by 4 in).
- b. [FOCR 311] Specimens shall be cleaned and dried to end-use specifications.

A.2.7.5.2 Adhesives and Coatings

[FOCR 312] Adhesives and coatings shall be applied in a thickness equivalent to normal use on aluminum foil or 304L stainless steel substrate and cured, if necessary, in accordance with the manufacturer's instructions.

A.2.7.5.3 Tapes

[FOCR 313] Tapes shall be applied in the as-received condition and thickness on aluminum foil or 304L stainless steel substrate or a nonreactive substrate, e.g., watch glass, glass Petri dish.

A.2.7.5.4 Greases and Gels

[FOCR 314] Greases and gels shall be applied on aluminum foil or 304L stainless steel substrate in a thickness equivalent to the normal use and cured, if required, in accordance with the manufacturer's instructions.

A.2.7.5.5 Complex Shapes

[FOCR 315] Complex shapes, e.g., O-rings, cables, and pipes, shall be tested in a cleanliness and configuration consistent with the intended use.

A.2.7.6 Test System

This test requires handling hazardous fluids and has the potential to produce energetic events such as fire, thermal runaway, and explosion.

- a. [FOCR 316] The test system for the reactivity and penetration test for films, fabrics, sheets, metals, adhesives, coatings, and composites shall consist of a nonreactive glass beaker in a fume hood approved for the handling of the test fluid.

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b. [FOCR 317] The test system for the reactivity test for all other materials shall consist of a nonreactive substrate (for example, watch glass, glass Petri dish) in a fume hood approved for the handling of the test fluid.

Optionally, the test system may include a thermocouple or resistance thermometer to monitor the temperature of the test fluid during the test.

A.2.7.7 Pre-Test Procedure

a. [FOCR 318] The fluid to be used in testing shall be analyzed and verified to meet the required use specifications before use.

b. [FOCR 319] All pertinent information for the test (for example, specimen identification, composition, pre-test mass and dimensions, and fluid analysis results) shall be recorded.

c. [FOCR 320] Reference/pre-test photographic images of the specimens shall be obtained.

d. [FOCR 321] The test system shall be visibly clean.

e. [FOCR 322] All the measuring devices shall be in current calibration.

f. [FOCR 323] As a precaution, initial exposure test of the specimen material shall be conducted in accordance with the screening test procedure of Test 15.

A.2.7.8 Test Procedure

A.2.7.8.1 Reactivity and Penetration of Films, Fabrics, Sheets, Metals, and Composites

a. [FOCR 324] The test material specimen shall be placed over a nonreactive beaker.

Optionally, position a sheathed thermocouple or resistance thermometer so that it touches the center of the test specimen and will be immersed in the test fluid.

b. [FOCR 325] The test fluid (typically 1 ml total) shall be added drop by drop, slowly at first, and observed for any evidence of gross reactivity.

Optionally, monitor the temperature of the test fluid.

c. [FOCR 326] If no gross reactivity is observed, the remainder of the test fluid shall be added to the center of the specimen, without exposing the edges of the specimen to the fluid.

During specimen preparation, consideration should be given to edge effects of the specimen, e.g., cut ends of composite structures, insufficient application of coatings.

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Specimens are prepared to a geometric surface area; however, actual surface area may be greater.

d. [FOCR 327] The specimen shall be continuously exposed to the test fluid for the specified exposure time.

Test fluid should be added as required to maintain a visible liquid film on the test specimen during the exposure time.

Optionally, monitor the temperature of the test fluid throughout the specified exposure time.

e. [FOCR 328] The specimen shall be observed for visible signs of reaction (table 4) (for example, burning, smoking, frothing, charring, solubility, swelling, fracture, or penetration of the specimen) throughout the test.

If the temperature was monitored, the starting temperature of the test fluid, the maximum fluid temperature observed during the test and the time of occurrence, and the test fluid temperature at the end of the test are recorded.

The test may be terminated at signs of gross reactivity.

f. [FOCR 329] After testing, each component shall be visually inspected for signs of degradation.

g. [FOCR 330] The time of the first instance of penetration (first drop of liquid in the beaker) shall be recorded.

h. [FOCR 331] For materials used for protective garments, the time of initial appearance of wetness on the underside of the test specimen shall be recorded.

Atmospheric condensation could form on the underside of the test specimen, giving a false indication of penetration; verification of penetration should be made.

A.2.7.8.2 Reactivity of All Other Materials

a. [FOCR 332] The specimen shall be placed on a nonreactive substrate, e.g., watch glass, glass Petri dish.

Optionally, position a sheathed thermocouple or resistance thermometer so that it touches the center of the test specimen and will be immersed in the test fluid.

b. [FOCR 333] The test fluid shall be added drop by drop, slowly at first, and observed for any evidence of gross reactivity.

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Optionally, monitor the temperature of the test fluid.

c. [FOCR 334] If no gross reactivity is observed, the remainder of the test fluid (typically 1 ml total) shall be added to the center of the specimen, without exposing the edges of the specimen to the fluid.

d. [FOCR 335] The test fluid shall be allowed to stand on the specimen for the specified exposure time.

Test fluid should be added as required to maintain a visible liquid film on the test specimen during the exposure time.

Optionally, monitor the temperature of the test fluid throughout the specified exposure time.

e. [FOCR 336] The specimen shall be observed for visible signs of reaction (table 4) (for example, burning, smoking, frothing, charring, solubility, swelling, fracture, and penetration of the specimen) throughout the test.

If the temperature was monitored, the starting temperature of the test fluid, the maximum fluid temperature observed during the test and the time of occurrence, and the test fluid temperature at the end of the test are recorded.

f. [FOCR 337] At the end of the specified exposure time, the liquid shall be blotted from the specimen.

(1) [FOCR 338] The specimen shall be rinsed with running water for 60 sec.

(2) [FOCR 339] The test specimen shall be allowed to air dry for 24 hr before final evaluation.

g. [FOCR 340] Greases or uncured materials shall not be rinsed or blotted.

(1) [FOCR 341] Any excess test liquid shall be poured off.

(2) [FOCR 342] The test specimen shall be allowed to stand in air for 24 hr before final evaluation.

A.2.7.8.3 Post-Test Material Analysis

[FOCR 343] Post-test material analysis shall include photographic documentation, visual observations of the material using the defined standard terms (section 7.15.9, table 4 in this NASA Technical Standard), and determination of the mass and dimensions when possible.

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A.2.7.9 Reporting

a. [FOCR 344] The test report shall include specimen identification, configuration, fluid identity, test conditions, and observations from the test.

b. [FOCR 345] Any reactivity observed during the exposure (for example, burning, smoking, frothing, charring, solubility, swelling, and fracture of the specimen) shall be recorded.

c. [FOCR 346] For penetration, the elapsed time at the first instance of penetration (or appearance of underside wetness for protective garments) shall be reported.

If the temperature was monitored, the starting temperature of the test fluid, the maximum fluid temperature observed during the test and the time of occurrence, and the test fluid temperature at the end of the test are reported.

d. [FOCR 347] Relevant post-test analysis results for the material shall be included.

e. [FOCR 348] Post-test photographic images shall be included, as required, to document condition of the material.

f. [FOCR 349] The test report shall be submitted to MAPTIS in a MAPTIS-compatible format.

A.2.7.10 Precision

[FOCR 350] Measurements shall be made to the following precision:

- a. Absolute pressure: ± 3 kPa (± 0.5 psi).
- b. Temperature: ± 3 °C (± 5 °F).
- c. Specimen dimensions: ± 2 mm (± 0.08 in).
- d. Specimen mass: ± 10 mg.

A.2.8 Autogenous Ignition Temperature

A.2.8.1 Purpose

The purpose of this test is to provide information on the temperature at which liquids or solids will ignite spontaneously in elevated-pressure oxygen environments. The test method is designed for high-pressure GOX but may be used in atmospheres from 0.5 percent to 100 percent oxygen.

A.2.8.2 Test Summary

The autogenous ignition temperature test consists of exposing a material in a reaction vessel to a pressurized environment containing 0.5 percent to 100 percent oxygen and heating at a predetermined rate. The minimum temperature required to cause the specimen to ignite

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spontaneously is denoted by a sudden temperature rise and is considered the autogenous ignition temperature.

A.2.8.3 Test Criteria

This test is used to determine the autogenous ignition temperature (commonly called the autoignition temperature) under specified conditions of the test; therefore, any test criteria are dependent upon the intended use of the material, as evaluated through hazards analyses through engineering assessments for safe system operations.

This procedure provides a numerical value for the temperature at the onset of ignition under controlled conditions. This is only one of a number of properties to be considered in assessing the overall ignitibility hazard of a material.

A.2.8.4 Definition

The definition of the relevant term used in the description of this test method is:

Autogenous ignition temperature: The lowest temperature at which a material will spontaneously ignite under specific test conditions.

A.2.8.5 Test Conditions

[FOCR 351] The test conditions shall be identical to those described in ASTM G72, Standard Test Method for Autogenous Ignition Temperature of Liquids and Solids in a High-Pressure Oxygen-Enriched Environment.

A.2.8.6 Test Systems and Procedures

- a. [FOCR 352] The test systems and procedures shall be identical to those described in ASTM G72.
- b. [FOCR 353] At least five specimens shall be tested under each test condition.

A.2.8.7 Reporting

- a. [FOCR 354] The test report shall include specimen identification, the method used, test conditions, ignition temperature, temperature rise on ignition, and observations from the test.
- b. [FOCR 355] The test report shall be submitted to MAPTIS in a MAPTIS-compatible format.

A.2.8.8 Precision

[FOCR 356] Measurements shall meet the ASTM G72 precision requirements.

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A.2.9 Heat of Combustion

A.2.9.1 Purpose

The purpose of this test is to provide information on the heat evolved per unit mass when a polymeric material is completely burned in GOX.

A.2.9.2 Test Summary

Heat of combustion is determined by burning a weighed specimen in a GOX environment in a bomb calorimeter under controlled conditions. The heat of combustion is calculated from temperature measurements before, during, and after combustion.

A.2.9.3 Test Criteria

This test is used to determine the heat of combustion of nonmetals. Any test criteria are dependent upon the intended use of the material, as evaluated through hazards analyses by engineering assessments to ensure safe systems operations. The heat of combustion is only one of a number of properties to be considered in assessing the overall oxygen compatibility of a material.

A.2.9.4 Definitions

Definitions of relevant terms used in the description of this test method are:

Gross heat of combustion: The quantity of heat released when a unit mass of fuel is burned in a constant volume enclosure, with the products being gaseous, other than water that is condensed to the liquid state.

Net heat of combustion: The quantity of heat released when a unit mass of fuel is burned at constant pressure, with all of the products, including water, being gaseous.

A.2.9.5 Test Conditions

[FOCR 357] The test conditions shall be identical to those described in ASTM D240, Standard Test Method for Heat of Combustion of Liquid Hydrocarbon Fuels by Bomb Calorimeter, or in ASTM D4809, Standard Test Method for Heat of Combustion of Liquid Hydrocarbon Fuels by Bomb Calorimeter (Precision Method).

A.2.9.6 Test Systems and Procedures

[FOCR 358] The test systems and procedures shall be identical to those described in ASTM D240 or ASTM D4809.

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The gross heat of combustion is determined experimentally; however, the net heat of combustion is the quantity more commonly used. The gross heat of combustion for polymers of interest (especially oxygen systems) is conservative when compared with the net heat of combustion by ~0.6 MJ/kg (~150 cal/g) or less. The gross heat of combustion is adequate for evaluating materials compatibility for oxygen compatibility applications.

A.2.9.7 Reporting

a. [FOCR 359] The test report shall include specimen identification, the method used, test conditions, gross heat of combustion, and observations from the test.

b. [FOCR 360] The test report shall be submitted to MAPTIS in a MAPTIS-compatible format.

A.2.9.8 Precision

[FOCR 361] Measurements shall meet the ASTM D240 or ASTM D4809 precision requirements.

A.2.10 Oxygen Index

A.2.10.1 Purpose

The purpose of this test is to provide information on the relative flammability of materials. It is a useful screen for determining whether previously untested nonmetallic materials are likely to be acceptable for use in oxygen systems, but its results cannot be used for absolute prediction of oxygen compatibility.

A.2.10.2 Test Summary

The oxygen index is determined by burning the material in a candle-like configuration and varying the oxygen concentration to find the lowest oxygen concentration that will support combustion.

A.2.10.3 Test Criteria

This test is used to determine the oxygen index of nonmetals. Any test criteria are dependent upon the intended use of the material, as evaluated through hazards analyses by engineering assessments to ensure safe systems operations. The oxygen index is only one of a number of properties to be considered in assessing the overall oxygen compatibility of a material.

A.2.10.4 Definitions

The definition of the relevant term used in the description of this test method is:

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Oxygen index: The minimum concentration of oxygen, expressed as a volume percent, in a mixture of oxygen and nitrogen that will just support flaming combustion of a material initially at room temperature under the conditions of ASTM D2863, Standard Test Method for Measuring the Minimum Oxygen Concentration to Support Candle-Like Combustion of Plastics (Oxygen Index).

A.2.10.5 Test Conditions

[FOCR 362] The test conditions shall be identical to those described in ASTM G125, Standard Test Method for Measuring Liquid and Solid Material Fire Limits in Gaseous Oxidants, or ASTM D2863.

A.2.10.6 Test Systems and Procedures

[FOCR 363] The test systems and procedures shall be identical to those described in ASTM G125 or ASTM D2863.

A.2.10.7 Reporting

- a. [FOCR 364] The test report shall comply with ASTM G125 or ASTM D2863.
- b. [FOCR 365] The test report shall be submitted to MAPTIS in a MAPTIS-compatible format.

A.2.11 Electrical Arc Test

A.2.11.1 Purpose

Electrical arc tests are designed to determine the voltage and current needed to produce ignition in polymeric materials.

A.2.11.2 Test Criteria

[FOCR 366] A material shall pass at the test conditions if there is no ignition in 60 tests.

Typically, the test current is varied to determine the threshold for ignition of the material.

A.2.11.3 Test Conditions

[FOCR 367] The following test conditions shall be determined by the OCA or by the test requester: test pressure, oxygen concentration, test specimen material/configuration, voltage, and current.

A.2.11.4 Specimen Preparation

[FOCR 368] The test specimen configuration shall be specified in the OCA or by the test requester.

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This preparation may include fraying the test materials with a noncontaminating wire brush.

A.2.11.5 Test System

Electrical arc tests are performed in a chamber pressurized with GOX or a specified gas mixture. The apparatus includes a single strand of wire in contact with the test material (figure 5, Electrical Arc Test Apparatus). The test chamber accommodates a gas inlet and outlet, a view window for normal and high-speed video recording, a test specimen mounting block (with built-in heat exchanger for temperature control), a thermocouple for temperature measurement, and power for the electrical arcing. The system is capable of a maximum test pressure of 689 kPa (100 psia). The power supply is designed to simulate batteries, while allowing flexibility for different voltages and currents during testing.

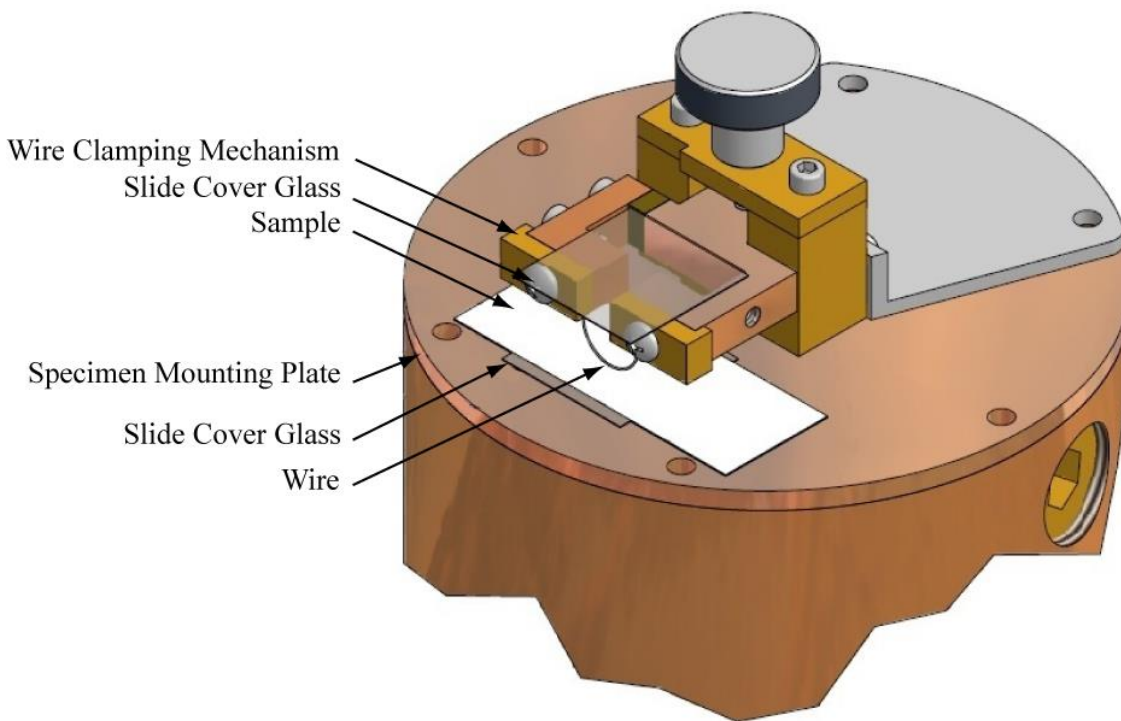


Figure 5—Electrical Arc Test Apparatus

A.2.11.6 Test Procedure

- a. Before testing begins, the test voltage is set to correspond to the predetermined test conditions, and the current is set slightly lower than the desired test conditions.*
- b. The appropriate wire size is determined, and the wire is clamped into place.*

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c. *The test chamber is sealed, purged, and pressurized with the specified pressure and oxygen concentration.*

d. *Power is then applied to the wire, and the test conductor manually increases the current until the wire breaks, creating an arcing event.*

e. *If an ignition occurs, testing is continued at a lower current level.*

f. *If no ignition occurs, high-speed video and visual inspection are used to verify that the wire broke in the desired location and was in contact with the test specimen. Each test consists of one wire-break event, which is observed for visual evidence of test material ignition.*

A.2.11.7 Reporting

a. [FOCR 369] The test report shall include specimen material and configuration, test pressure, oxygen concentration, and current/voltage required for ignition.

b. [FOCR 370] The test report shall be submitted to MAPTIS in a MAPTIS-compatible format.

A.2.12 Frictional Heating

A.2.12.1 Purpose

The GOX and LOX frictional heating tests are designed to evaluate a material's susceptibility to ignition by frictional heating. These tests include rotational friction tests and reciprocal friction tests. Frictional heating tests can be used to accomplish the following objectives:

a. *To assess the ability of a pair of materials to convert mechanical energy to thermal energy during rubbing.*

b. *To determine the coefficient of friction and wear.*

c. *For rotational friction tests, to determine the minimum contact pressure (P) required to ignite a test material specimen at a given linear surface velocity (v). This product (Pv_{ign}) is used to rank the relative ignition resistance of materials.*

d. *For reciprocal friction tests, to determine whether the materials will ignite at the test frequency profile and displacement of movement.*

A.2.12.2 Test Criteria

[FOCR 371] The test criteria shall be defined in the OCA.

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For both rotational and reciprocal friction tests, ignition is detected typically by a significant and rapid temperature increase, displacement, or a change in pressure (an initial pressure drop followed by a pressure increase). After termination of the test, the test specimens are inspected visually for consumed material.

A.2.12.3 Definitions

Definitions of relevant terms used in the description of this test method are:

P: Minimum contact pressure.

v: Linear surface velocity.

A.2.12.4 Test Conditions

[FOCR 372] The test parameter values shall be determined by the OCA or by the test requester.

The test parameters for frictional heating tests are pressure, oxygen concentration (for GOX tests), rotational speed (for rotational friction tests), frequency profile and displacement of movement (for reciprocal friction tests), normal force, and test specimen materials.

Most of the existing friction data was generated using rotational friction at the following standard set of conditions, which may be useful for comparing data:

- a. The oxygen pressure is maintained constant at 6.9 MPa (1000 psig) for GOX tests and 2.1 MPa (300 psig) for LOX tests.*
- b. The rotational speed is maintained constant at 17,000 revolutions per minute (rpm).*
- c. The normal load is increased from 0 at a rate of 1.6 to 1.7 N/sec (7 to 7.5 lbf/sec) to the point of specimen failure or to the maximum load capacity of the test system.*

Any of these test conditions may be changed to meet the recommendations in the OCA.

A.2.12.5 Specimen Preparation

- a. [FOCR 373] The test specimen configuration shall be specified in the OCA or by the test requester.
- b. [FOCR 374] Before testing, the specimens shall be prepared according to the end-use application or by washing with any warm liquid detergent solution, rinsing with deionized water, and drying with GN₂.
- c. [FOCR 375] The specimens shall then be sealed in clean bags until test.

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d. [FOCR 376] From this point, the specimens shall be handled with clean, powder-free, lint-free gloves to maintain cleanliness.

Most of the existing rotational friction data was generated with specimens configured as follows, which may be useful for comparing data:

For each rotational friction test, one test specimen remained stationary and one specimen rotated. The test apparatus was configured as shown in figure 6, Frictional Heating Test System. Figure 7, Rotational Frictional Heating Stationary Test Specimen, shows test specimen configuration. The stationary test specimen had two thermocouple holes (0.1 cm (0.039 in) in diameter) on its radial side. The rotational test specimens were fabricated similarly but without the 0.1-cm (0.039-in) diameter holes.

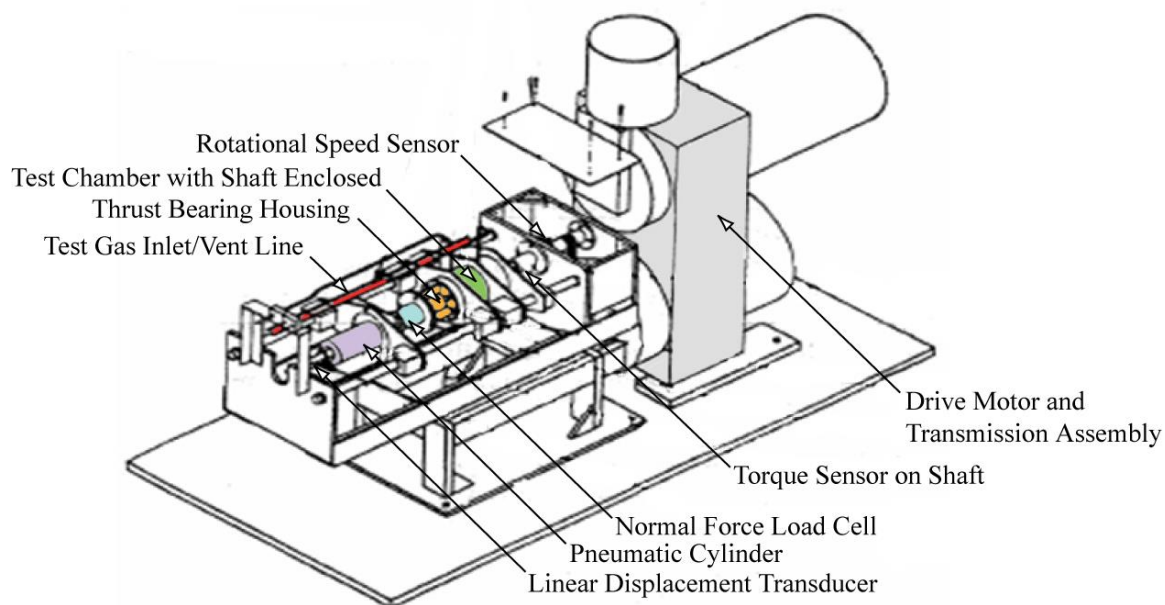


Figure 6—Frictional Heating Test System

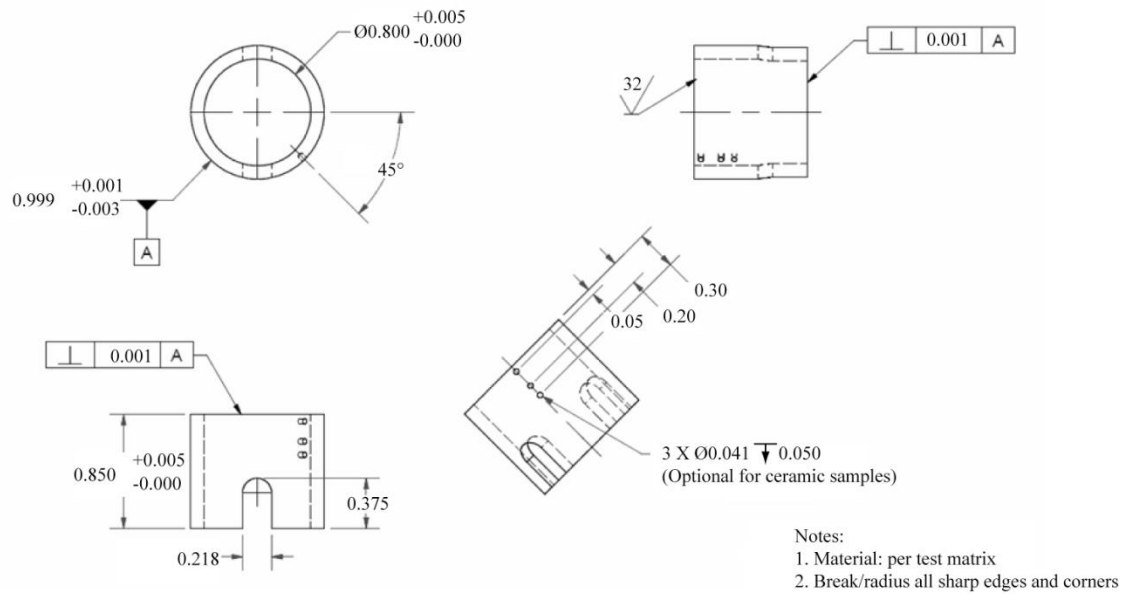


Figure 7—Rotational Frictional Heating Stationary Test Specimen
 (all dimensions in inches)

A.2.12.6 Test System

The test system varies, depending on which test is used and where the test is performed.

Most of the existing rotational friction data was generated using an apparatus similar to that shown in figures 6 and 8, High-Pressure Test Chamber. This apparatus consists of a high-pressure test chamber, an electrical motor, a transmission assembly, and a pneumatic actuation cylinder. The chamber contains a rotating shaft that extends through the chamber by a series of bearings and seals. The shaft is connected at one end to a drive motor/transmission assembly that is capable of rotating the shaft at rotational speeds from 1000 to 27,000 rpm. The other end of the shaft is connected to a pneumatically actuated cylinder that allows axial movement of the shaft and provides the capability of applying a normal load of up to 4450 N (1000 lbf) to the test specimen. The rotating test specimen is mounted on the shaft, and the stationary test specimen is affixed to the test chamber.

The rotating test specimen has an average linear surface velocity, relative to the stationary specimen, of approximately 20.5 m/sec (67.2 ft/sec) when the shaft rotates at 17,000 rpm.

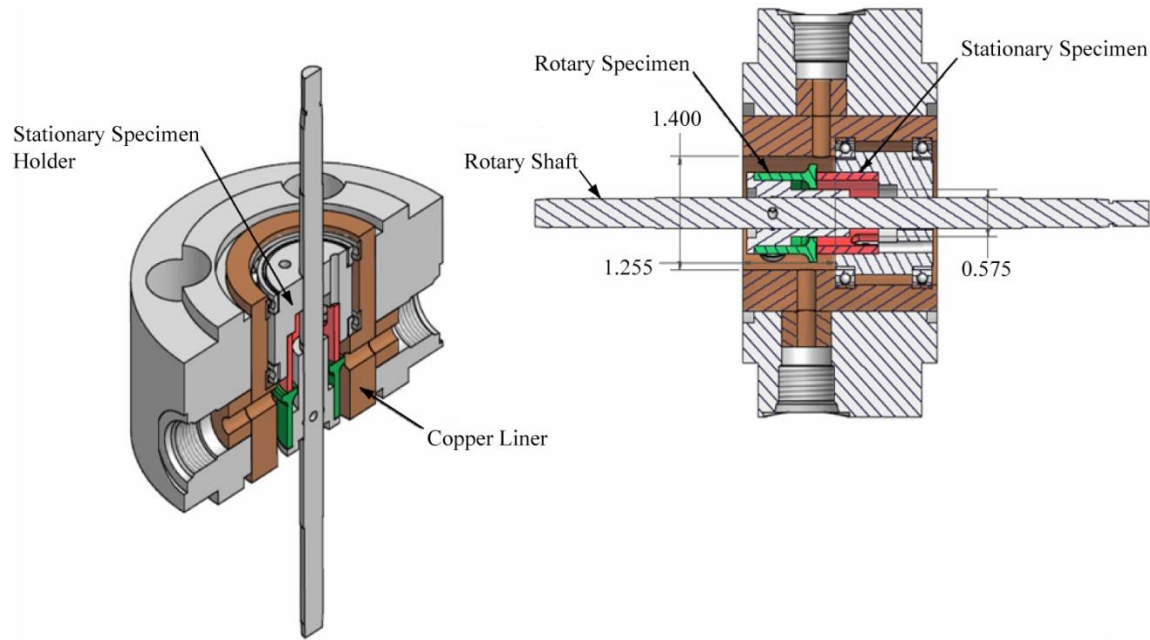


Figure 8—High-Pressure Test Chamber

A.2.12.7 Pre-Test and Test Procedures

The pre-test and test procedures vary, depending on which test is used and where the test is performed.

A.2.12.8 Reporting

- a. [FOCR 377] Results of frictional heating tests shall be summarized in a report that includes documentation of the specimen materials and configuration, the test conditions and apparatus, and test results.
- b. [FOCR 378] If the coefficient of friction is calculated from the data, this value shall be reported.
- c. [FOCR 379] Pre- and post-test photographic images of the test specimens shall be included in the report.
- d. [FOCR 380] The test report shall be submitted to MAPTIS in a MAPTIS-compatible format.

A.2.13 Particle Impact Test

A.2.13.1 Purpose

Particle impact tests are designed to evaluate the susceptibility of materials or components to ignition.

Particle impact tests can be used to accomplish the following objectives:

- a. To determine the conditions at which ignition and consumption of a material occurs when impacted by single or multiple particles entrained in a flow of GOX.*
- b. To determine if a specific material or component is subject to ignition and sustained combustion in a given flow environment when impacted by single or multiple particles entrained in a flow of GOX.*

A.2.13.2 Test Criteria

- a. [FOCR 381] The test criteria shall be defined in the OCA or by the test requester.*

The following criteria are typically applied:

For testing materials:

No ignition: Characterized by no evidence of combustion. A material that does not ignite may show one or more particle indentations on the surface. These indentations may include localized erosion.

Ignition: A portion of or the entire specimen is consumed, and the target is not recoverable.

For testing components:

No ignition: Characterized by no evidence of combustion. A component that does not ignite may show one or more particle indentations on the inside surface of the component. These indentations may include localized erosion.

Ignition: Characterized by obvious consumption of the component, which may include burnout.

Typically for materials, the test conditions are varied to determine the threshold for ignition.

- b. [FOCR 382] For component tests, a minimum of 60 tests shall be performed at a single set of conditions on a single component.*

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- (1) [FOCR 383] If there are no ignitions in 60 tests, the component shall pass the test at the test conditions.
- (2) [FOCR 384] One ignition in 60 tests shall constitute failure of the test at the test conditions.

A.2.13.3 Definitions

Definitions of relevant terms used in the description of this test method are:

High-velocity particle impact: Tests are performed using the supersonic particle impact test fixture. Gas stream velocities in this test system are supersonic ($> \text{Mach } 1$). Gas velocities vary with the temperature and pressure of the GOX system.

Low-velocity particle impact: Tests are performed with the subsonic particle impact test fixture. Gas stream velocities in this system are up to 180 m/sec (600 ft/sec).

A.2.13.4 Test Conditions

a. [FOCR 385] The following test conditions shall be determined by the OCA or by the test requester: gas/target temperature, target pressure, gas velocity, target material/configuration, and particle configuration (size, material, and quantity).

b. [FOCR 386] Components shall be tested at the worst-case conditions as determined by the OCA or by the test requester.

Typically, material tests are performed with 2000- μm (0.08-in) diameter aluminum 2017 particles at supersonic velocity and an upstream test pressure of 27.6 (± 0.7) MPa (4000 (± 100) psig). These conditions may be considered the standard test parameters with which to rank materials against each other for resistance to ignition by particle impact. The majority of the supersonic data generated was generated at this standard set of conditions.

A.2.13.5 Specimen Preparation

a. [FOCR 387] The test specimen configuration shall be specified in the OCA or by the test requester.

b. [FOCR 388] Before testing, material specimens shall be prepared according to the end-use conditions or by washing with any liquid detergent solution, rinsing with deionized water, and drying with GN_2 .

c. [FOCR 389] The specimens shall then be sealed in clean bags until testing.

d. [FOCR 390] From this point on, the targets shall be handled with powder-free, lint-free gloves to maintain cleanliness.

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e. [FOCR 391] The configurations of test targets for the low- and high-velocity testing shall be as follows:

(1) Material Specimens.

A. High-Velocity Particle Impact Material Specimen Preparation (figure 9, High-Velocity Particle Impact Target Specimen Configuration):

Specimens are cup-shaped with an outside diameter of 0.99 cm (0.390 in) and a target surface thickness of 1.5 cm (0.060 in). Each specimen is press fit onto a copper specimen holder for testing.

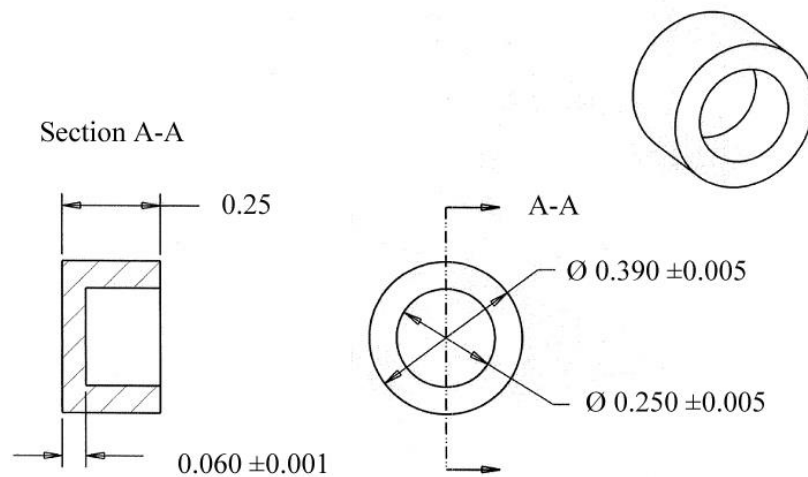


Figure 9—High-Velocity Particle Impact Target Specimen Configuration
(all dimensions are in inches)

B. Low-Velocity Particle Impact Material Specimen Preparation (figure 10, Low-Velocity Particle Impact Target Specimen Configuration):

Specimens are flat discs, 1.5 cm (0.060 in) thick, with holes drilled radially to allow flow through the outer edge of the target.

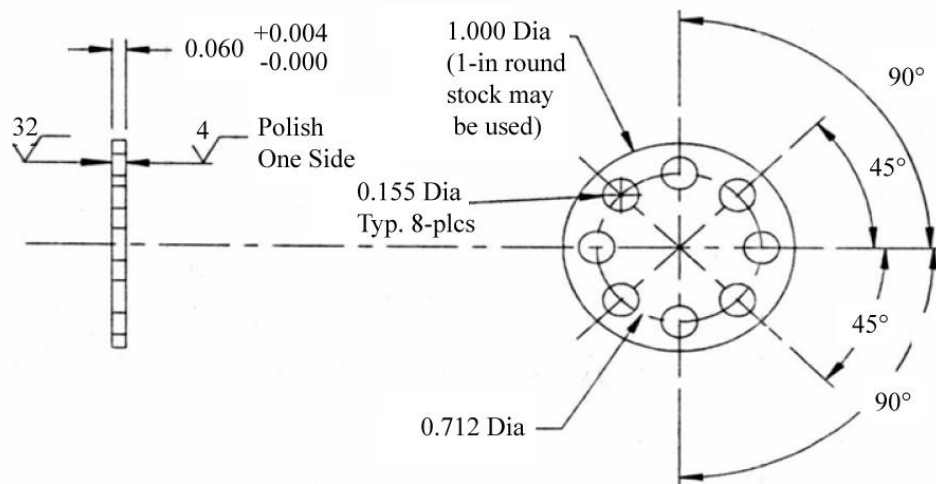


Figure 10—Low-Velocity Particle Impact Target Specimen Configuration
(all dimensions are in inches)

- (2) Component Testing: Components shall be configured in their end-use configuration.

Components can be adapted to either the high- or low-velocity test systems.

A.2.13.6 Test System

a. [FOCR 392] Particle impact tests shall be conducted in a facility capable of supplying heated GOX at the desired test pressure.

b. [FOCR 393] High-velocity particle impact test system: The test specimen shall be placed on a mounting post directly in line with the gas flow stream.

The high-velocity particle impact test system (figure 11, High-Velocity Particle Impact Test System) consists of four major sections: the particle injector assembly, gas inlet and flow straightener, converging-diverging nozzle, and specimen holder. GOX enters the system at subsonic velocities. As the gas enters the converging portion of the nozzle and continues into the throat, it accelerates to a velocity of ~Mach 1. The gas expands as it enters the diverging portion of the nozzle and accelerates to supersonic velocities. The gas stream impinges on the specimen face, creating a stagnation pressure, and flows around the specimen. (This stagnation pressure is lower than the upstream pressure.)

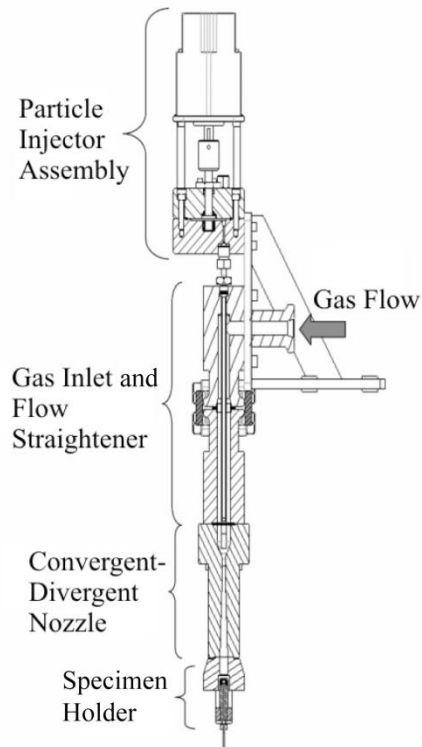


Figure 11—High-Velocity Particle Impact Test System

The following describes testing steps for an example test facility configuration; other facility configurations may lend themselves to variations of this description:

- (1) A bare wire thermocouple is sandwiched between the back face of the test specimen and the specimen post.*
 - (2) The post is placed in a quick-change holder, which mounts on the outlet of the convergent-divergent nozzle.*
 - (3) The upstream GOX pressure is measured by two bonded strain-gauge pressure transducers located upstream of the inlet.*
 - (4) The flow stream (GOX) temperature is measured by a Type-T thermocouple located in approximately the same location.*
 - (5) The bare wire thermocouple, mounted behind the specimen, measures the test specimen temperature. This temperature is reported as the test temperature.*
- c. Low-velocity particle impact test system:

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The low velocity test system delivers GOX at velocities up to 183 m/sec (600 ft/sec), and it consists of the particle injector, the flow straightener, and the impact chamber (figure 12, Low-Velocity Particle Impact Test System).

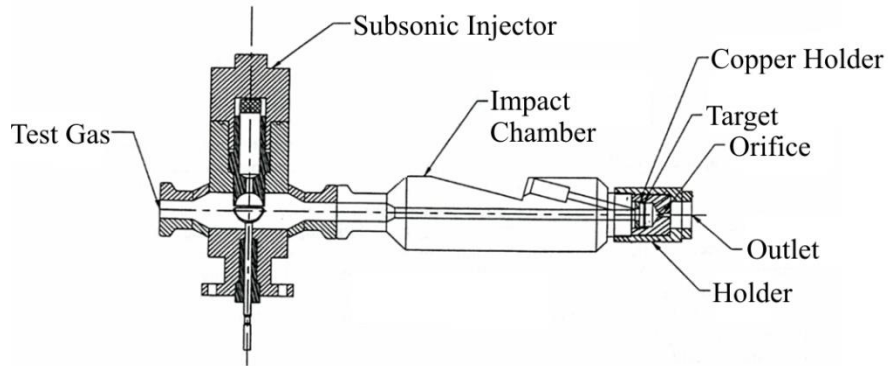


Figure 12—Low-Velocity Particle Impact Test System

- (1) [FOCR 394] A flow orifice, sized to produce the desired gas velocity, shall be placed immediately downstream of either the material specimen or component.
- (2) [FOCR 395] GOX shall enter the particulate injector at subsonic velocities and flow through the flow straightener section and around the target specimen in the impact chamber.

The following describes testing steps for an example test facility configuration; other facility configurations may lend themselves to variations of this description.

- A. *The upstream GOX pressure is measured by a bonded strain-gauge pressure transducer mounted in the flow straightener of the test fixture, located upstream of the inlet.*
- B. *The flow stream (GOX) temperature is measured by a Type-T thermocouple located in approximately the same location. This temperature is reported as the test temperature.*

A.2.13.7 Test Procedure

The following steps describe the testing procedure for an example test facility configuration; other facility configurations may lend themselves to variations of this description:

- a. *Particles are loaded into the particle injector, and the test material or component is configured for testing.*
- b. *A high-speed video camera is recommended to record the test.*

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- c. GOX at the desired test pressure and temperature is allowed to flow until the temperature of the target specimen or component is achieved and the gas flow stabilizes.*
- d. The particle(s) are injected into the flow stream.*
- e. After impact, the GOX flow is terminated, and the test system is allowed to vent to ambient pressure.*
- f. To verify that particle impacts have occurred, the presence of a flash is noted, and the target specimen or component is inspected visually.*
- g. At the completion of test data storage and specimen inspection, the procedure is repeated until the specimen has either passed or failed testing.*

A.2.13.8 Reporting

- a. [FOCR 396] Results of particle impact tests shall be summarized in a report that includes documentation of the specimen target material or component, the particle material and size, the test conditions, and the response of each target to impact.
- b. [FOCR 397] Pre- and post-test photographic images of the test specimens shall be included in the report.
- c. [FOCR 398] The test report shall be submitted to MAPTIS in a MAPTIS-compatible format.

APPENDIX B

CERTIFICATION OF FACILITIES

B.1 Purpose and/or Scope

The purpose of this Appendix is to provide requirements for certification of facilities.

B.2 [FOCR 399] Certification of test facilities shall be obtained for the performance of any of the required tests included in this document.

Certification is necessary because data from these tests will be presented for materials selection approval.

B.3 [FOCR 400] Certification shall be the responsibility of a panel appointed by the NASA Office of the Chief Engineer.

B.4 [FOCR 401] The certification panel shall consist of M&P representatives from NASA-certified facilities and be supported by specialists in appropriate disciplines to evaluate specific test methods.

B.5 [FOCR 402] An application for certification shall be submitted according to requirements listed below:

a. [FOCR 403] The test facility applying for certification shall prepare and submit detailed written procedures to perform all aspects of the subject test methods to the certification panel for review.

b. [FOCR 404] Procedures shall include provisions for instrument calibration, specimen preparation, test chamber certification, test atmosphere analysis or verification, data analysis, report preparation, quality control provisions, recording and archiving of test data, materials control, and control of flight articles.

c. [FOCR 405] The test facility procedures shall also include the definition of the minimum acceptable personnel qualifications, training requirements, and personnel certification procedures.

d. [FOCR 406] Changes to these procedures shall be maintained and made available at the request of the certification panel.

e. [FOCR 407] The test facility also shall supply documentation demonstrating that operations performed in the subject test methods present no hazards to personnel or flight hardware.

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Facilities certified to ISO/IEC 17025, General requirements for competence of testing and calibration laboratories, are considered to have met the requirements of section A.5.

B.6 [FOCR 408] The certification panel shall perform an on-site inspection to ensure that the test facility can perform the methods from receipt of a material to issuance of a final report.

B.7 [FOCR 409] The certification panel shall recommend to the NASA Office of the Chief Engineer certification of the test facility on the basis of compliance with this document.

The NASA Office of the Chief Engineer has the authority to certify the test facility.

B.8 [FOCR 410] The certification panel and test facility shall maintain certification records.

B.9 [FOCR 411] NASA certification of a test facility shall be maintained by meeting the following requirements:

a. [FOCR 412] The test facility shall perform the required test method(s) for which it is certified at least once in an 18-month period from its certification date and report the data to MAPTIS.

b. [FOCR 413] The test facility shall have participated with other NASA-certified test facilities in the most recent round-robin testing for the test method.

c. [FOCR 414] Round-robin tests for Test 1, Test 4, and Test 7 shall be conducted at least every 2 years.

Round-robin tests are recommended for other tests in this NASA Technical Standard.

d. [FOCR 415] The test data shall have been reviewed and approved by the certification panel or the panel's designees.

e. [FOCR 416] Facility test procedures shall be available for audit by the certification panel.

B.10 [FOCR 417] All instrumentation used in the test shall be in current calibration and, where available, bear the appropriate documentation to validate traceability to the appropriate national, international, or intrinsic measurement standards.

B.11 [FOCR 418] The test facility shall ensure and certify that all testing is accomplished in accordance with approved test plans and procedures and that the data records and test results are complete and accurate.

B.12 [FOCR 419] Complete test records shall be prepared by the test facility for each material or system tested.

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B.13 [FOCR 420] The test facility shall maintain certification and test data records for the life of the facility.

B.14 [FOCR 421] Records and data shall be submitted to MAPTIS in a MAPTIS-compatible format.

The NASA Chief Engineer reserves the right to conduct an on-site audit of any certified test facility at any time.

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APPENDIX C

REFERENCE DOCUMENTS

C.1 Purpose and/or Scope

The purpose of this Appendix is to provide guidance made available in the reference documents listed below.

C.2 Reference Documents

ASTM

ASTM D92 Standard Test Method for Flash and Fire Points by Cleveland Open Cup Tester

ASTM D1310 Standard Test Method for Flash Point and Fire Point of Liquids by Tag Open-Cup Apparatus

ISO

ISO 14624-4 Space systems — Safety and compatibility of materials — Part 4: Determination of upward flammability of materials in pressurized gaseous oxygen or oxygen-enriched environments

ISO/IEC 17025 General requirements for competence of testing and calibration laboratories

NASA

JSC 29353 Flammability Configuration Analysis for Spacecraft Applications

NPR 7120.5 NASA Space Flight Program and Project Management Requirements

APPENDIX D

SUBMITTING MATERIAL FOR TESTING

D.1 Purpose and/or Scope

The purpose of this Appendix is to provide requirements for submitting materials for testing.

The quality of the data obtained from the test material or assembled article is dependent on the proper identification and packaging of the material/assembled article for submission for test.

Failure to follow good practice may result in unnecessary delays or a loss in confidence in the data obtained.

D.2 Submitting Material

D.2.1 [FOCR 422] The test material shall be free of flaws and visual contamination.

D.2.2 *The minimum quantity of material to be submitted for each test is shown in table 3.*

D.2.3 *Solid material should be packaged in a visibly clean, sealed bag (for example, zip-locked, heat-sealed, taped) that is compatible with the specimen material. The test material name, as specified by the manufacturer, should appear on the exterior of the bag; the material itself should not be marked. Non-solid materials may be packaged in the original manufacturing container.*

D.2.4 *Electrostatic-discharge-sensitive items and hardware should be packaged in accordance with their end-use specification.*

D.2.5 Documentation

a. [FOCR 423] The MSDS shall be provided.

b. [FOCR 424] If the test facility is to prepare material specimens, the manufacturer's literature or preparation instructions, at a minimum, shall be provided.

c. [FOCR 425] A completed test request form (section D.3 in this NASA Technical Standard) shall be submitted.

Table 6, Key for Completing "Check Test(s) to be Performed" Section, provides a key for completing the Test Information section of the test request form.

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Table 6—Key for Completing “Check Test(s) to be Performed” Section

Test Number	Test Title
1	Upward Flammability
2	Heat and Visible Smoke Release Rate
3	Flash Point of Liquids
4	Electrical Wire Insulation Flammability
6	Odor Assessment
7	Determination of Offgassed Products
10	Configurational Flammability
13a	Mechanical Impact for Materials in Ambient Pressure LOX
13b	Mechanical Impact for Materials in Variable Pressure GOX and LOX
15	Reactivity of Materials in Hydrazine, Monomethylhydrazine, Unsymmetrical Dimethylhydrazine, Aerozine 50, Nitrogen Tetroxide, and Ammonia
17	Upward Flammability of Materials in GOX
18	Arc Tracking
AIT	Autogenous Ignition Temperature
EA	Electrical Arc
FH	Frictional Heating
GFI	Gaseous Fluid Impact for Components
HC	Heat of Combustion
IE	Reactivity and Penetration of Materials due to Incidental Exposure to Hydrazine, Monomethylhydrazine, Unsymmetrical Dimethylhydrazine, Aerozine 50, Nitrogen Tetroxide, and Ammonia
OCA	Oxygen Compatibility Assessment
OI	Oxygen Index
PI	Particle Impact
Special	Other tests (not governed by NASA-STD-6001B), e.g., Volatile Condensable Material

D.3 Test Request Form

Figure 13, Representative Test Request Form, is an example test form. The exact format may vary, depending on the specific test facility.

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Requester Information		Facility to Complete										
*Name:	Name:	Facility I.D. No.										
*Organization:	Organization:											
*Address:	Address:	Requester Tracking No.										
E-Mail:	E-Mail:	Request Date										
*Phone:	Phone:	Material Code:										
Material/Assembly Information												
*MSDS Attached? Yes ☐ No ☐		*Manufacturing Literature Attached? Yes ☐ No ☐										
*Material/Assembly Name/Designation:		Part Number:	Serial Number:									
Batch/Lot #:		Chemical Class:										
Specification:		*Program:	Flight Hardware									
Cleaning Specification/Method:		*Project:	Yes ☐ No ☐									
Other (date of manufacture, composition, color, acronyms, etc.)												
Manufacturer Information		Supplier Information										
*Name:		Name:										
*Address:		Address:										
*Phone:	Web Address:	Phone:	Web Address:									
*Test Information												
Check test(s) to be performed:												
1	2	3	4	6	7	10	13a	13b	15	17	18	AIT
☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
EA	FH	GFI	HC	IE	OCA	OI	PI	Special:				
☐	☐	☐	☐	☐	☐	☐	☐	☐				
*Number of Specimens Provided:		*Use Atmosphere/Fluid:		*Use Thickness:		Intended Application:						
*Test Atmosphere				*Test Pressure				*Test Temperature				
Mass:		Dimensions:		*Photographic Coverage: Video ☐ Stills ☐ None ☐								
*Is test data/information restricted? Yes ☐ No ☐							Restriction:					
Cure Information												
*Is cure required? Yes ☐ No ☐							*Has requester performed cure? Yes ☐ No ☐					
*Provide cure method, whether performed by requester or to be performed by test facility, in the following table:												
Step	Time (hr)	Mix Ratio (if applicable)					Temperature (°F)			Pressure (psia)		
1.		Weight ☐ Volume ☐										
2.		Weight ☐ Volume ☐										
3.		Weight ☐ Volume ☐										
Notes/Remarks/Special Instructions												

* Required information

Figure 13—Representative Test Request Form

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APPENDIX E

REQUIREMENTS COMPLIANCE MATRIX

E.1 Purpose

This Appendix provides a listing of requirements contained in this NASA Technical Standard for selection and verification of requirements by programs and projects. (**Note:** Enter “Yes” to describe the requirement’s applicability to the program or project; or enter “No” if the intent is to tailor, and enter how tailoring is to be applied in the “Rationale” column.)

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Section	Description	Requirement in this Standard	Applicable (Yes or No)	If No, Enter Rationale
1.2.1	Applicability	[FOCR 1] Materials intended for use in space vehicles, specified test facilities, and contractually specified Ground Support Equipment (GSE) shall meet the requirements of this NASA Technical Standard.		
1.2.2	Applicability	[FOCR 2] Responsible NASA Centers’ materials and processes (M&P) organizations shall include applicable requirements of this NASA Technical Standard in their materials control programs.		
1.3	Tailoring	[FOCR 3] Tailoring of this NASA Technical Standard for application to a specific program or project shall be formally documented as part of program or project requirements and approved by the responsible Technical Authority in accordance with NPR 7120.5, NASA Space Flight Program and Project Management Requirements.		
2.1.1	Applicable Documents, General	[FOCR 4] The latest issuances of cited documents shall apply unless specific versions are designated.		
2.1.2	Applicable Documents, General	[FOCR 5] Non-use of specifically designated versions shall be approved by the responsible Technical Authority.		
2.4.2	Order of Precedence	[FOCR 6] Conflicts between this NASA Technical Standard and other requirements documents shall be resolved by the responsible Technical Authority.		

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Section	Description	Requirement in this Standard	Applicable (Yes or No)	If No, Enter Rationale
4. Evaluation and Testing of Materials				
4.1a	Material Evaluation Requirements	[FOCR 7] Materials located in habitable areas of spacecraft, including the materials of the spacecraft, stowed equipment, and experiments, shall be evaluated for flammability and offgassing.		
4.1b	Material Evaluation Requirements	[FOCR 8] All materials located in other areas shall be evaluated for flammability.		
4.1c	Material Evaluation Requirements	[FOCR 9] Materials that are exposed to LOX, GOX, and other reactive fluids shall be evaluated for compatibility with the fluid in their use application.		
4.1d	Material Evaluation Requirements	[FOCR 10] Materials exposed to pressurized breathing gases also shall be evaluated for offgassing.		
4.1e	Material Evaluation Requirements	[FOCR 11] Electrical insulation materials also shall be evaluated for arc tracking.		
4.1f	Material Evaluation Requirements	[FOCR 12] The worst-case anticipated use environment (most hazardous pressure, temperature, material thickness, and fluid exposure conditions) shall be used in the evaluation of material suitability.		
4.2a	Materials Usage Agreement (MUA)	[FOCR 13] Systems containing materials that have not been tested or do not meet the criteria of the required tests shall be verified to be acceptable in the use configuration by analysis or configurational testing through the use of an MUA.		
4.2b	Materials Usage Agreement (MUA)	[FOCR 14] An MUA shall be prepared in accordance with the MUA system described in NASA-STD-6016, Standard Materials and Processes Requirements for Spacecraft, for flight hardware, or the latest version of NASA-STD-5005, Standard for the Design and Fabrication of Ground Support Equipment, for GSE.		
4.2c	Materials Usage Agreement (MUA)	[FOCR 15] The MUA shall document the acceptance rationale.		
4.2d	Materials Usage Agreement (MUA)	[FOCR 16] The MUA shall be submitted for approval as described in NASA-STD-6016 in accordance with the Program/Project Materials and Processes Selection, Control, and Implementation Plan.		

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Section	Description	Requirement in this Standard	Applicable (Yes or No)	If No, Enter Rationale																																			
5. Applicable Material Tests																																							
5.a	Applicable Material Tests	[FOCR 17] Required tests shall be conducted on materials in their worst-case exposure conditions and representative use thicknesses and product forms.																																					
5.b	Applicable Material Tests	[FOCR 18] To be considered acceptable for use without restriction, the materials shall pass the required tests.																																					
5, Table 1	Required Tests for Each Material Use	Table 1, Required Tests for Each Material Use, lists the required tests for each material use. Descriptions of these test methods are provided in section 7 in this NASA Technical Standard. <table><tr><th colspan="3">Table 1—Required Tests for Each Material Use</th></tr><tr><th>ENVIRONMENT</th><th>TEST NUMBER</th><th>TITLE</th></tr><tr><td rowspan="4">Habitable Flight Compartments</td><td>1</td><td>Upward Flame Propagation</td></tr><tr><td>4</td><td>Electrical Wire Insulation Flammability</td></tr><tr><td>7</td><td>Determination of Offgassed Products¹</td></tr><tr><td>18</td><td>Arc Tracking</td></tr><tr><td rowspan="3">Other Areas²</td><td>1</td><td>Upward Flame Propagation</td></tr><tr><td>4</td><td>Electrical Wire Insulation Flammability</td></tr><tr><td>18</td><td>Arc Tracking</td></tr><tr><td rowspan="2">LOX and GOX Environments^{3,4}</td><td>1</td><td>Upward Flame Propagation⁵</td></tr><tr><td>17</td><td>Upward Flammability of Materials in GOX⁶</td></tr><tr><td rowspan="2">Breathing Gases⁴</td><td>1</td><td>Upward Flame Propagation</td></tr><tr><td>7</td><td>Determination of Offgassed Products</td></tr><tr><td>Reactive Fluids</td><td>15</td><td>Reactivity of Materials in Hydrazine, Monomethylhydrazine, Unsymmetrical Dimethylhydrazine, Aerozine 50, Nitrogen Tetroxide, and Ammonia</td></tr></table> ¹ Not required for materials inside hermetically sealed containers. ² Includes all areas outside the habitable flight compartment. ³ [FOCR 19] Materials used in LOX or GOX environments shall be evaluated as described in section 6.3 in this NASA Technical Standard.	Table 1—Required Tests for Each Material Use			ENVIRONMENT	TEST NUMBER	TITLE	Habitable Flight Compartments	1	Upward Flame Propagation	4	Electrical Wire Insulation Flammability	7	Determination of Offgassed Products ¹	18	Arc Tracking	Other Areas ²	1	Upward Flame Propagation	4	Electrical Wire Insulation Flammability	18	Arc Tracking	LOX and GOX Environments ^{3,4}	1	Upward Flame Propagation ⁵	17	Upward Flammability of Materials in GOX ⁶	Breathing Gases ⁴	1	Upward Flame Propagation	7	Determination of Offgassed Products	Reactive Fluids	15	Reactivity of Materials in Hydrazine, Monomethylhydrazine, Unsymmetrical Dimethylhydrazine, Aerozine 50, Nitrogen Tetroxide, and Ammonia		
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Section	Description	Requirement in this Standard	Applicable (Yes or No)	If No, Enter Rationale
		⁴ [FOCR 20] The need for an OCA, as described in section 6.3 in this NASA Technical Standard, shall be addressed for compressed air systems and pressurized systems containing enriched oxygen. ⁵ Required for nonmetals for pressures ≤ 345 kPa (≤ 50 psia). ⁶ Required for nonmetals for pressures > 345 kPa (> 50 psia) and for all metals used in GOX and LOX.		
5.c	Applicable Material Tests	[FOCR 21] Materials that do not meet the criteria of the required tests and that are to remain candidates for use shall be verified to be acceptable in the use configuration by analysis or test and approved through the MUA process as described in section 4.2 in this NASA Technical Standard.		
5.d	Applicable Material Tests	[FOCR 22] The responsible NASA program/project M&P organization shall determine when batch/lot testing is required.		
6. System Evaluations				
6.a	System Evaluations	[FOCR 23] System evaluations shall demonstrate the acceptability of configurations resulting from the use of flammable, incompatible, or offgassing materials.		
6.b	System Evaluations	[FOCR 24] System evaluations shall address single-barrier failures that can expose materials to reactive fluids.		
6.1.1	System Flammability in Low-Pressure Air or Oxygen-Enriched Environments, Including Habitable Environments	[FOCR 25] A system flammability evaluation shall be conducted if materials that fail to meet the criteria of Test 1 are proposed for use in spacecraft, contractually specified test facilities, or contractually specified GSE.		
6.1.2	System Flammability in Low-Pressure Air or Oxygen-Enriched Environments, Including Habitable Environments	[FOCR 26] If the analytical approach is used, the following fire control guidelines shall be addressed and the results documented and submitted to the responsible NASA program/project M&P organization for evaluation: a. The amount and arrangement of flammable materials are controlled to prevent the spread of fire in the vehicle; any potential fire is isolated in defined areas by firebreaks or other techniques (described in JSC 29353), without propagation paths to other areas. b. The design ensures that ignition sources are eliminated or controlled. c. The effects of an isolated fire on personnel in spacecraft crew-habitable modules are considered. d. The rate and magnitude of pressure and temperature increase in any potential fire will not cause structural damage to or compromise the pressure integrity of the vehicle.		

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6.1.3	System Flammability in Low-Pressure Air or Oxygen-Enriched Environments, Including Habitable Environments	[FOCR 27] If the test approach is used, materials that do not meet the criteria of Test 1 shall be evaluated by a configuration test, such as Test 10, which evaluates the worst possible failure modes in the worst-case operating conditions.		
6.1.4	System Flammability in Low-Pressure Air or Oxygen-Enriched Environments, Including Habitable Environments	[FOCR 28] If the results of the evaluation are inconclusive and a major assembly flammability test, such as a crew module test, is required, the hardware developer shall submit a test plan to the responsible NASA program/project M&P organization for approval before test.		
6.3.1	Material and Component Compatibility in LOX and GOX Environments	[FOCR 29] Materials, components, and systems used in LOX and GOX environments shall be evaluated using the approach shown in figure 1, Approach for Selecting Materials, Components, and Systems used in LOX and GOX Environments.		
6.3.2	Material and Component Compatibility in LOX and GOX Environments	[FOCR 30] If the materials are determined to be flammable, an OCA shall be performed in accordance with NASA/TM-2007-213740, Guide for Oxygen Compatibility Assessments on Oxygen Components and Systems.		
6.3.3a	Compressed Air and Enriched Oxygen Systems	[FOCR 31] The need for an OCA shall be addressed for compressed air systems and pressurized systems containing enriched oxygen (greater than 21 percent oxygen by volume).		
6.3.3b	Compressed Air and Enriched Oxygen Systems	[FOCR 32] If it is determined that an OCA is not required, concurrence shall be obtained from the responsible program/project M&P organization.		

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Section	Description	Requirement in this Standard	Applicable (Yes or No)	If No, Enter Rationale
6.3.4a	Compressed Air and Enriched Oxygen Systems	FOCR 33] Pre-Selection of Materials. As a minimum, previously untested polymeric materials that fail Test 1 or Test 17 shall be subjected to autogenous ignition temperature (Appendix A.2.8) and heat of combustion testing (Appendix A.2.9).		
6.3.4b	Compressed Air and Enriched Oxygen Systems	[FOCR 34] Flammability Assessment. Data used to assess material flammability shall be applicable to the specific environment and part/component configuration.		
6.3.4b(1)	Compressed Air and Enriched Oxygen Systems	[FOCR 35] If the flammability or configuration of the materials is unknown, the materials shall be considered flammable.		
6.3.4b(2)	Compressed Air and Enriched Oxygen Systems	[FOCR 36] The responsible NASA program/project M&P organization shall approve that the data used to make flammability assessments are applicable to the specific design configuration.		
6.3.4c(1)	OCA	[FOCR 37] The results of the OCA shall be documented and submitted to the responsible NASA program/project M&P organization for concurrence.		
6.3.4c(2)	OCA	[FOCR 38] An MUA, if applicable, shall be used to submit the OCA results.		
6.3.4c(3)	OCA	[FOCR 39] The hardware developer shall verify that the materials assessed in the OCA reflect the as-built design.		
6.4.1	Materials Compatibility in Systems Containing Reactive Fluids	[FOCR 40] A system evaluation shall be conducted for all systems containing reactive fluids.		
6.4.2	Materials Compatibility in Systems Containing Reactive Fluids	[FOCR 41] Materials used in system construction shall be evaluated for compatibility with the fluid to which they are to be exposed, as well as compatibility with the fluid decomposition products.		
6.4.3	Materials Compatibility in Systems Containing Reactive Fluids	[FOCR 42] Evaluations shall be supported by analysis, by Test 15 data, and when necessary, by configuration-specific testing.		

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Section	Description	Requirement in this Standard	Applicable (Yes or No)	If No, Enter Rationale
6.4.4	Materials Compatibility in Systems Containing Reactive Fluids	[FOCR 43] The analytical approach shall examine possible sources of uncontrolled exothermic reactions, develop control methods, and document the results.		
6.4.5	Materials Compatibility in Systems Containing Reactive Fluids	[FOCR 44] The rationale for use of materials that fail to meet the criteria of Test 15 shall be documented and submitted to the responsible NASA program/project M&P organization for approval.		
7. Test Methods				
7.a	Test Methods	[FOCR 45] Tests shall be performed by test facilities certified in accordance with Appendix B in this NASA Technical Standard.		
7.a(1)	Test Methods	[FOCR 46] The test facility shall demonstrate good laboratory practices to produce accurate and repeatable test data.		
7.a(2)	Test Methods	[FOCR 47] Good laboratory practices shall include calibration and maintenance procedures.		
7.a(3)	Test Methods	[FOCR 48] At least every 2 years, test facilities shall demonstrate testing proficiency to maintain certification in accordance with Appendix B in this NASA Technical Standard.		
7.b	Test Methods	[FOCR 49] Properly identified material for testing shall be provided by the responsible NASA Center or hardware developer and accompanied by a test request form similar to that provided in Appendix D in this NASA Technical Standard.		
7.c(1)	Material and Specimen Receipt and Preparation	[FOCR 50] Materials shall be accompanied by Material Safety Data Sheets (MSDSs) to comply with materials-handling requirements defined by the Occupational Safety and Health Administration (OSHA).		
7.c(2)	Material and Specimen Receipt and Preparation	[FOCR 51] Material specimens shall be inspected visually and any flaws or contamination noted in the test report.		
7.c(3)	Material and Specimen Receipt and Preparation	[FOCR 52] Specimens shall be prepared in the proper dimensions.		

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7.c(4)	Material and Specimen Receipt and Preparation	[FOCR 53] Specimens shall be weighed and identified individually.		
7.c(5)	Material and Specimen Receipt and Preparation	[FOCR 54] Specimens shall be cleaned to end-use specifications.		
7.c(6)	Material and Specimen Receipt and Preparation	[FOCR 55] Specimens shall be inspected after preparation to ensure suitability for the specific test requested.		
7.c(7)	Material and Specimen Receipt and Preparation	[FOCR 56] As a minimum, all fluids used for testing shall meet or exceed user specifications.		
7.d	Material and Specimen Receipt and Preparation	[FOCR 57] When there is a deviation from standard test parameters, such as nonstandard specimen preparation or test conditions, the test shall be identified as nonstandard and the results annotated as nonstandard.		
7, Table 3	Minimum Quantities of Materials Required for Testing for Each Atmosphere or Fluid, Note 1	<i>Table 3, Minimum Quantities of Materials Required for Testing for Each Atmosphere or Fluid, lists the minimum quantities of material required to perform each test properly.</i>		
		Table 3—Minimum Quantities of Materials Required for Testing for Each Atmosphere or Fluid		
		MATERIAL TEST FORM¹	TEST	MINIMUM QUANTITIES
		Solids (sheet, rod, etc.)	1	6 specimens, 30 by 6.4 cm (12 by 2.5 in) by required thickness [thin films, 30 by 7.5 cm (12 by 3.0 in) by required thickness ²]
			2	18 specimens, 10 by 10 cm (4 by 4 in) by required thickness
			6	1500 cm ² (232.5 in ²) total area or 15 g (0.5 oz)
			7	4000 cm ² (620 in ²) total area or 40 g (1.4 oz)

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			13	1000 cm ² (155 in ²) total area, 0.64 cm (0.25 in) maximum thickness or individual specimen diameter of 1.75 cm (0.69 in)		
			15	50 cm ² (7.75 in ²) total area or 20 g (0.7 oz) by required thickness		
			17	15 specimens, 10.2-cm (4-in) (minimum) by 0.32-cm (0.125-in) diameter rods		
		Coatings (paints, inks, adhesives, etc.)	1	Sufficient to cover [paints, 6 specimens, 30 by 6.4 cm (12 by 2.5 in) by required thickness]		
			2	18 specimens, 10 by 10 cm (4 by 4 in) by required thickness		
			6	1500 cm ² (232.5 in ²) total area or 500 ml (16.9 oz)		
			7	4000 cm ² (620 in ²) total area or 500 ml (16.9 oz)		
			13	500 ml (16.9 oz)		
			15	500 ml (16.9 oz)		
		Foams	1	6 specimens, 30 by 6.4 cm (12 by 2.5 in) by required thickness		
			2	18 specimens, 10 by 10 cm (4 by 4 in) by required thickness		
			6	1500 cm ² (232.5 in ²) total area		
			7	4000 cm ² (620 in ²) total area		
			13	1000 cm ² (155 in ²) total area [0.64 cm (0.25 in) maximum thickness]		
			15	50 cm ² (7.75 in) total area or 20 g (0.7 oz)		
		Insulated Wires	1	2 m (6.6 ft) in length		
			4	10 m (33 ft) in length		
			6	15 g (0.5 oz)		
			7	40 g (1.4 oz)		
			13	250 cm (98 in) in length		
			15	1 m (3.3 ft) in length		
			18	70 m (230 ft) in length		

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		Special Requirement	3	As recommended by selected ASTM test method		
			Gaseous Fluid Impact	As recommended by the OCA or by the test requester		
			Autogenous Ignition Temperature	10 g (0.35 oz)		
			Heat of Combustion	10 g (0.35 oz)		
			Oxygen Index	As recommended by ASTM G125; when using ASTM D2863, 25 specimens, 15.2 cm by 6.4 cm (6 by 2.5 in) or equivalent amount of material		
			Frictional Heating	As recommended by the OCA or by the test requester		
			Particle Impact	As recommended by the OCA or by the test requester		
		Notes: ¹ [FOCR 58] Actual test configurations and material quantities for material forms other than those listed, e.g., O-rings and seals, shall be established and approved by the responsible NASA program/project M&P organization. ² When influence of test frame on results is of concern, optional thin films of 30 by 20 cm (12 by 8 in) by required thickness may be used, subject to the approval of the responsible NASA program/project M&P organization.				
7.1.2a	Test Summary	[FOCR 59] Test 1 shall evaluate the candidate material in the worst-case atmosphere to which the material could be exposed.				
7.1.2b	Test Summary	[FOCR 60] Five test specimens, each with dimensions of 30 cm by 6.5 cm (12 in by 2.5 in) and tested one at a time, shall be mounted vertically in a test stand.				
7.1.2c	Test Summary	[FOCR 61] The specimens shall be ignited at the bottom by an ignition system and allowed to burn until each self-extinguishes (figure 2, Flammability Post-Test Specimen).				
7.1.2d	Test Summary	[FOCR 62] The burn length, i.e., the linear distance of consumed specimen, shall be measured and recorded in the test report.				
7.1.2e	Test Summary	[FOCR 63] Whether or not ignition of the paper occurred by a transfer of burning debris shall be recorded on the test report.				
7.1.3a	Test Criteria	[FOCR 64] The material shall meet the test acceptance criteria and be considered self-extinguishing if both of the following conditions are met: (1) No test specimen of the five standard-sized specimens burns >15 cm (6 in).				

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		(2) No test specimen propagates a flame by the transfer of burning debris.		
7.1.4	Test Method	[FOCR 65] The test method defined in ISO 14624-1:2003, Space systems — Safety and compatibility of materials — Part 1: Determination of upward flammability of materials, shall be followed for this test, with the following exceptions, clarifications, and additions: <i>The relevant ISO 14624-1:2003 section is included in parentheses.</i> a. Exception (throughout): One test shall consist of five replicate, identical specimen ignitions. <i>The term “specimen” is used to indicate one standard-sized section of a material used for testing. The Upward Flame Propagation Test requires five specimens to produce one standard test.</i> b. Clarification (section 3.1): <u>Burn length</u>: the distance from the bottom of the specimen to the farthest evidence of damage to the test specimen caused by flame consumption as determined by visual observation, post-test examination, video of burn, and/or other means. c. Addition (section 3.8): <u>Maximum Oxygen Concentration</u>: The maximum oxygen concentration at which at least five specimens pass the acceptance criteria of this test. d. Exception, addition (section 4): (1) In addition, the ignited specimens shall not ignite the paper (produce flaming combustion) below the test specimens, which would indicate that the transfer of burning debris would have sufficient energy to ignite adjacent materials. (2) If, during a test, the paper ignites because of burning debris, subsequent burns during the same material test shall be conducted without the paper. <i>This is done because the burning paper can inhibit the burning of the test specimen. Individual specimen test results performed without the paper should be identified in the test report.</i> <i>Failure of any one specimen constitutes failure of the material in that test environment.</i>		

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		e. Addition (section 4): Materials shall be tested in the worst-case configuration. f. Additions (section 5.1): (1) The test atmosphere shall consist of a mixture of GOX and GN₂, with the oxygen level being the highest that the material could witness in use conditions. <i>If the worst-case environment is uncertain, determination of the maximum oxygen concentration is recommended.</i> (2) The test gases shall be mixed thoroughly before testing a specimen. <i>The gases may be premixed before introduction into the test chamber or mixed inside the test chamber.</i> g. Additions (section 5.1): (1) When gases are mixed in the chamber, they shall be circulated with a fan until a homogeneous mixture is attained, as determined by a gas analyzer. (2) The test gases shall be verified for conformity with the specification (including accuracy) for oxygen concentration to within +1.0 percent -0 percent of reading. h. Addition (section 6.1): <i>Testing may be conducted in a fume hood in air if the above conditions can be met and the test results are verified against test chamber testing results.</i> i. Addition (section 6.1): Air shall not be allowed to flow during tests. j. Clarification (section 6.3): <i>The term "ambient conditions" refers to an oxygen concentration of 20.9 percent, a pressure of 101.4 kPa (14.7 psia), and a temperature of 23 (±5) °C (73 (±9) °F).</i> k. Exception (section 6.4): Bare nickel-chromium wire 0.8 to 1.0 mm in diameter, with a nominal resistivity of 2 to 2.5 Ω·m and of sufficient length to wrap three equally spaced turns around the chemical igniter.		

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		l. Exception (section 6.8): The paper surface density shall be between 150 g/m² and 250 g/m². m. Exception (section 7.1, table 1): The minimum quantity of materials required for testing shall be as indicated in table 3 of this document. n. Exception (section 8.1b): All specimens shall be video recorded during testing. o. Exception (section 8.2.1): Except when testing in air, the test specimen shall be subjected to vacuum for no less than 1 min but for no more than 3 min. p. Exception (section 8.2.1): The igniter shall be retracted from the test specimen once the igniter extinguishes. q. Additions (section 8.2.2): (1) Any flames emanating from the paper below the specimen shall be observed and recorded. (2) The paper shall be supported by a non-flammable, non-conducting screen. (3) Flame jets and sparks emanating from the specimen during combustion shall be observed and recorded. r. Addition: The test report shall be submitted to MAPTIS in a MAPTIS-compatible format.		
7.4.2a	Test Summary	[FOCR 66] The test shall evaluate the candidate wire insulation material in the worst-case atmosphere to which the wire exposure is anticipated.		
7.4.2b	Test Summary	[FOCR 67] Five test specimens, each a wire equivalent to the candidate wire and having a length of 1.2 m (48 in), with the active area being 30 cm (12 in), shall be tested one at a time and mounted at a 15-degree angle from vertical in a test stand.		
7.4.2c	Test Summary	[FOCR 68] The specimens shall be ignited at the bottom by an ignition system and allowed to burn until each self-extinguishes.		
7.4.2d	Test Summary	[FOCR 69] The burn length, i.e., the linear distance of the consumed specimen, shall be measured and recorded.		

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7.4.2e	Test Summary	[FOCR 70] Whether ignition of the paper occurred by a transfer of burning debris shall be recorded on the test report.		
7.4.3a	Test Criteria	[FOCR 71] The material shall meet the test acceptance criteria and be considered self-extinguishing, if both of the following conditions are met: (1) No test specimen of the five tested burns >15 cm (6 in). (2) No test specimen propagates a flame by the transfer of burning debris.		
7.4.4	Test Method	[FOCR 72] The test method described in ISO 14624-2:2003, Space systems — Safety and compatibility of materials — Part 2: Determination of flammability of electrical-wire insulation and accessory materials, shall be followed for this test, with the following exceptions, clarifications, and additions: <i>The relevant ISO 14624-2:2003 sections are included in parentheses.</i> a. Exception (throughout): One test shall consist of five replicate, identical specimen ignitions. b. Exception (throughout): The minimum quantity of material required for testing is listed in table 3 of this document. c. Clarification (section 4.1): <u>Burn length</u>: the distance from the bottom of the specimen as it sits in the test stand to the farthest evidence of damage to the specimen's test insulation material caused by flame consumption as determined by visual observation, post-test examination, video of burn, and/or other means. d. Addition (section 4.3): <u>Transfer of burning debris</u>: movement of burning particles from a burning specimen to adjacent materials with sufficient energy to ignite the paper below the test specimen. e. Addition (section 4): <u>Maximum Oxygen Concentration</u>: The maximum oxygen concentration at which at least five specimens pass the acceptance criteria of this test. f. Exception (section 6.1): A standard test shall consist of five wire specimens tested. g. Addition (section 6.1): In addition, the ignited wire test specimens shall not ignite the paper (produce flaming combustion) below the test specimens, which would indicate that the transfer of burning debris would have sufficient energy to ignite adjacent materials.		

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		h. Addition (section 6.1): If, during a test, the paper ignites because of burning debris, subsequent burns during testing of the same wire specimens shall be conducted without the paper. <i>This is done because the burning paper can inhibit the burning of the wire test specimen.</i> i. Additions (section 6.1): (1) Failure of any one specimen shall constitute failure of the wire in that test environment. (2) Wires shall be tested in the worst-case configuration. j. Additions (section 6.2.1): (1) The test atmosphere shall consist of a mixture of GOX and GN₂, with the oxygen level being the highest that the material could witness in use conditions. <i>If the worst-case environment is uncertain, determination of the maximum oxygen concentration is recommended.</i> (2) The test gases shall be mixed thoroughly before testing a specimen. <i>The gases may be premixed before introduction into the test chamber or may be mixed inside the chamber.</i> k. Additions (section 6.2.1): (1) When gases are mixed in the chamber, they shall be circulated with a fan until a homogeneous mixture is attained, as determined by a gas analyzer. (2) The test gases shall be verified for conformity with the specification (including accuracy) for oxygen concentration to within +1.0 percent -0 percent of reading. l. Addition (section 6.3.1): <i>Testing may be conducted in a fume hood if the above conditions can be met and the test results are verified against test chamber testing results.</i>		

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		m. Addition (section 6.3.1): Air shall not be allowed to flow during tests. n. Addition (section 6.3.3): <i>The term “ambient conditions” refers to an oxygen concentration of 20.9 percent, a pressure of 101.4 kPa (14.7 psia), and a temperature of 23 (±5) °C (73 (±9) °F).</i> o. Exception (section 6.3.5): Bare nickel-chromium wire 0.8 to 1.0 mm in diameter, with a nominal resistivity of 2 to 2.5 Ω·m and of sufficient length to wrap three equally spaced turns around the chemical igniter. p. Exception (section 6.3.8): The paper shall have the following characteristics: (1) Dimensions: (200 mm ±50 mm) × (300 mm ±50 mm). (2) Surface density: between 150 g/m² and 250 g/m². (3) Type: chemical wood processed. (4) Color: uniformly white. (5) Condition: clean, free from dirt spots, oil spots, and foreign matter (lint, fuzz, etc.); free from holes, tears, cuts, folds, and scuff marks; and containing no splices. q. Exception (section 6.5.1.4): All specimens shall be video recorded during testing. r. Addition (section 6.5.2.1): The test specimen shall be subjected to vacuum for no less than 1 min but for no more than 3 min. s. Exception (section 6.5.2.1): The igniter shall be retracted from the test specimen once the igniter extinguishes. t. Additions (section 6.5.2.2): (1) Flames emanating from the paper below the wire specimen shall be observed and recorded.		

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		(2) The paper shall be supported by a non-flammable, non-conducting screen. (3) Flame jets and sparks emanating from the specimen during combustion shall be observed and recorded. u. Exception (section 7): Alternate Wire Insulation Flammability Test in a Gas Flow Environment, Test B, shall not be performed as Test 4. v. Addition: The test report shall be submitted to MAPTIS in a MAPTIS-compatible format.		
7.7 Determination of Offgassed Products (Test 7)				
7.7.2a	Test Summary	[FOCR 73] Test 7 shall determine the offgassing characteristics under standardized conditions for materials and assembled articles to be located within habitable environments.		
7.7.2b	Test Summary	[FOCR 74] Specimens shall be placed into certified-clean sealed containers and thermally conditioned for 72 (± 1) hr at 50 (± 3) °C (122 (± 5) °F).		
7.7.2c	Test Summary	[FOCR 75] After the thermal conditioning, the atmosphere inside the specimen container shall be analyzed to determine the offgassed compounds.		
7.7.2d	Test Summary	[FOCR 76] Using the SMAC for each offgassed compound, the overall toxicity rating shall be determined.		
7.7.2e	Test Summary	[FOCR 77] The overall rating of each material or assembled article shall determine the quantity of each material or the number of assembled articles that can be flown concurrently.		
7.7.2f	Test Summary	[FOCR 78] For a material or assembled article, the total Toxic Hazard Index (T) values for all volatile offgassed products shall be less than 0.5.		
7.7.3a	Test Criteria	[FOCR 79] For a material or assembled article, the total T values for all volatile offgassed products shall be less than 0.5.		
7.7.3b	Test Criteria	[FOCR 80] All materials used in habitable flight compartments shall meet the offgassing requirements, using one of the following methodologies:		
7.7.3b(1)	Test Criteria	[FOCR 81] Assembled article: Summation of T values (total concentration in mg/m ³ /SMAC in mg/m ³) of all offgassed constituent products shall not exceed 0.5.		
7.7.3b(2)A	Test Criteria	[FOCR 82] Individual materials used to make up an assembled article shall be evaluated based on the actual or estimated mass of the material used in the hardware component.		
7.7.3b(2)B	Test Criteria	[FOCR 83] The total T value for all materials used to make up the assembled article shall be less than 0.5.		

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7.7.3b(3)	Test Criteria	[FOCR 84] More than one assembled article: If a single assembled article is tested or evaluated for toxicity but more than one is to be flown, the T value obtained for one unit times the number of flight units shall be less than 0.5.		
7.7.3b(4)	Test Criteria	[FOCR 85] Bulk materials and other materials not inside a sealed container: All materials shall be evaluated individually using the ratings in the MAPTIS database (http://maptis.nasa.gov/).		
7.7.4	Test Method	[FOCR 86] The test method described in the baseline ISO 14624-3:2005, Space systems — Safety and compatibility of materials — Part 3: Determination of offgassed products from materials and assembled articles, shall be followed for this test, with the following exceptions, clarifications, and additions: <i>The relevant ISO 14624-3:2005 sections are included in parentheses.</i> a. Addition (section 3.4 note): <i>A current listing of SMAC values is maintained in MAPTIS at http://maptis.nasa.gov/.</i> b. Addition (section 3): <u>Primary Gas Standard</u>: gas mixtures that have gravimetric or analytical traceability and to which all measurements are ultimately compared. c. Addition (section 3): <u>Trace</u>: the result reported when the identified offgassed compound is present in less than reportable quantities. This category includes compounds that are labeled as unidentified components, because the concentration is too low for the spectral information to allow for identification. It does not include compounds that have adequate spectral information but are labeled as unidentified components because suitable standard spectra for identification are not available. d. Addition (section 3): <u>Reportable quantities</u>: This amount shall be determined by each analytical laboratory and based on analyzed concentrations of the specific compound. <i>Compounds that have been identified, but for which the specific compound is unavailable as a standard, may have reportable quantities based on analyzed concentrations of a representative compound.</i> e. Addition (section 3.5): T Value Calculations: <i>The calculation of the toxic hazard index uses the free volume (spacecraft volume) of the habitable area of the craft under consideration. For the Space Shuttle Orbiter, this volume is 65 m³ (2300 ft³); for the</i>		

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		<i>International Space Station (ISS), the total volume has increased as assembly has proceeded, but the volume used in T value calculation is set at 118 m³ (4170 ft³), which is the volume of the US Laboratory (Destiny) Module; and for the Multipurpose Crew Vehicle Orion vehicle, this volume is anticipated to be 15 m³ (530 ft³). For habitable volumes not defined herein, contact the responsible NASA program/project M&P organization.</i> <i>SMAC is the maximum concentration of an offgassed product that is allowed in the habitable area of the spacecraft for a specified duration.</i> <i>SMAC values for each offgassed constituent are reported in mg/m³. (Unless otherwise specified, all calculations use 7-day SMACs as identified by the NASA JSC Toxicology Office.)</i> (1) Calculations for 45.36 kg (100 lb) of material shall be: Q = Amount for each offgassed constituent in micrograms M = Mass of material tested (g) TX1 = Reported amount for each offgassed constituent in micrograms of constituent per gram of material tested (µg/g) TX2 = Amount for each offgassed constituent in milligrams per cubic meter (mg/m³) for 45.36 kg (100 lb) of material TC = T value for each offgassed constituent for 45.36 kg (100 lb) of material (dimensionless) T100 = Summation of the TC values of all offgassed constituents detected for 45.36 kg (100 lb) of material (dimensionless) MLW_{kg} = Maximum Limit Weight in kilograms of material MLW_{lb} = Maximum Limit Weight in pounds of material Calculation of TX1: $TX1 \left(\frac{\mu g}{g} \right) = \frac{Q(\mu g)}{M(g)} \quad (Eq. 1)$ Calculation of TX2: Convert µg/g to mg/m³		

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		$TX2\left(\frac{\text{mg}}{\text{m}^3}\right) = TX1\left(\frac{\mu\text{g}}{\text{g}}\right) \times \left(\frac{45.36 \text{ kg}}{\text{Spacecraft volume (m}^3\text{)}}\right) \quad (\text{Eq. 2})$ Simplification yields: $TX2\left(\frac{\text{mg}}{\text{m}^3}\right) = TX1\left(\frac{\mu\text{g}}{\text{g}}\right) \times \left(\frac{45.36 \left(\text{kg} = \left(\frac{\text{g} \times \text{mg}}{\mu\text{g}}\right)\right)}{\text{Spacecraft volume (m}^3\text{)}}\right) \quad (\text{Eq. 3})$ $TX2\left(\frac{\text{mg}}{\text{m}^3}\right) = \left(\frac{(TX1 \times 45.36) (\text{mg})}{\text{Spacecraft volume (m}^3\text{)}}\right) \quad (\text{Eq. 4})$ Calculation of TC: $TC = \frac{TX2\left(\frac{\text{mg}}{\text{m}^3}\right)}{SMAC\left(\frac{\text{mg}}{\text{m}^3}\right)} \quad (\text{Eq. 5})$ Calculation of T100: $T100 = \sum_{n=1}^n TC_n, \text{ for } n \text{ TC values} \quad (\text{Eq. 6})$ Calculation of the maximum limit weight (MLW): $MLW_{\text{kg}} = \frac{0.5}{T100} \times 45.36 \text{ kg} \quad (\text{Eq. 7})$		

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		$MLW_{lb} = \frac{0.5}{T100} \times 100 \text{ lb} \quad (\text{Eq. 8})$ (2) Calculation for assembled articles shall be: <i>All assembled articles should be tested in the intended flight configuration.</i> Q = Amount for each offgassed constituent in micrograms TC = T value for each offgassed constituent (dimensionless) T = Summation of the TC values of all offgassed constituents detected (dimensionless) Calculation of the TC: $TC = \left(\frac{Q(\mu\text{g}) \left(\frac{1(\text{mg})}{1000(\mu\text{g})} \right)}{\text{Spacecraft volume (m}^3\text{)}} \right) \times \left(\frac{1}{\text{SMAC} \left(\frac{\text{mg}}{\text{m}^3} \right)} \right) \quad (\text{Eq. 9})$ Calculation of the T: $T = \sum_{n=1}^n TC_n, \text{ for } nTC \text{ values} \quad (\text{Eq. 10})$ Calculation of the maximum number of assembled articles (round down): $\text{Total Number of Assembled Articles} = \frac{0.5}{T} \quad (\text{Eq. 11})$ f. Exception (section 3.11): <i>Room temperature is equal to 23 (±5) °C (73 (±9) °F).</i>		

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		g. Clarification (section 6.1): <i>The test atmosphere should be at least a volume fraction of 20.9 (± 2) percent for GOX with the balance GN_2 or argon, and the test pressure should be ± 15 kPa (± 2.18 psi) of the test facility ambient pressure.</i> h. Addition (section 6.1 note): <i>Batteries or assembled articles containing batteries should be tested in an inert atmosphere to reduce the risk of generating an explosive gas mixture. The batteries or assembled articles containing batteries tested in an inert atmosphere do not need to be tested again in an oxygen atmosphere for toxic offgassing.</i> i. Clarification, addition (section 7.3): <i>Analytical instrumentation is not specified; however, it should be capable of the identification and quantification of all offgassed products at, or below, their SMAC concentrations when tested at a test-material-mass-to-specimen-container-volume ratio of 5.0 (± 0.25) g/l.</i> (1) If the instrumentation cannot achieve this sensitivity, the minimum reportable concentration (reporting limit) for those offgassed products (except for formaldehyde) shall be reported. (2) For formaldehyde, the analytical technique shall be capable of detecting a concentration of 0.1 ppm or current SMAC. <i>The recommended analytical instruments include a gas chromatograph, primarily using a flame ionization detector, a gas chromatograph/mass spectrometer, and an infrared spectrophotometer. Some analytical compounds may be more difficult to determine; therefore, special methods may be required to identify and quantify these compounds. For example, the determination of formaldehyde may be performed using the proposed method of trapping on 2,4-dinitrophenylhydrazine cartridges for derivation and subsequent analysis by High Performance Liquid Chromatography.</i> <i>In some cases, the reporting limit for a compound is greater than the SMAC, but the ability to meet the SMAC requirement is known to be crucial. To determine if the compound offgasses at the SMAC, it is necessary to test more than the standard test-material-mass-to-specimen-container-volume ratio of 5.0 (± 0.25) g/l. The quantity of test material should be increased proportionally from the standard test-material-mass-to-specimen-container-volume ratio to a quantity that will allow the analysis to meet the SMAC requirement.</i>		

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		<i>Example:</i> <i>Benzene SMAC = 100 ppb</i> <i>Reporting limit = 500 ppb</i> <i>Standard material weight per chamber volume = 5 g/l</i> $\frac{5 \text{ g/l}}{100 \text{ ppb}} = \frac{X \text{ g/l}}{500 \text{ ppb}}$ <i>X = Necessary material weight per chamber volume = 25 g/l</i> j. Addition (section 8.2.1): <i>Materials should meet a test-material-mass-to-specimen-container-volume ratio of 5.0 (±0.25) g/l. If the specimen weight cannot be met, the maximum practical quantity of specimen, at least 750 (±50) cm²/l of test chamber volume is tested, with the actual specimen weight and surface area reported.</i> k. Additions (section 8.2): (1) Unless the requester specifies a post-cure duration beyond the cure duration specified in the manufacturer's procedure, the material shall be aged by performing a post-cure under open-air, ambient conditions for a minimum of 14 days. (2) The duration of the post-cure shall be reported. l. Additions (section 8.2.3): (1) Liquids and semi-solids shall be placed in non-reactive vessels. (2) The approximate total specimen surface area shall be reported. m. Addition (section 8.3): Pre- and post-test photographic images shall be acquired when needed or requested by the test requester or test lead.		

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		n. Additions (section 9.4): (1) The methods of quantitative analysis shall be traceable to primary gas standards. (2) When available, standards used to quantify specific compounds shall be traceable to the national, international, or intrinsic standard. o. Exception (section 10.2): The specimen container shall be cooled to 23 (±5) °C (73 (±9) °F). p. Addition (section 10.2): If strong odors are detected during the Test 7 procedures, the test lead shall note this and recommend that Test 6 (Odor Assessment) be performed. q. Addition: The test report shall be submitted to MAPTIS in a MAPTIS-compatible format. r. Exception (section 13.1): Ethyl alcohol, methyl alcohol, tetrachloroethylene, tetrachloromethane, and acrylonitrile shall be measured to within 30 percent of the specified concentrations. s. Exception, addition (table 1): <i>The analysis of furfural in Mix B (as defined in ISO 14624-3:2005) is not required; however, this compound is useful as a diagnostic tool because it presents a meaningful challenge to the analytical system.</i> t. Exception (Annex A.1): Laboratories should be accredited to perform the determination of offgassed products from materials and assembled articles test method contained in ISO 14624-3:2005. u. Exception (Annex B): <i>Annex B is not applicable to Test 7.</i> v. Clarification (Annex C.3): Ratings shall be based on the MLW: <u>Materials:</u> K = MLW of 45 kg (100 lb) or greater H = MLW of 22.5 to 45 kg (50 to 100 lb)		

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		A = MLW of 4.5 to 22.5 kg (10 to 50 lb) V = MLW of 2.3 to 4.5 kg (5 to 10 lb) X = MLW of 0 to 2.3 kg (0 to 5 lb) <u>Assemblies:</u> A = Summation of $T \leq 0.5$ X = Summation of $T > 0.5$		
7.15 Reactivity of Materials in Hydrazine, Monomethylhydrazine, Unsymmetrical Dimethylhydrazine, Aerozine 50, Nitrogen Tetroxide, and Ammonia (Test 15)				
7.15.1	Purpose	[FOCR 87] Appropriate long-term tests shall be conducted for materials with long-term exposure to fuels, oxidizers, and other hazardous fluids.		
7.15.4a(1)	Test Conditions, Screening Test	[FOCR 88] The standard test conditions for the screening test shall be the test facility ambient temperature and pressure.		
7.15.4a(2)	Test Conditions, Screening Test	[FOCR 89] If a material is used above the test temperature, additional testing at the maximum use temperature shall be performed.		
7.15.4b(1)	Test Conditions, Immersion Test	[FOCR 90] For hydrazine, monomethylhydrazine (MMH), unsymmetrical dimethylhydrazine (UDMH), aeroxine 50, and nitrogen tetroxide and the mixed oxides of nitrogen, the standard test conditions for the immersion test shall be 71 (± 3) °C (160 (± 5) °F) and the vapor pressure of the fluid at that temperature.		
7.15.4b(2)	Test Conditions, Immersion Test	[FOCR 91] For ammonia, the standard test conditions for the immersion test shall be 30 (± 3) °C [86 (± 5) °F], and the vapor pressure at that temperature.		
7.15.5a	Test Fluids	[FOCR 92] Hydrazine shall meet the requirements of MIL-PRF-26536, Propellant, Hydrazine, high purity grade.		
7.15.5b	Test Fluids	[FOCR 93] MMH shall meet the requirements of MIL-PRF-27404, Propellant, Monomethylhydrazine.		
7.15.5c	Test Fluids	[FOCR 94] UDMH shall meet the requirements of MIL-PRF-25604, Propellant, uns-Dimethylhydrazine.		
7.15.5d	Test Fluids	[FOCR 95] Aerozine 50 shall meet the requirements of MIL-PRF-27402, Propellant, Hydrazine-uns-Dimethylhydrazine (50% N ₂ H ₄ – 50% UDMH).		
7.15.5e	Test Fluids	[FOCR 96] Nitrogen tetroxide and mixed oxides of nitrogen shall meet the requirements of MIL-PRF-26539, Propellants, Dinitrogen Tetroxide, type MON-3.		
7.15.5f	Test Fluids	[FOCR 97] Ammonia shall be procured with a minimum purity of 99.99 percent (liquid phase).		
7.15.6a	Specimen Preparation	[FOCR 98] Solid and foam materials shall be cut to obtain three specimens 4 by 1 by 0.15 cm (1.6 by 0.4 by 0.06 in) to obtain a total surface area measuring approximately 25 (± 5) cm ² (3.9 (± 0.8) in ²).		

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7.15.6b	Specimen Preparation	[FOCR 99] Greases, semisolids, and liquids shall be applied onto both sides of a tared, wedge-shaped 304L stainless steel substrate.		
7.15.6c	Specimen Preparation	[FOCR 100] The tare weight of the 304L stainless steel substrate shall be determined to an accuracy of 1.0×10^{-4} g.		
7.15.6d	Specimen Preparation	[FOCR 101] Electrical insulation from wires and cables shall be stripped to prepare a specimen measuring $25 (\pm 5) \text{ cm}^2$ ($3.9 (\pm 0.8) \text{ in}^2$) specimen of the insulation.		
7.15.6e(1)	Specimen Preparation, Inks, adhesives, coatings, and primers	[FOCR 102] Test materials shall be applied in the end-use thickness onto abraded 4- by 1- by 0.15-cm (1.6- by 0.4- by 0.06- in) 304L stainless steel substrates.		
7.15.6e(2)	Specimen Preparation, Inks, adhesives, coatings, and primers	[FOCR 103] Requested cures shall be performed.		
7.15.6e(3)	Specimen Preparation, Inks, adhesives, coatings, and primers	[FOCR 104] A default thickness of 0.013 cm (0.005 in) shall be used if the adhesive end-use thickness is not specified.		
7.15.6e(4)	Specimen Preparation, Inks, adhesives, coatings, and primers	[FOCR 105] Sufficient substrates shall be coated to achieve a $25- (\pm 5-) \text{ cm}^2$ ($3.9- (\pm 0.8-) \text{ in}^2$) specimen.		
7.15.6f(1)	Specimen Preparation, Heat-shrinkable tubing	[FOCR 106] Tubing shall be pre-shrunk in accordance with manufacturer's instructions to simulate actual-use conditions.		
7.15.6f(2)	Specimen Preparation, Heat-shrinkable tubing	[FOCR 107] Tubing shall be cut to achieve a surface area of $25 (\pm 5) \text{ cm}^2$ ($3.9 (\pm 0.8) \text{ in}^2$).		

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7.15.6g	Specimen Preparation	[FOCR 108] Pressure-sensitive tapes shall be applied to both sides of three 4- by 1- by 0.15-cm (1.6- by 0.4- by 0.06-in) 304L stainless steel substrates to achieve a surface area of 25 (± 5) cm ² (3.9 (± 0.8) in ²).		
7.15.6h	Specimen Preparation	[FOCR 109] Segments shall be cut or an appropriate number of O-rings and irregularly shaped items be used to achieve a surface area of 25 (± 5) cm ² (3.9 (± 0.8) in ²).		
7.15.6i	Specimen Preparation	[FOCR 110] Specimens shall be cleaned to end-use specifications.		
7.15.6j	Specimen Preparation	[FOCR 111] The specimen's mass shall be determined and the approximate surface area reported.		
7.15.6k	Specimen Preparation	[FOCR 112] Flaws or residual contamination shall be reported.		
7.15.6l	Specimen Preparation	[FOCR 113] A reference specimen shall be prepared for the immersion test.		
7.15.7a	Test System	[FOCR 114] The test system for the immersion test (figure 3, Immersion Test System for Test 15) shall consist of one reference and one specimen immersion tube, each constructed of borosilicate glass and instrumented with a pressure transducer.		
7.15.7b	Test System	[FOCR 115] The total ullage volume above the fill line shall be sufficient for accurate pressure measurement and recorded.		
7.15.7c	Test System	[FOCR 116] The remainder of the system shall be constructed of 304L stainless steel.		
7.15.7d	Test System	[FOCR 117] The immersion tubes shall be maintained in a temperature-controlled water bath instrumented with at least one thermocouple positioned in the bath near the immersion tubes.		
7.15.7d(1)	Test System	[FOCR 118] Thermocouples shall not touch the immersion bath container sides or bottom.		
7.15.7d(2)	Test System	[FOCR 119] The water bath level shall always be above the specimen and reference tube fill levels.		
7.15.7e	Test System	[FOCR 120] Test data, consisting of time, temperature, and pressure, shall be recorded at appropriate intervals (depending on the reaction rate) throughout the test.		
7.15.7f	Test System	[FOCR 121] The immersion system shall have provisions for inert atmosphere purging, filling operations, and cleaning.		
7.15.8a	Pre-Test Procedure	[FOCR 122] The fluid to be used in testing shall be analyzed and verified to meet the required specifications before use.		
7.15.8b	Pre-Test Procedure	[FOCR 123] All pertinent information for the test (for example, specimen identification, composition, pre-test mass and dimensions, and fluid analysis results) shall be recorded.		
7.15.8c	Pre-Test Procedure	[FOCR 124] Reference/pre-test photographic images of the specimens shall be acquired and retained.		

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7.15.8d	Pre-Test Procedure	[FOCR 125] The test system shall be clean and all the measuring devices in current calibration.		
7.15.9a(1)	Screening Test Procedure	[FOCR 126] A test specimen weighing ≤ 0.25 g shall be placed in the glass container.		
7.15.9a(2)	Screening Test Procedure	[FOCR 127] The test fluid shall be added drop by drop with observation for any evidence of gross reactivity.		
7.15.9a(3)	Screening Test Procedure	[FOCR 128] If no gross reactivity is observed, the remainder of the test fluid (typically 10 ml total) shall be added.		
7.15.9a(4)	Screening Test Procedure	[FOCR 129] The specimen shall be observed for visible signs of reaction at the beginning and appropriate intervals (depending on the reaction rate) throughout the test.		
7.15.9a(5)A	At end of 2-hr exposure	[FOCR 130] The specimen shall be removed from the fluid.		
7.15.9a(5)B	At end of 2-hr exposure	[FOCR 131] Observations shall be recorded.		
7.15.9a(5)C	At end of 2-hr exposure	[FOCR 132] Post-exposure mass and dimensions shall be recorded for the remaining material.		
7.15.9a(5)D	At end of 2-hr exposure	[FOCR 133] Post-test visual observations shall be made using defined standard terms (table 4, Standard Descriptive Terms and Definitions for Test 15 Post-Test Analysis).		
7.15.9b(1)	Immersion Test Procedure	[FOCR 134] Both the test specimen and the reference material specimen shall be exposed to the same fluid in their respective containers.		
7.15.9b(2)	Immersion Test Procedure	[FOCR 135] For polymeric specimens, polytetrafluoroethylene (ASTM D3294 Type 1, Grade 1, and a minimum of class B) shall be used as the reference material.		
7.15.9b(3)	Immersion Test Procedure	[FOCR 136] For metallic specimens, 304L stainless steel shall be the reference material.		
7.15.9b(4)	Immersion Test Procedure	[FOCR 137] The specimen and the reference materials shall be placed in the appropriate immersion tube and then purged with GN_2 to remove air.		
7.15.9b(5)	Immersion Test Procedure	[FOCR 138] The system shall be checked for leaks and sufficient test fluid (~25 ml) added up to the fill line to cover the specimen completely to produce the same ullage space in the specimen and the reference tubes.		
7.15.9b(6)	Immersion Test Procedure	[FOCR 139] The pressure and temperature recording system shall be activated.		
7.15.9b(7)	Immersion Test Procedure	[FOCR 140] The immersion tubes shall be placed in the water bath with the water bath level above the specimen and reference tube fill levels.		

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7.15.9b(8)	Immersion Test Procedure	[FOCR 141] Heating of the bath shall be ~ 0.25 °C (~ 0.32 °F)/min.		
7.15.9b(9)	Immersion Test Procedure	[FOCR 142] When not in use, the system shall be maintained under dry GN ₂ purge.		
7.15.9b(9)A	Immersion Test Procedure	[FOCR 143] The closed system shall be held at 71 (±3) °C (160 (±5) °F).		
7.15.9b(9)B	Immersion Test Procedure	[FOCR 144] Pressure shall be recorded as a function of time.		
7.15.9b(9)C	Immersion Test Procedure	[FOCR 145] At the end of the test, the immersion tubes shall be vented, removed from the water bath, and allowed to cool.		
7.15.9b(9)D	Immersion Test Procedure	[FOCR 146] The test fluid shall be removed from the immersion tubes using inert atmosphere transfer and submitted for post-test analysis.		
7.15.9b(9)E	Immersion Test Procedure	[FOCR 147] The immersion tubes shall be removed from the system, inspected, rinsed with deionized water, and allowed to dry overnight.		
7.15.9b(9)Fi	Immersion Test Procedure	[FOCR 148] Post-test analyses for hydrazine fuels shall include assay or purity, carbon dioxide (CO ₂), and NVR.		
7.15.9b(9)Fii	Immersion Test Procedure	[FOCR 149] Nitrogen tetroxide and ammonia post-test analyses shall include NVR.		
7.15.9b(9)G	Immersion Test Procedure	[FOCR 150] Post-test material analyses shall include photographic documentation, visual observations using the standard terms (table 4), and determination of the material mass and dimensions after a documented drying time.		
7.15.9b(9)H	Immersion Test Procedure	[FOCR 151] Post-test DSC and FTIR analyses shall be performed on specimens tested in nitrogen tetroxide as a screen for the formation of potentially energetic materials produced by nitration.		
7.15.10a	Reporting	[FOCR 152] The test report shall include specimen identification, method of preparation, configuration, fluid identity and specification (including grade), test conditions, data, and observations from the test, including visual observations, photographic images, gas evolution rate or system pressure with time, and results of post-test analyses.		
7.15.10b	Reporting	[FOCR 153] The test report shall be submitted to MAPTIS in a MAPTIS-compatible format.		
7.15.10c	Reporting	[FOCR 154] If there is a deviation from standard test conditions (for example, test duration, temperature, and specimen surface area), the test shall be identified as nonstandard.		
7.15.11	Precision	[FOCR 155] Measurements shall be made to the following precision: a. Absolute pressure: ± 3 kPa (± 0.5 psi). b. Temperature: ± 3 °C (± 5 °F). c. Specimen dimensions: ± 2 mm (± 0.08 in).		

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		d. Specimen Mass: ± 1 mg. e. Time: ± 10 sec.		
7.15.12	Failure Criteria	[FOCR 156] The material shall be considered to have failed this test if any of the following occur in either the screening test or the immersion test: a. Burning, charring, or fire. b. Frothing. c. The material dissolves in the test fluid. d. The material crumbles, becomes friable, or generates particulate. e. The material changes shape or physical dimensions by more than 20 percent. f. The material suffers more than 20 percent degradation of the physical properties for which it was selected. g. Visible changes in appearance of the test fluid (color and/or clarity). h. For nitrogen tetroxide, a pressure rise >68.9 kPa (>10 psi). i. For polymers exposed to nitrogen tetroxide, evidence of material nitration as indicated by DSC and/or FTIR.		
7.15.13a	Quality Control	[FOCR 157] Based on historical data, the pressure rise on the system reference side shall be <0.4 kPa (<0.06 psia)/hr for nitrogen tetroxide and ammonia.		
7.15.13b	Quality Control	[FOCR 158] In the case of the hydrazine fuels, the gas evolution rate shall be $<7.0 \times 10^{-3}$ standard $\text{cm}^3/(\text{cm}^2 \text{ hr})$.		
7.17.2a	Test Summary	[FOCR 159] Test 17 shall determine the flammability of materials at the use pressure in GOX at ambient or elevated temperatures.		
7.17.2b	Test Summary	[FOCR 160] The test specimens shall be 0.32-cm (0.126-in) diameter rods with a minimum length of 10.2 cm (4 in).		
7.17.2c	Test Summary	[FOCR 161] The specimens shall be mounted vertically in a test chamber, ignited at the bottom by an ignition system, and allowed to burn until each self-extinguishes.		
7.17.2d	Test Summary	[FOCR 162] The burn length, e.g., the length of the specimen that has been consumed by combustion, shall be measured.		
7.17.2e	Test Summary	[FOCR 163] To determine a pressure at which the material is not flammable, at least 10 specimens shall be tested without consumption >3 cm (>1.2 in).		
7.17.3a	Test Criteria	[FOCR 164] To determine a pressure at which the material is not flammable, at least 10 specimens shall be tested without consumption greater than >3 cm (>1.2 in).		
7.17.3b	Test Criteria	[FOCR 165] The test shall be considered valid only if the promoter ignites, burns, and detaches from the specimen.		

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7.17.4	Test Method	[[FOCR 166] The test method defined in ASTM G124, Standard Test Method for Determining the Combustion Behavior of Metallic Materials in Oxygen-Enriched Atmospheres, shall be followed for this test, with the following exceptions, clarifications, and additions. <i>The relevant ASTM G124 section is included in parentheses.</i> <i>ISO 14624-4, Space systems — Safety and compatibility of materials — Part 4: Determination of upward flammability of materials in pressurized GOX or oxygen-enriched environments, describes a similar test procedure.</i> a. Exception (throughout): The specimen shall be tested at ambient temperature if its use temperature is ≤ 260 °C (≤ 500 °F). b. Exception (section 8.2): For metallic materials, the promoter shall consist of a sufficient quantity of aluminum or magnesium to release at least 3.0 kJ (717 cal). c. Exception (section 8.2): For polymeric materials, the promoter shall be sufficient for ignition of the material, as determined by the responsible NASA program/project M&P organization. d. Exception (section 10.1): The test specimens shall be a minimum of 10.2 cm (4 in) (± 5 percent) in length. <i>A minimum length of 15 cm (6 in) (± 5 percent) is preferable.</i> e. Exception (section 10.1): If a material cannot be obtained or prepared in the cylindrical form, a nonstandard specimen shall be prepared. <i>The nonstandard specimen should have 0.32-cm (0.126-in) (± 5 percent) sides and should be a minimum of 10.2 cm (4 in) (± 5 percent) in length.</i> f. Addition (section 16): The sample temperature measurement shall have precision of ± 5 percent of reading.		

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		g. Addition (section 16): The test report shall be submitted to MAPTIS in a MAPTIS-compatible format. h. Addition (section 16): If there is a deviation from the standard test parameters (for example, nonstandard specimen preparation, orientation, configuration, ignition source, and test pressure), the test shall be identified as nonstandard. i. Addition (section 16.1.6): The pressure measurements shall have a precision of ± 1 percent of reading. j. Addition (section 16.1.9): The measurement of the length of sample consumed shall have a precision of ± 0.25 cm (± 0.1 in).		
7.18 Arc Tracking (Dry Arc Propagation Resistance) (Test 18)				
7.18.3a	Test Criteria	[FOCR 167] To determine the arc propagation resistance of the wire configuration, at least 15 seven-wire bundles shall be tested.		
7.18.3b	Test Criteria	[FOCR 168] A wire insulation material shall be considered to have met the acceptance criteria of this test if all the following requirements are met:		
7.18.3b(1)	Test Criteria	[FOCR 169] At least 64 wires shall pass the dielectric test.		
7.18.3b(2)	Test Criteria	[FOCR 170] Three wires or less shall fail the dielectric test in any one bundle.		
7.18.3b(3)	Test Criteria	[FOCR 171] Actual damage to the wire shall not be more than 7.6 cm (3 in) in any one bundle.		
7.18.5	Test Conditions	[FOCR 172] Test conditions shall be as defined in either MIL-STD-2223, Test Methods for Insulated Electric Wire, Method 3007, Dry Arc-Propagation Resistance, or SAE AS4373, Test Methods for Insulated Electric Wire, Method 508, Dry Arc Propagation Resistance.		
7.18.6a	Specimen Preparation	[FOCR 173] A test specimen shall be a bundle of seven wires and of sufficient length (35.6 cm (14 in) minimum) to allow the bundle to be installed in the test fixture.		
7.18.6b	Specimen Preparation	[FOCR 174] Fifteen bundles shall be required for a full test.		
7.18.6c	Specimen Preparation	[FOCR 175] The specimens shall be prepared according to MIL-STD-2223, Method 3007, or SAE AS4373, Method 508.		
7.18.7	Test System	[FOCR 176] The test system shall be that described in MIL-STD-2223, Method 3007, or SAE AS4373, Method 508.		
7.18.8	Pre-Test Procedure	[FOCR 177] The pre-test procedures shall be those described in MIL-STD-2223, Method 3007, or SAE AS4373, Method 508.		

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7.18.9	Test Procedure	[FOCR 178] The test procedures shall be those described in MIL-STD-2223, Method 3007, or SAE AS4373, Method 508.		
7.18.10a	Reporting	[FOCR 179] The test report shall include specimen identification, test conditions, total number of wires that pass the dielectric test, number of wires from each bundle that pass the dielectric test, and the length of physical damage to each individual wire in the bundle.		
7.18.10b	Reporting	[FOCR 180] The test report shall be submitted to MAPTIS in a MAPTIS-compatible format.		
Appendix A Supplemental Tests				
A.2a	Test Methods for Providing Supplemental Data	[FOCR 181] Tests shall be performed by test facilities certified in accordance with Appendix B in this NASA Technical Standard.		
A.2a(1)	Test Methods for Providing Supplemental Data	[FOCR 182] The test facility shall demonstrate GLPs to produce accurate and repeatable test data.		
A.2a(2)	Test Methods for Providing Supplemental Data	[FOCR 183] GLPs shall include calibration and maintenance procedures.		
A.2a(3)	Test Methods for Providing Supplemental Data	[FOCR 184] At least every 2 years, test facilities shall demonstrate testing proficiency to maintain certification in accordance with Appendix B in this NASA Technical Standard.		
A.2b	Test Methods for Providing Supplemental Data	[FOCR 185] Properly identified material for testing shall be provided by the responsible NASA Center or contractor hardware supplier and accompanied by a test request form similar to that provided in Appendix D in this NASA Technical Standard.		
A.2c(1)	Test Methods for Providing Supplemental Data	[FOCR 186] Materials shall be accompanied by MSDSs to comply with materials-handling requirements defined by OSHA.		
A.2c(2)	Material and Specimen Receipt	[FOCR 187] Material specimens shall be visually inspected and any flaws or contamination noted in the test report.		

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	and Preparation			
A.2c(3)	Material and Specimen Receipt and Preparation	[FOCR 188] Specimens shall be prepared in the proper dimensions.		
A.2c(4)	Material and Specimen Receipt and Preparation	[FOCR 189] Specimens shall be weighed and identified individually.		
A.2c(5)	Material and Specimen Receipt and Preparation	[FOCR 190] Specimens shall be cleaned to end-use specifications.		
A.2c(6)	Material and Specimen Receipt and Preparation	[FOCR 191] Specimens shall be inspected after preparation to ensure suitability for the specific test requested.		
A.2c(7)	Material and Specimen Receipt and Preparation	[FOCR 192] As a minimum, all fluids used for testing shall meet or exceed user specifications.		
A.2d	Material and Specimen Receipt and Preparation	[FOCR 193] When there is a deviation from standard test parameters, e.g., nonstandard specimen preparation or test conditions, the test shall be identified as nonstandard.		
A.2.1.3	Test Criteria	[FOCR 194] The test shall be conducted on specimens that are representative of those in the intended use environment and at worst-case conditions.		
A.2.1.5	Test Conditions	[FOCR 195] Test conditions (incident heat flux and oxygen concentration) shall simulate the worst-case conditions in which ignition and combustion of the material are most likely to occur.		
A.2.1.6a	Specimen Preparation	[FOCR 196] Specimens shall be prepared in accordance with ASTM E1354, Standard Test Method for Heat and Visible Smoke Release Rates for Materials and Products Using an Oxygen Consumption Calorimeter.		
A.2.1.6b	Specimen Preparation	[FOCR 197] Coatings or any viscous materials that cannot be mounted for testing without a substrate shall be applied to the end-use substrate material in the worst-case thickness.		

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Section	Description	Requirement in this Standard	Applicable (Yes or No)	If No, Enter Rationale
A.2.1.6c(1)	Specimen Preparation, Cleaning of Specimens	[FOCR 198] Contamination on the surfaces of solid, nonporous specimens shall be removed by washing with deionized water and any detergent, rinsing with deionized water, and drying with filtered GN ₂ .		
A.2.1.6c(2)	Specimen Preparation, Cleaning of Specimens	[FOCR 199] Particulate on the surfaces of solid, porous specimens shall be removed with filtered GN ₂ .		
A.2.1.7	Test System	[FOCR 200] The test system shall be as described in ASTM E1354 but with additional capabilities that will allow testing in GOX/GN ₂ mixtures other than air.		
A.2.1.8	Pre-Test Procedure	[FOCR 201] The pre-test procedures shall be as described in ASTM E1354.		
A.2.1.9	Test Procedure	[FOCR 202] The test procedures shall be as described in ASTM E1354, with the following exceptions: a. The specimen shall be placed on the load cell in the test chamber and exposed to the test atmosphere by allowing the test gases to flow through the test chamber. b. The initial oxygen concentration shall be measured.		
A.2.1.10	Reporting	[FOCR 203] The test report content shall be as described in ASTM E1354.		
A.2.2.5	Selection of a Flash Point Test Method	[FOCR 204] The appropriate flash point test method for the application shall be selected by following the recommendations of ASTM E502, Standard Test Method for Selection and Use of ASTM Standards for the Determination of Flash Point of Chemicals by Closed Cup Methods.		
A.2.2.6	Test Systems and Procedures	[FOCR 205] The test systems and procedures shall be identical to those described in the selected ASTM test method(s).		
A.2.2.7	Reporting	[FOCR 206] The test report shall include specimen identification, the method used, test conditions, the flash point of the liquid corrected to 101.3 kPa (14.7 psia), and observations from the test.		
A.2.2.8	Precision	[FOCR 207] Measurements shall meet the precision requirements of the ASTM method(s) used.		
A.2.3.4a	Test Conditions	[FOCR 208] The test atmosphere shall be at least a volume fraction of 20.9 (±2) percent for GOX, with the balance being GN ₂ or argon.		
A.2.3.4b	Test Conditions	[FOCR 209] Test pressure shall be ±3.5 kPa (±0.5 psi) of the test facility ambient pressure.		
A.2.3.4c	Test Conditions	[FOCR 210] The maximum volume fraction limits (expressed as a volume fraction in µl/l) for impurities in the compressed gases shall be: (1) Carbon monoxide: 1.0.		

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		(2) Carbon dioxide: 3.0. (3) Total hydrocarbons, as methane: 0.1. (4) Halogenated compounds: 0.5. (5) Water: 7.0.		
A.2.3.4d	Test Conditions	[FOCR 211] The specimen shall be subject to a thermal exposure for 72 (± 1) hr at 50 (± 3) °C (122 (± 5) °F).		
A.2.3.5a	Specimen Preparation	[FOCR 212] Test specimens shall be prepared from either materials or assembled articles.		
A.2.3.5b	Specimen Preparation	[FOCR 213] Handling of test specimens shall be in a manner that preserves the integrity of the specimen surface without adding contaminants.		
A.2.3.5c	Specimen Preparation	[FOCR 214] All materials shall meet the requirement of test-material-mass-to-specimen-container-volume ratio of 5.0 (± 0.25) g/l.		
A.2.3.5d(1)A	Specimen Preparation	[FOCR 215] Materials that are essentially two-dimensional and require application to a substrate (for example, coatings, primers, inks, paints, adhesives, tapes, and thin film lubricants) shall be applied at their thickness of use to clean aluminum substrates.		
A.2.3.5d(1)Ai	Specimen Preparation	[FOCR 216] A sufficient number of substrates with applied specimen material shall be prepared to provide a net test-material-mass-to-specimen-container-volume ratio of 5.0 (± 0.25) g/l.		
A.2.3.5d(1)Aii	Specimen Preparation	[FOCR 217] The approximate total specimen surface area shall be recorded.		
A.2.3.5d(1)B	Specimen Preparation	[FOCR 218] Materials that are essentially two-dimensional and are not applied to a substrate (for example, fabrics, photographic film plastic, plastic film, elastomers, and non-adhesive tape) shall be cut to convenient test dimensions.		
A.2.3.5d(1)Bi	Specimen Preparation	[FOCR 219] Heat-shrinkable tubing shall be shrunk to simulate actual use configuration.		
A.2.3.5d(1)Bii	Specimen Preparation	[FOCR 220] A sufficient quantity of specimens shall be prepared to provide a test-material-mass-to-specimen-container-volume ratio of 5.0 (± 0.25) g/l.		
A.2.3.5d(1)C	Specimen Preparation	[FOCR 221] Non-curing liquids shall be placed in suitable non-reactive vessels and the approximate total specimen surface area reported.		
A.2.3.5d(1)D	Specimen Preparation	Specialized items and materials that do not meet the above requirements and that require special handling, e.g., non-homogeneous materials.		
A.2.3.5d(1)Di		[FOCR 222] Such materials shall be tested in the manner designated by the responsible NASA program/project M&P organization.		

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A.2.3.5d(1)Dii	Specimen Preparation	[FOCR 223] The manner of testing and specimen preparation shall be reported.		
A.2.3.5d(2)A	Specimen Preparation for Flight Articles	[FOCR 224] Specimens that are assembled articles shall be inspected for parts that are not designated for flight, e.g., dust covers, tape, or test leads.		
A.2.3.5d(2)Ai	Specimen Preparation for Flight Articles	[FOCR 225] These items shall be removed before testing.		
A.2.3.5d(2)Aii	Specimen Preparation for Flight Articles	[FOCR 226] The absence of items (for example, batteries or photographic film) that will be included during flight but that are not included with the specimen shall be recorded.		
A.2.3.5d(2)B	Specimen Preparation for Flight Articles	[FOCR 227] The ratio of test-material-mass-to-specimen-container-volume shall be approximately 1:3.		
A.2.3.5d(2)C	Specimen Preparation for Flight Articles	[FOCR 228] When the offgas test atmosphere is preserved for odor assessment, the odor assessment shall be initiated within 14 days from the start of the specimen gas preservation.		
A.2.3.5e	Specimen Preparation	[FOCR 229] Specimens shall be cleaned in accordance with section 8.4 of ISO 14624-3:2005.		
A.2.3.6a	Test System	[FOCR 230] The test system shall consist of the specimen container, the test chamber with controlled temperature, and volunteer odor panel personnel.		
A.2.3.6b	Test System	[FOCR 231] The specimen container, including any soft goods, shall have an odor rating of 0 (Undetectable), as defined in section A.2.3.8.g in this NASA Technical Standard, when an empty chamber is tested in accordance with the procedure in section A.2.3.8 in this NASA Technical Standard.		
A.2.3.6c	Test System	[FOCR 232] The test chamber shall have the capability to maintain the test temperature to within ± 3 °C (± 5.4 °F) for the duration of the test.		
A.2.3.6d	Test System	[FOCR 233] The test chamber instrumentation shall have the capability to record the temperature continuously.		

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A.2.3.6e	Test System	[FOCR 234] Glass syringes (30 cm ³ (1.8 in ³) capacity) shall be used for measuring and administering the odor specimen.		
A.2.3.6f	Test System	[FOCR 235] When administering the specimen, a low-odor non-obtrusive mask that covers the nose and mouth shall be used.		
A.2.3.6g	Test System	[FOCR 236] The syringes and masks shall have an odor rating of 0 (Undetectable), as defined in section A2.3.8.g in this NASA Technical Standard, when an empty chamber or blank air specimen is tested in accordance with the procedure in section A.2.3.8 in this NASA Technical Standard.		
A.2.3.7a	Pre-Test Procedure	[FOCR 237] A NASA or NASA contractor volunteer shall perform the following: (1) Read, understand, and sign a consent form as required in NPR 7100.1, Protection of Human Research Subjects, and NPD 7100.8, Protection of Human Research Subjects. (2) Be examined and qualified by a registered medical professional. (3) Be odor qualified by distinguishing the seven primary standards (table 5, Seven Primary Standards for Detection of Odor) from three odorless standards.		
A.2.3.7b	Pre-Test Procedure	[FOCR 238] An odor panel shall test each new compressed gas that is used for test atmospheres before use.		
A.2.3.7c	Pre-Test Procedure	[FOCR 239] Gases used for test atmospheres shall be judged as being odorless.		
A.2.3.7d	Pre-Test Procedure	[FOCR 240] The specimen containers shall be cleaned by washing with a diluted soap solution and rinsing with deionized water, allowed to dry, and then purged with filtered air or GN ₂ .		
A.2.3.7e	Pre-Test Procedure	[FOCR 241] Before use, each container shall be certified to be free of odor by the odor panel.		
A.2.3.7f	Pre-Test Procedure	[FOCR 242] Before odor testing, materials or assembled articles shall be tested for offgassed products (Test 7).		
A.2.3.7g	Pre-Test Procedure	[FOCR 243] If the gas from the specimen container has been determined to be unsafe to test, based on the offgassing analysis, the calculated toxic level and reasons for not performing the test shall be recorded.		
A.2.3.7h	Pre-Test Procedure	[FOCR 244] Duplicate specimens of materials shall be offgassed—one specimen for offgassing analysis and the other for odor assessment.		
A.2.3.7i	Pre-Test Procedure	[FOCR 245] If the specimen cannot be provided in duplicate, the same specimen shall be used in both offgassing analysis and odor assessment.		
A.2.3.7j	Pre-Test Procedure	[FOCR 246] Before each test, members of the odor panel shall receive a nose and throat examination (for nasal irritation or other abnormal conditions) by a medical staff member.		

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A.2.3.7k	Pre-Test Procedure	[FOCR 247] Results of the examination shall be recorded.		
A.2.3.7l	Pre-Test Procedure	[FOCR 248] Each odor panel member shall identify an odorless solution from two of the seven primary standards.		
A.2.3.7m	Pre-Test Procedure	[FOCR 249] Members of the odor panel shall not be allowed to participate if their sense of smell has been affected in any manner, such as by recent smoking, ingestion of highly flavored foods, or exposure to pungent vapors.		
A.2.3.8a	Test Procedure	[FOCR 250] The specimen shall be weighed and placed in the specimen container.		
A.2.3.8b	Test Procedure	[FOCR 251] The atmosphere in the specimen container shall be replaced with the test atmosphere, either by purging or by evacuation.		
A.2.3.8b(1)	Test Procedure	[FOCR 252] The requesting organization shall indicate if the specimen can or cannot withstand a vacuum.		
A.2.3.8b(2)	Test Procedure	[FOCR 253] Exposure of any specimen to vacuum shall be less than 3 min.		
A.2.3.8c	Test Procedure	[FOCR 254] The specimen container, with the test atmosphere, shall be at the specified test pressure when the test temperature is achieved.		
A.2.3.8d	Test Procedure	[FOCR 255] The specimen shall be placed in the test chamber and heated to the test temperature of 50 (±3) °C (122 (±5) °F), unless otherwise specified.		
A.2.3.8d(1)	Test Procedure	[FOCR 256] The test temperature shall be maintained for 72 (±1) hr.		
A.2.3.8d(2)	Test Procedure	[FOCR 257] Then, the specimen container shall be cooled to ambient temperature and the pressure recorded.		
A.2.3.8e	Test Procedure	[FOCR 258] The odor panel shall be convened.		
A.2.3.8f	Test Procedure	[FOCR 259] At least five qualified members shall be administered at least 30 cm ³ of the gas from the specimen container.		
A.2.3.8g	Test Procedure	[FOCR 260] Using the following scale, each odor panel member shall assign an odor characteristic to the gas from the specimen container: Undetectable: 0 Barely Detectable: 1 Easily Detectable: 2 Objectionable: 3 Revolting: 4		
A.2.3.8h	Test Procedure	[FOCR 261] Between 2 and 4 hr after the test, the panel members shall be examined medically for nasal irritation or other abnormal conditions and the results of this examination recorded.		

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A.2.3.8i	Test Procedure	[FOCR 262] Each odor panel member shall be exposed to only one odor specimen in 48 hr.		
A.2.3.9a	Reporting	[FOCR 263] The test report shall include specimen identification, configuration, test conditions, and the average rating from the odor panel members.		
A.2.3.9b	Reporting	[FOCR 264] Any nasal irritation or other abnormal conditions from the medical examination shall be recorded.		
A.2.3.9c	Reporting	[FOCR 265] If the material is determined to be unsafe for testing (based on the offgassing analysis from Test 7), the reasons for the determination not to perform Test 6 on the specimen shall be included.		
A.2.3.9d	Reporting	[FOCR 266] The test report shall be submitted to MAPTIS in a MAPTIS-compatible format.		
A.2.3.10	Precision	[FOCR 267] Measurements shall be made to the following precision: a. Absolute pressure: ± 1 percent of reading. b. Temperature: ± 3 °C (± 5.4 °F). c. Oxygen concentration: ± 0.5 percent of reading. d. Mass: ± 0.01 g.		
A.2.3.11	Odor Panel Member Requalification	[FOCR 268] Each odor panel member shall be requalified every 4 months or before conducting an odor test, using the procedure described in section A.2.3.7.a in this NASA Technical Standard.		
A.2.4.2	Configurational Flammability (Test 10), Test Criteria	[FOCR 269] Test 10 shall be performed in accordance with the test procedure described in Test 1.		
A.2.4.3a	Configurational Flammability (Test 10), Test Method	[FOCR 270] A test plan specific for the worst-case use environment (maximum temperature, pressure, and oxygen concentration), placement of ignition source, use configuration, and application shall be developed by the test organization in conjunction with the test requester.		
A.2.4.3b	Configurational Flammability (Test 10), Test Method	[FOCR 271] The acceptance criteria shall be determined by the responsible NASA program/project M&P organization.		
A.2.5 Mechanical Impact for Materials in Ambient Pressure LOX (Test 13A) and Mechanical Impact for Materials in Variable Pressure GOX and LOX (Test 13B)				
A.2.5.2a	Test Summary	[FOCR 272] The test specimens shall be 17.5 mm (0.69 in) in diameter.		
A.2.5.2b	Test Summary	[FOCR 273] Materials shall be tested in their use thickness for thicknesses up to 6.35 mm (0.25 in).		
A.2.5.2c	Test Summary	[FOCR 274] If the use thickness is greater than 6.35 mm (0.25 in), the test specimen thickness shall be to 6.35 (± 0.13) mm (0.250 (± 0.006) in).		

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A.2.5.2d	Test Summary	[FOCR 275] The specimens shall be subjected to mechanical impact at the desired energy and inspected for evidence of a reaction.		
A.2.5.3a	Test Criteria	[FOCR 276] When impacted at 98 J (72 ft-lb), 20 specimens shall not react.		
A.2.5.3b	Test Criteria	[FOCR 277] If 1 specimen out of 20 reacts, 40 additional specimens shall be tested without any reactions.		
A.2.5.5a	Test Conditions	[FOCR 278] For ambient pressure LOX tests, the test conditions (pressure and temperature) shall be the test facility ambient pressure and the LOX boiling point at that pressure.		
A.2.5.5b	Test Conditions	[FOCR 279] For variable pressure LOX and GOX tests, the test conditions (pressure and temperature) shall be determined for each test as recommended in the OCA.		
A.2.5.6a	Specimen Preparation	[FOCR 280] Materials shall be prepared as stated in ASTM G86, Standard Test Method for Determining Ignition Sensitivity of Materials to Mechanical Impact in Ambient Liquid Oxygen and Pressurized Liquid and Gaseous Oxygen Environments.		
A.2.5.6b	Specimen Preparation	[FOCR 281] The test material dimensions specified in ASTM G86 shall be used, unless otherwise specified by the responsible NASA program/project M&P organization.		
A.2.5.6c	Specimen Preparation	[FOCR 282] The specimen shall be inspected and flaws noted.		
A.2.5.6d	Specimen Preparation	[FOCR 283] If the flaws result from specimen preparation at the test facility, new specimens shall be prepared.		
A.2.5.6e	Specimen Preparation	[FOCR 284] When testing materials to determine relative ranking, specimen preparation (including cleaning) and condition (for example, thickness, surface finish, product form) shall be identical.		
A.2.5.7	Test System	[FOCR 285] The test system shall be identical to that described in ASTM G86.		
A.2.5.8	Pre-Test Procedure	[FOCR 286] The pre-test procedure, including calibration of the test system, shall be identical to that described in ASTM G86.		
A.2.5.9	Test Procedure	[FOCR 287] The test procedure shall be identical to that described in ASTM G86, with the exception that testing is performed at the requester-specified pressure, temperature, and impact energy.		
A.2.5.10a	Reporting	[FOCR 288] The test report shall include specimen identification, configuration, test conditions, number of reactions, and observations from the test.		
A.2.5.10b	Reporting	[FOCR 289] The test report shall be submitted to MAPTIS in a MAPTIS-compatible format.		
A.2.5.11	Precision	[FOCR 290] Precision of measurements (for example, specimen dimensions, drop height, and time) shall be identical to ASTM G86.		
A.2.6 Gaseous Fluid Impact for Components				
A.2.6.2a	Test Summary	[FOCR 291] The test specimens shall be components in their normal use condition, including the end-use cleanliness level.		

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Section	Description	Requirement in this Standard	Applicable (Yes or No)	If No, Enter Rationale
A.2.6.2b	Test Summary	[FOCR 292] The test specimens shall be subjected to gaseous fluid impact at 1.25 times Maximum Operating Pressure (MOP), followed by functional tests and inspection for evidence of ignition.		
A.2.6.3a	Test Criteria	[FOCR 293] At least one component shall be subjected to a minimum of 60 pneumatic impacts.		
A.2.6.3b	Test Criteria	[FOCR 294] Pressurization shall be through the ports specified by an OCA.		
A.2.6.5a	Test Conditions	[FOCR 295] Test pressure shall be 1.25 times MOP.		
A.2.6.5b	Test Conditions	[FOCR 296] Test gas temperature shall be 60 (±3) °C (140 (±5.4) °F).		
A.2.6.5c	Test Conditions	[FOCR 297] Pressurization rate to within 95 percent of the test pressure shall be 20 (+0/-5) msec to enhance reaction probability in the component.		
A.2.6.6	Specimen Preparation	[FOCR 298] Components shall be tested in their normal use condition, including the end-use cleanliness level.		
A.2.6.7	Test System	[FOCR 299] The test system shall be as described in ASTM G74, Standard Test Method for Ignition Sensitivity of Materials to Gaseous Fluid Impact, with the exception that the test chamber subassembly is replaced by the component to be tested.		
A.2.6.8	Test Procedure	[FOCR 300] The test procedure for testing components shall be as described in ASTM G74 with the following exceptions: a. The test chamber subassembly shall be replaced with the component to be tested. b. The configuration of the component to be tested shall be as recommended in the OCA. c. The component shall be subjected to impacts at intervals of approximately 30 sec in each test configuration recommended in the OCA. d. After testing, each component shall be subjected to functional tests and inspected for signs of ignition.		
A.2.6.9	Reporting	[FOCR 301] The test report shall include specimen identification, configuration, test conditions, number of reactions, and observations from the test.		
A.2.6.10	Precision	[FOCR 302] Measurements shall be made to the following precision: a. Absolute pressure: ±1 percent of reading. b. Temperature: ±3 °C (±5.4 °F).		
A.2.7 Reactivity and Penetration of Materials due to Incidental Exposure to Hydrazine, Monomethylhydrazine, Unsymmetrical Dimethylhydrazine, Aerozine 50, Nitrogen Tetroxide, and Ammonia				
A.2.7.4a	Test Conditions	[FOCR 303] Unless otherwise specified, the test conditions shall be the test facility ambient temperature and pressure.		
A.2.7.4b(1)	Test Conditions, Test Fluids	[FOCR 304] Hydrazine shall meet the requirements of MIL-PRF-26536.		
A.2.7.4b(2)	Test Conditions,	[FOCR 305] UDMH shall meet the requirements of MIL-PRF-25604.		

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Section	Description	Requirement in this Standard	Applicable (Yes or No)	If No, Enter Rationale
	Test Fluids			
A.2.7.4b(3)	Test Conditions, Test Fluids	[FOCR 306] MMH shall meet the requirements of MIL-PRF-27404.		
A.2.7.4b(4)	Test Conditions, Test Fluids	[FOCR 307] Aerozine 50 shall meet the requirements of MIL-PRF-27402.		
A.2.7.4b(5)	Test Conditions, Test Fluids	[FOCR 308] Nitrogen tetroxide and mixed oxides of nitrogen shall meet the requirements of MIL-PRF-26539.		
A.2.7.4b(6)	Test Conditions, Test Fluids	[FOCR 309] Ammonia shall be procured with a stated purity of 99.99 percent (liquid phase).		
A.2.7.5.1a	Specimen Preparation, Films, Fabrics, Sheets, Metals, and Composites	[FOCR 310] Preparation of a specimen of the proper mass or dimensions: specimens shall be prepared to a geometric surface area of 10.2 by 10.2 cm (4 by 4 in).		
A.2.7.5.1b	Specimen Preparation, Films, Fabrics, Sheets, Metals, and Composites	[FOCR 311] Specimens shall be cleaned and dried to end-use specifications.		
A.2.7.5.2	Specimen Preparation, Adhesives and Coatings	[FOCR 312] Adhesives and coatings shall be applied in a thickness equivalent to normal use on aluminum foil or 304L stainless steel substrate and cured, if necessary, in accordance with the manufacturer's instructions.		
A.2.7.5.3	Specimen Preparation, Tapes	[FOCR 313] Tapes shall be applied in the as-received condition and thickness on aluminum foil or 304L stainless steel substrate or a nonreactive substrate, e.g., watch glass, glass Petri dish.		
A.2.7.5.4	Specimen Preparation, Greases and Gels	[FOCR 314] Greases and gels shall be applied on aluminum foil or 304L stainless steel substrate in a thickness equivalent to the normal use and cured, if required, in accordance with the manufacturer's instructions.		
A.2.7.5.5	Specimen Preparation, Complex Shapes	[FOCR 315] Complex shapes, e.g., O-rings, cables, and pipes, shall be tested in a cleanliness and configuration consistent with the intended use.		

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Section	Description	Requirement in this Standard	Applicable (Yes or No)	If No, Enter Rationale
A.2.7.6a	Test System	[FOCR 316] The test system for the reactivity and penetration test for films, fabrics, sheets, metals, adhesives, coatings, and composites shall consist of a nonreactive glass beaker in a fume hood approved for the handling of the test fluid.		
A.2.7.6b	Test System	[FOCR 317] The test system for the reactivity test for all other materials shall consist of a nonreactive substrate (for example, watch glass, glass Petri dish) in a fume hood approved for the handling of the test fluid.		
A.2.7.7a	Pre-Test Procedure	[FOCR 318] The fluid to be used in testing shall be analyzed and verified to meet the required use specifications before use.		
A.2.7.7b	Pre-Test Procedure	[FOCR 319] All pertinent information for the test (for example, specimen identification, composition, pre-test mass and dimensions, and fluid analysis results) shall be recorded.		
A.2.7.7c	Pre-Test Procedure	[FOCR 320] Reference/pre-test photographic images of the specimens shall be obtained.		
A.2.7.7d	Pre-Test Procedure	[FOCR 321] The test system shall be visibly clean.		
A.2.7.7e	Pre-Test Procedure	[FOCR 322] All the measuring devices shall be in current calibration.		
A.2.7.7f	Pre-Test Procedure	[FOCR 323] As a precaution, initial exposure test of the specimen material shall be conducted in accordance with the screening test procedure of Test 15.		
A.2.7.8 Test Procedure				
A.2.7.8.1a	Reactivity and Penetration of Films, Fabrics, Sheets, Metals, and Composites	[FOCR 324] The test material specimen shall be placed over a nonreactive beaker.		
A.2.7.8.1b	Reactivity and Penetration of Films, Fabrics, Sheets, Metals, and Composites	[FOCR 325] The test fluid (typically 1 ml total) shall be added drop by drop, slowly at first, and observed for any evidence of gross reactivity.		
A.2.7.8.1c	Reactivity and Penetration of	[FOCR 326] If no gross reactivity is observed, the remainder of the test fluid shall be added to the center of the specimen, without exposing the edges of the specimen to the fluid.		

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Section	Description	Requirement in this Standard	Applicable (Yes or No)	If No, Enter Rationale
	Films, Fabrics, Sheets, Metals, and Composites			
A.2.7.8.1d	Reactivity and Penetration of Films, Fabrics, Sheets, Metals, and Composites	[FOCR 327] The specimen shall be continuously exposed to the test fluid for the specified exposure time.		
A.2.7.8.1e	Reactivity and Penetration of Films, Fabrics, Sheets, Metals, and Composites	[FOCR 328] The specimen shall be observed for visible signs of reaction (table 4) (for example, burning, smoking, frothing, charring, solubility, swelling, fracture, or penetration of the specimen) throughout the test.		
A.2.7.8.1f	Reactivity and Penetration of Films, Fabrics, Sheets, Metals, and Composites	[FOCR 329] After testing, each component shall be visually inspected for signs of degradation.		
A.2.7.8.1g	Reactivity and Penetration of Films, Fabrics, Sheets, Metals, and Composites	[FOCR 330] The time of the first instance of penetration (first drop of liquid in the beaker) shall be recorded.		
A.2.7.8.1h	Reactivity and Penetration of Films, Fabrics, Sheets,	[FOCR 331] For materials used for protective garments, the time of initial appearance of wetness on the underside of the test specimen shall be recorded.		

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Section	Description	Requirement in this Standard	Applicable (Yes or No)	If No, Enter Rationale
	Metals, and Composites			
A.2.7.8.2a	Reactivity of All Other Materials	[FOCR 332] The specimen shall be placed on a nonreactive substrate, e.g., watch glass, glass Petri dish.		
A.2.7.8.2b	Reactivity of All Other Materials	[FOCR 333] The test fluid shall be added drop by drop, slowly at first, and observed for any evidence of gross reactivity.		
A.2.7.8.2c	Reactivity of All Other Materials	[FOCR 334] If no gross reactivity is observed, the remainder of the test fluid (typically 1 ml total) shall be added to the center of the specimen, without exposing the edges of the specimen to the fluid.		
A.2.7.8.2d	Reactivity of All Other Materials	[FOCR 335] The test fluid shall be allowed to stand on the specimen for the specified exposure time.		
A.2.7.8.2e	Reactivity of All Other Materials	[FOCR 336] The specimen shall be observed for visible signs of reaction (table 4) (for example, burning, smoking, frothing, charring, solubility, swelling, fracture, and penetration of the specimen) throughout the test.		
A.2.7.8.2f	Reactivity of All Other Materials	[FOCR 337] At the end of the specified exposure time, the liquid shall be blotted from the specimen.		
A.2.7.8.2f(1)	Reactivity of All Other Materials	[FOCR 338] The specimen shall be rinsed with running water for 60 sec.		
A.2.7.8.2f(2)	Reactivity of All Other Materials	[FOCR 339] The test specimen shall be allowed to air dry for 24 hr before final evaluation.		
A.2.7.8.2g	Reactivity of All Other Materials	[FOCR 340] Greases or uncured materials shall not be rinsed or blotted.		
A.2.7.8.2g(1)	Reactivity of All Other Materials	[FOCR 341] Any excess test liquid shall be poured off.		
A.2.7.8.2g(2)	Reactivity of All Other Materials	[FOCR 342] The test specimen shall be allowed to stand in air for 24 hr before final evaluation.		
A.2.7.8.3	Post-Test Material Analysis	[FOCR 343] Post-test material analysis shall include photographic documentation, visual observations of the material using the defined standard terms (section 7.15.9, table 4 in this NASA Technical Standard), and determination of the mass and dimensions when possible.		
A.2.7.9a	Reporting	[FOCR 344] The test report shall include specimen identification, configuration, fluid identity, test conditions, and observations from the test.		
A.2.7.9b	Reporting	[FOCR 345] Any reactivity observed during the exposure (for example, burning, smoking, frothing, charring, solubility, swelling, and fracture of the specimen) shall be recorded.		

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Section	Description	Requirement in this Standard	Applicable (Yes or No)	If No, Enter Rationale
A.2.7.9c	Reporting	[FOCR 346] For penetration, the elapsed time at the first instance of penetration (or appearance of underside wetness for protective garments) shall be reported.		
A.2.7.9d	Reporting	[FOCR 347] Relevant post-test analysis results for the material shall be included.		
A.2.7.9e	Reporting	[FOCR 348] Post-test photographic images shall be included, as required, to document condition of the material.		
A.2.7.9f	Reporting	[FOCR 349] The test report shall be submitted to MAPTIS in a MAPTIS-compatible format.		
A.2.7.10	Precision	[FOCR 350] Measurements shall be made to the following precision: a. Absolute pressure: ± 3 kPa (± 0.5 psi). b. Temperature: ± 3 °C (± 5 °F). c. Specimen dimensions: ± 2 mm (± 0.08 in). d. Specimen mass: ± 10 mg.		
A.2.8 Autogenous Ignition Temperature				
A.2.8.5	Test Conditions	[FOCR 351] The test conditions shall be identical to those described in ASTM G72, Standard Test Method for Autogenous Ignition Temperature of Liquids and Solids in a High-Pressure Oxygen-Enriched Environment.		
A.2.8.6a	Test Systems and Procedures	[FOCR 352] The test systems and procedures shall be identical to those described in ASTM G72.		
A.2.8.6b	Test Systems and Procedures	[FOCR 353] At least five specimens shall be tested under each test condition.		
A.2.8.7a	Reporting	[FOCR 354] The test report shall include specimen identification, the method used, test conditions, ignition temperature, temperature rise on ignition, and observations from the test.		
A.2.8.7b	Reporting	[FOCR 355] The test report shall be submitted to MAPTIS in a MAPTIS-compatible format.		
A.2.8.8	Precision	[FOCR 356] Measurements shall meet the ASTM G72 precision requirements.		
A.2.9 Heat of Combustion				
A.2.9.5	Test Conditions	[FOCR 357] The test conditions shall be identical to those described in ASTM D240, Standard Test Method for Heat of Combustion of Liquid Hydrocarbon Fuels by Bomb Calorimeter, or in ASTM D4809, Standard Test Method for Heat of Combustion of Liquid Hydrocarbon Fuels by Bomb Calorimeter (Precision Method).		
A.2.9.6	Test Systems and Procedures	[FOCR 358] The test systems and procedures shall be identical to those described in ASTM D240 or ASTM D4809.		
A.2.9.7a	Reporting	[FOCR 359] The test report shall include specimen identification, the method used, test conditions, gross heat of combustion, and observations from the test.		
A.2.9.7b	Reporting	[FOCR 360] The test report shall be submitted to MAPTIS in a MAPTIS-compatible format.		

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Section	Description	Requirement in this Standard	Applicable (Yes or No)	If No, Enter Rationale
A.2.9.8	Precision	[FOCR 361] Measurements shall meet the ASTM D240 or ASTM D4809 precision requirements.		
A.2.10 Oxygen Index				
A.2.10.5	Test Conditions	[FOCR 362] The test conditions shall be identical to those described in ASTM G125, Standard Test Method for Measuring Liquid and Solid Material Fire Limits in Gaseous Oxidants, or ASTM D2863.		
A.2.10.6	Test Systems and Procedures	[FOCR 363] The test systems and procedures shall be identical to those described in ASTM G125 or ASTM D2863.		
A.2.10.7a	Reporting	[FOCR 364] The test report shall comply with ASTM G125 or ASTM D2863.		
A.2.10.7b	Reporting	[FOCR 365] The test report shall be submitted to MAPTIS in a MAPTIS-compatible format.		
A.2.11 Electrical Arc Test				
A.2.11.2	Test Criteria	[FOCR 366] A material shall pass at the test conditions if there is no ignition in 60 tests.		
A.2.11.3	Test Conditions	[FOCR 367] The following test conditions shall be determined by the OCA or by the test requester: test pressure, oxygen concentration, test specimen material/configuration, voltage, and current.		
A.2.11.4	Specimen Preparation	[FOCR 368] The test specimen configuration shall be specified in the OCA or by the test requester.		
A.2.11.7a	Reporting	[FOCR 369] The test report shall include specimen material and configuration, test pressure, oxygen concentration, and current/voltage required for ignition.		
A.2.11.7b	Reporting	[FOCR 370] The test report shall be submitted to MAPTIS in a MAPTIS-compatible format.		
A.2.12 Frictional Heating				
A.2.12.2	Test Criteria	[FOCR 371] The test criteria shall be defined in the OCA.		
A.2.12.4	Test Conditions	[FOCR 372] The test parameter values shall be determined by the OCA or by the test requester.		
A.2.12.5a	Specimen Preparation	[FOCR 373] The test specimen configuration shall be specified in the OCA or by the test requester.		
A.2.12.5b	Specimen Preparation	[FOCR 374] Before testing, the specimens shall be prepared according to the end-use application or by washing with any warm liquid detergent solution, rinsing with deionized water, and drying with GN ₂ .		
A.2.12.5c	Specimen Preparation	[FOCR 375] The specimens shall then be sealed in clean bags until test.		
A.2.12.5d	Specimen Preparation	[FOCR 376] From this point, the specimens shall be handled with clean, powder-free, lint-free gloves to maintain cleanliness.		
A.2.12.8a	Reporting	[FOCR 377] Results of frictional heating tests shall be summarized in a report that includes documentation of the specimen materials and configuration, the test conditions and apparatus, and test results.		

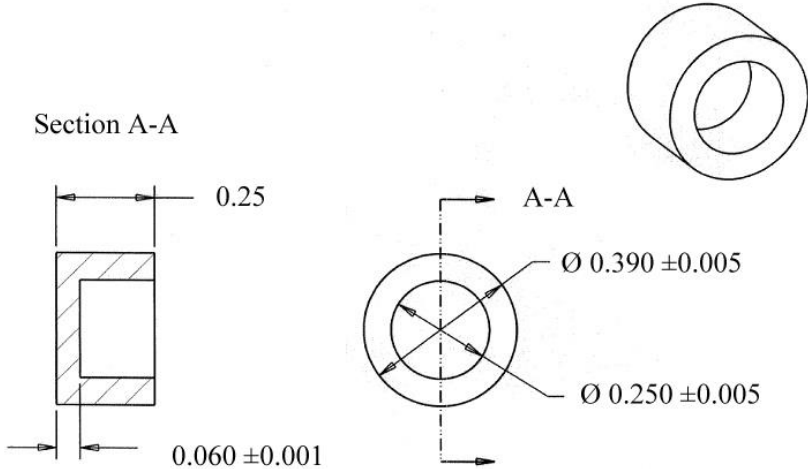
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Section	Description	Requirement in this Standard	Applicable (Yes or No)	If No, Enter Rationale
A.2.12.8b	Reporting	[FOCR 378] If the coefficient of friction is calculated from the data, this value shall be reported.		
A.2.12.8c	Reporting	[FOCR 379] Pre- and post-test photographic images of the test specimens shall be included in the report.		
A.2.12.8d	Reporting	[FOCR 380] The test report shall be submitted to MAPTIS in a MAPTIS-compatible format.		
A.2.13 Particle Impact Test				
A.2.13.2a	Test Criteria	[FOCR 381] The test criteria shall be defined in the OCA or by the test requester.		
A.2.13.2b	Test Criteria	[FOCR 382] For component tests, a minimum of 60 tests shall be performed at a single set of conditions on a single component.		
A.2.13.2b(1)	Test Criteria	[FOCR 383] If there are no ignitions in 60 tests, the component shall pass the test at the test conditions.		
A.2.13.2b(2)	Test Criteria	[FOCR 384] One ignition in 60 tests shall constitute failure of the test at the test conditions.		
A.2.13.4a	Test Conditions	[FOCR 385] The following test conditions shall be determined by the OCA or by the test requester: gas/target temperature, target pressure, gas velocity, target material/configuration, and particle configuration (size, material, and quantity).		
A.2.13.4b	Test Conditions	[FOCR 386] Components shall be tested at the worst-case conditions as determined by the OCA or by the test requester.		
A.2.13.5a	Specimen Preparation	[FOCR 387] The test specimen configuration shall be specified in the OCA or by the test requester.		
A.2.13.5b	Specimen Preparation	[FOCR 388] Before testing, material specimens shall be prepared according to the end-use conditions or by washing with any liquid detergent solution, rinsing with deionized water, and drying with GN ₂ .		
A.2.13.5c	Specimen Preparation	[FOCR 389] The specimens shall then be sealed in clean bags until testing.		
A.2.13.5d	Specimen Preparation	[FOCR 390] From this point on, the targets shall be handled with powder-free, lint-free gloves to maintain cleanliness.		
A2.13.5e	Specimen Preparation	[FOCR 391] The configurations of test targets for the low- and high-velocity testing shall be as follows: (1) Material Specimens. A. High-Velocity Particle Impact Material Specimen Preparation (figure 9, High-Velocity Particle Impact Target Specimen Configuration):		

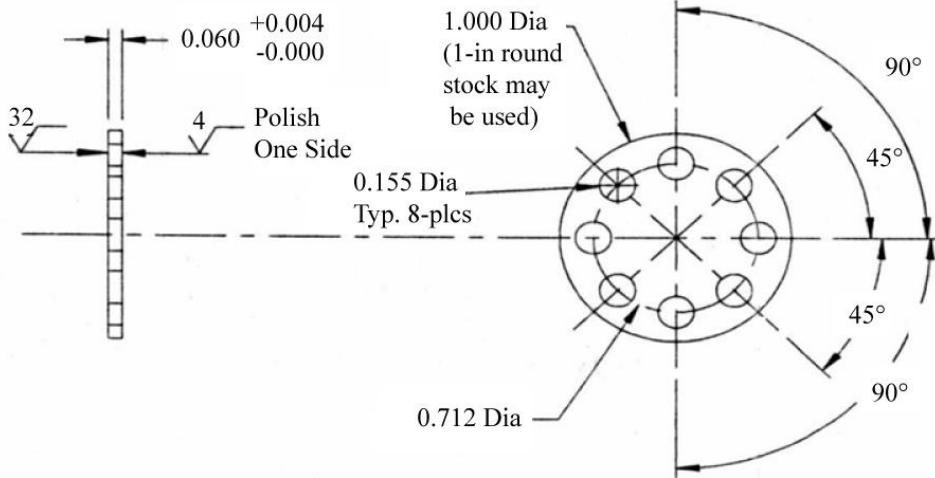
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Section	Description	Requirement in this Standard	Applicable (Yes or No)	If No, Enter Rationale
		<i>Specimens are cup-shaped with an outside diameter of 0.99 cm (0.390 in) and a target surface thickness of 1.5 cm (0.060 in). Each specimen is press fit onto a copper specimen holder for testing.</i>  Figure 9—High-Velocity Particle Impact Target Specimen Configuration (all dimensions are in inches) B. Low-Velocity Particle Impact Material Specimen Preparation (figure 10, Low-Velocity Particle Impact Target Specimen Configuration): <i>Specimens are flat discs, 1.5 cm (0.060 in) thick, with holes drilled radially to allow flow through the outer edge of the target.</i>		

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Section	Description	Requirement in this Standard	Applicable (Yes or No)	If No, Enter Rationale
		 Figure 10—Low-Velocity Particle Impact Target Specimen Configuration (all dimensions are in inches) (2) Component Testing: Components shall be configured in their end-use configuration.		
A.2.13.6a	Test System	[FOCR 392] Particle impact tests shall be conducted in a facility capable of supplying heated GOX at the desired test pressure.		
A.2.13.6b	Test System	[FOCR 393] High-velocity particle impact test system: The test specimen shall be placed on a mounting post directly in line with the gas flow stream.		
A.2.13.6c(1)	Test System	[FOCR 394] A flow orifice, sized to produce the desired gas velocity, shall be placed immediately downstream of either the material specimen or component.		
A.2.13.6c(2)	Test System	[FOCR 395] GOX shall enter the particulate injector at subsonic velocities and flow through the flow straightener section and around the target specimen in the impact chamber.		
A.2.13.8a	Reporting	[FOCR 396] Results of particle impact tests shall be summarized in a report that includes documentation of the specimen target material or component, the particle material and size, the test conditions, and the response of each target to impact.		

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Section	Description	Requirement in this Standard	Applicable (Yes or No)	If No, Enter Rationale
A.2.13.8b	Reporting	[FOCR 397] Pre- and post-test photographic images of the test specimens shall be included in the report.		
A.2.13.8c	Reporting	[FOCR 398] The test report shall be submitted to MAPTIS in a MAPTIS-compatible format.		
Appendix B Certification of Facilities				
B.2	Purpose	[FOCR 399] Certification of test facilities shall be obtained for the performance of any of the required tests included in this document.		
B.3	Purpose	[FOCR 400] Certification shall be the responsibility of a panel appointed by the NASA Office of the Chief Engineer.		
B.4	Purpose	[FOCR 401] The certification panel shall consist of M&P representatives from NASA-certified facilities and be supported by specialists in appropriate disciplines to evaluate specific test methods.		
B.5	Purpose	[FOCR 402] An application for certification shall be submitted according to requirements listed below:		
B.5a	Purpose	[FOCR 403] The test facility applying for certification shall prepare and submit detailed written procedures to perform all aspects of the subject test methods to the certification panel for review.		
B.5b	Purpose	[FOCR 404] Procedures shall include provisions for instrument calibration, specimen preparation, test chamber certification, test atmosphere analysis or verification, data analysis, report preparation, quality control provisions, recording and archiving of test data, materials control, and control of flight articles.		
B.5c	Purpose	[FOCR 405] The test facility procedures shall also include the definition of the minimum acceptable personnel qualifications, training requirements, and personnel certification procedures.		
B.5d	Purpose	[FOCR 406] Changes to these procedures shall be maintained and made available at the request of the certification panel.		
B.5e	Purpose	[FOCR 407] The test facility also shall supply documentation demonstrating that operations performed in the subject test methods present no hazards to personnel or flight hardware.		
B.6	Purpose	[FOCR 408] The certification panel shall perform an on-site inspection to ensure that the test facility can perform the methods from receipt of a material to issuance of a final report.		
B.7	Purpose	[FOCR 409] The certification panel shall recommend to the NASA Office of the Chief Engineer certification of the test facility on the basis of compliance with this document.		
B.8	Purpose	[FOCR 410] The certification panel and test facility shall maintain certification records.		
B.9	Purpose	[FOCR 411] NASA certification of a test facility shall be maintained by meeting the following requirements:		
B.9a	Purpose	[FOCR 412] The test facility shall perform the required test method(s) for which it is certified at least once in an 18-month period from its certification date and report the data to MAPTIS.		
B.9b	Purpose	[FOCR 413] The test facility shall have participated with other NASA-certified test facilities in the most recent round-robin testing for the test method.		

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Section	Description	Requirement in this Standard	Applicable (Yes or No)	If No, Enter Rationale
B.9c	Purpose	[FOCR 414] Round-robin tests for Test 1, Test 4, and Test 7 shall be conducted at least every 2 years.		
B.9d	Purpose	[FOCR 415] The test data shall have been reviewed and approved by the certification panel or the panel's designees.		
B.9e	Purpose	[FOCR 416] Facility test procedures shall be available for audit by the certification panel.		
B.10	Purpose	[FOCR 417] All instrumentation used in the test shall be in current calibration and, where available, bear the appropriate documentation to validate traceability to the appropriate national, international, or intrinsic measurement standards.		
B.11	Purpose	[FOCR 418] The test facility shall ensure and certify that all testing is accomplished in accordance with approved test plans and procedures and that the data records and test results are complete and accurate.		
B.12	Purpose	[FOCR 419] Complete test records shall be prepared by the test facility for each material or system tested.		
B.13	Purpose	[FOCR 420] The test facility shall maintain certification and test data records for the life of the facility.		
B.14	Purpose	[FOCR 421] Records and data shall be submitted to MAPTIS in a MAPTIS-compatible format.		
Appendix D Submitting Material for Testing				
D.2.1	Submitting Material	[FOCR 422] The test material shall be free of flaws and visual contamination.		
D.2.5a	Documentation	[FOCR 423] The MSDS shall be provided.		
D.2.5b	Documentation	[FOCR 424] If the test facility is to prepare material specimens, the manufacturer's literature or preparation instructions, at a minimum, shall be provided.		
D.2.5c	Documentation	[FOCR 425] A completed test request form (section D.3 in this NASA Technical Standard) shall be submitted.		

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